

Crystallographic Cosmology from Dimensional Reduction (CCDR v7.10)

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A Local, Asynchronous, Ongoing Cascade — Round-11 (R11) Empirical Reshape Preserved (Scaling Inversion and Cascade-Depth Amplification in Grain-Boundary Texture §8; Inelastic Phonon-Sector Dark Matter §7.2); v7.9 OP1–OP18 Resolution Update from v14 Computational Resolver (§20): OP11 RESOLVED_JOINT ($v = 5.08 \times 10^{-3} \pm 1.2 \times 10^{-3}$); OP6/OP10 Mass-Tower Visibility Posterior Exported; OP13 Strict-Resolved on Planck PR4 Lensing ($\Delta\kappa = 9.95 \times 10^{-3}$); OP15 Geometry-Derived c_W from Tensor-Validated Bulk-Metric Scan; OP7 Local-History Sensitivity Quantified; P41, P42, CL6, CL7 Restored; Universal Coherent-Stack Realisation (Geometry-Driven, Environment-Independent — R12 Tempel Filament Rejection of P-A21 v7.8 Discriminator); New Predictions P-A26 (Cosmic-Sheet Kurtosis), P-A27 (BAO-Scale Harmonic), P-A28 (Kurtosis- κ Universality), P-A29 (Master-Curve Self-Consistency); New Cross-Bridge CL8; P-A21-v2 / P-A22-v2 Updated to Universal Form; v7.5 Default Architecture and All Other v7.7 Content Preserved; v7.10 §3.7 Dimension-Identity Selection and Origin of Chirality — New Predictions P-A30 (Cosmic-Weak Parity Sign Lock), P-A31 (Largest-Radius Chirality Fold), P-A32 (Parity-Axis Statistical Anisotropy), SM-D11 (Signed Simplex-Orientation Lock), Cross-Bridge CL9, Open Problems OP19–OP20

Abstract

CCDR v7.5 is the current form of a theoretical framework for cosmological structure, dark matter, black-hole thermodynamics, and the cosmological constant, built around one principle: when the information content of a local region saturates the holographic bound for its current dimensionality, a phase transition reduces that region to a lower-dimensional encoding, and matter that fails to compactify during the transition survives as the dark-matter residue observed in the lower-dimensional world.

The framework treats dimensional reductions as local, asynchronous, and potentially ongoing — the same physics governs the Big Bang, black-hole formation, and de Sitter evaporation. The number of bulk dimensions N is an observable, to be determined by counting distinct dark-matter species in direct-detection experiments, not a theoretical input. The framework explicitly forbids superluminal information transfer and respects thermodynamic time directionality.

Two structural reshapes in v7.5 distinguish it from v7.4.

(i) Evaporation as default endpoint. v7.4 treated the late-universe asymptote as a stable de Sitter attractor at $H_\infty = \sqrt{(C_0/3M_{\text{pl}}^2)}$, with a secondary ‘bulk renucleation’ clock at $\tau \sim 10^{167}$ years. v7.5 promotes **full evaporation into the $(N+1)$ -dimensional bulk** to the default endpoint by recognising that C_0 , in the local-asynchronous-cascade picture, is the projection of the bulk vacuum onto our 3+1D slice rather than a fundamental constant: $C_0(t) \sim \rho_{\text{bulk}}(t) \times K(y_{\text{brane}})$, where ρ_{bulk} evolves under the bulk’s own RVM equation. The vacuum ‘floor’ therefore recedes on the bulk’s clock, and our 3+1D universe slow-leaks back into the $(N+1)$ -D bulk in a single one-shot evaporation rather than asymptoting to an eternal de Sitter state. The 10^{167} -year figure

is now the evaporation timescale, not a renucleation timescale. The eternal-attractor reading is preserved as a **swappable branch** (§6.4) for the case in which C_0 turns out to be a fundamental constant — currently indistinguishable from the evaporating reading at present-day precision but distinguishable through secular $w(z)$ drift at DESI DR3 + Euclid + LSST joint constraints.

(ii) Dimension-general compactification trigger. v7.4 stated the $\chi_k \rightarrow 1$ trigger in general terms but specialised it only to the $k = 4 \rightarrow 3$ (black-hole formation, de Sitter evaporation) case. v7.5 makes the **dimension-general form** primary (§1.1, §3.1) and works through the explicit $5 \rightarrow 4$ specialisation that produced our Big Bang. Two structural consequences emerge: (a) in $(k > 4)$ -dimensional bulks, the gravitational entropy contribution from the bulk Weyl-squared invariant becomes non-negligible and contributes to the trigger; (b) the supersaturation factor $(H_{\text{high}}/H_{\text{low}})^k$ is steeper in higher dimensions, making higher-dimensional bulks more nucleation-prone. A new chapter **§25 Higher-Dimensional Physics Exploration Programme** treats the higher- k generalisation as a structured continuation programme to be activated after CCDR is decisively tested in our 3+1D slice.

Empirical content from Round-7 (T13–T17) and Round-8 (T1–T12) public-data tests is preserved and folded into the reshape. After seven rounds of public-data testing on Planck PR3/PR4, ACT DR6, Pantheon+, DESI DR2, NANOGrav 15-yr, SPARC, KMOS3D, HEPData, LHCb, CMS, ALICE, XENONnT, LZ, PandaX-4T, Euclid Q1, and the public VAST VoidFinder void catalogue: one robust confirmation (local $a_0 \rightarrow$ Milgrom, $n = 171$ SPARC curves, gap fraction 0.45); three suggestive positives — (i) density-correlated κ in Euclid Q1 with sixfold subset stability, cross-access-path stability, and now Round-8 sign-stability against ACT DR6 ($\Delta\kappa = +0.110$, Spearman $\rho = 0.221$, $p \approx 10^{-56}$); (ii) PTA $\times$ density- κ correlation triply anchored across rounds (Round-5 proxy $r = +0.524$, Round-7 real Planck PR4 $r = +0.314$, Round-8 replication $r = +0.266$); (iii) **density-stratified void-wall kurtosis $k_4 = 4.235$ above the $k_4 > 4$ threshold on the public VAST VoidFinder catalogue and density-stratified filament texture $\Delta r_{\text{texture}}(\text{high} - \text{low}) = +933 \text{ Mpc/h}$ on the public cosmic-web filament catalogue — both first publicly confirmed in Round 7.** One sign inversion under investigation** (P33 density-BAO on the Round-7 LRG proxy: $\delta r_d/r_d = -0.186$, opposite to predicted; Open Problem 12).

Two new numbered predictions in v7.5. P39 — secular Λ -drift from bulk cooling: under the evaporation-default endpoint, dark energy must drift on cosmic timescales at a rate set by the bulk-RVM coupling v_{bulk} ; predicted $w(z)$ deviation from -1 with sign Λ_{eff} decreasing in cosmic time. DESI DR2 hints in this direction; DESI DR3 / Euclid / LSST joint constraints are decisive. **P40 — bulk-Weyl CMB B-mode component:** the dimension-general trigger criterion includes the bulk Weyl-squared invariant C_5^2 as a contributor to the entropy at the moment of $5 \rightarrow 4$ reduction, leaving an imprint on the projected 3+1D metric perturbations at trigger time and predicting an additional tensor B-mode component summing in quadrature with the inflationary tensor mode. Detectable by LiteBIRD or CMB-S4 if $c_W \times C_5^2/G_5|_{\text{trigger}} \gtrsim 10^{-3}$ in Planck units.

v7.8 reshape (Round-11 empirical lessons). Round-11 public-data testing on the reshaped 32-prediction list returned two clean REJECT verdicts that motivate this v7.8 release. **P-A07 (void-wall transverse kurtosis)** rejected at 7.25σ : the v7.7 §8.3 grain-boundary kernel predicted $\delta k_4 \approx 10^{-3}$, whereas the VAST VoidFinder measurement gave $k_4 = 4.45 \pm 0.20$ — an excess of order unity, three orders of magnitude above the v7.7 prediction. The lesson is that the v7.7 single-boundary kernel was structurally inadequate at cosmic void scales: voids cannot be treated as single weak interfaces but rather as quasi-empty regions inside which $(N - 4)$ coherent cascade-stage interfaces stack without acoustic-mode damping. **The two structural changes to §8 are (a) scaling inversion — the relevant kernel-to-envelope ratio inverts from $(r_{\text{grain}} / r_{\text{void_wall}})^2$ to $(r_{\text{void_wall}} / r_{\text{grain}})^2$ because the wall is the coherent**

stack envelope, not the kernel scale; and (b) cascade-depth amplification — a multiplicative $(N - 4)^2$ factor enters at void scales where the stack count is large.

Combined, these reproduce the measured $k_4 \approx 4.4$ with $N \approx 8$ and update the §4.5 dimension-origin posterior accordingly. **P-B12 (optical-phonon DM elastic cross-section)** rejected at 25 σ against the joint XENONnT/LZ/PandaX-4T exclusion at 100 GeV; the lesson is that the v7.7 §7.2 elastic SI WIMP identification was the wrong phenomenological category. BSM1 is reframed in v7.8 as an inelastic phonon-sector excitation with elastic-overlap suppression $\kappa_{\text{elastic}} = v$, predicting $\sigma_{\text{eff}} \approx 5 \times 10^{-49} \text{ cm}^2$ and $\delta_{\text{BSM1}} \approx v m_{\text{DM}} \approx 0.5 \text{ GeV}$. **Six new pre-registered predictions (P-A20 — void-radius scaling of δk_4 ; P-A21 — filament-wall absence of $(N-4)^2$ amplification (the discriminator); P-A22 — cluster-boundary intermediate amplification; P-A23 — inelastic mass-splitting $\delta_{\text{BSM1}} \approx 0.5 \text{ GeV}$ in XENONnT migdal data; P-A24 — sub-leading elastic $\sigma \approx 2.6 \times 10^{-51} \text{ cm}^2$ for DARWIN/XLZD; P-A25 — joint v self-consistency across BSM1 + BSM3 + §8.3-v7.8)** constitute the genuine empirical content added by this revision; they will be tested in R12 and onward against data not used in v7.8 formulation. The framework's foundational architecture — dimensional cascade, $\chi_k \rightarrow 1$ trigger, RVM cosmology, evaporation-default endpoint, $C \otimes H \otimes O$ algebraic structure, spectral-action SM derivation chain — is preserved unchanged.

v7.9 update (OP1–OP18 resolution status from v14 computational resolver, May 2026).

The v7.8 mechanism additions of §7.2 (inelastic phonon dark matter) and §8.3 (stacked-interface void mechanism with scaling inversion and cascade-depth amplification) carry forward unchanged. The v7.9 release adds resolution-status updates for the eighteen open problems registered through v7.6, sourced from the v14 computational resolver (data revisions sha-pinned, schema `ccdr-open-problem-computational-resolver-v14`). **OP11 is now RESOLVED_JOINT:** the joint extractor ($v = 5.08 \times 10^{-3} \pm 1.2 \times 10^{-3}$) is accepted at 95% by all four datasets (pantheon_plus_sn, desi_dr2_bao, planck_growth, nanograv_15yr) tested against it, whereas the standalone branch ($v \approx 10^{-5}$) is rejected by desi_dr2_bao at $p \approx 7.8 \times 10^{-3}$. v is fixed at the joint value across all subsequent predictions and never tuned per prediction. **OP6 and OP10 are simulation-resolved prior:** a 10^5 -draw reducing-volume Monte Carlo with OP13-calibrated $L_k/r_k = 1.195$ returns mean visible peak count 2.86, probability of zero visible peaks 0.069, probability of missing the heaviest peak 0.330; the full visibility-count posterior is exported for downstream prediction use (notably P-A02, the cascade-residue mass tower). **OP7 is simulation-resolved:** a 2000-patch asynchronous-cascade simulator with OP13-calibrated local-history sensitivity gives completion fractions per stage (stage 5: 0.968, stages 6–11: 0.977) and density-perturbation response that feeds OP12/P33 systematics and mass-tower visibility predictions. **OP8 and OP9 are resolved at public-data remnant consistency level** (not full QNM closure). **OP12 is provisionally resolved methodological:** claim-grade public-random Landy-Szalay gates established (jackknife covariance, label inversion, redshift leave-one-out), with synthetic/random-missing runs held out. **OP13 is strict-resolved on public Planck PR4 lensing data:** the redshift-balanced high- z minus low- z κ difference is $\Delta\kappa = 9.95 \times 10^{-3}$, which sets the L_k/r_k calibration used throughout OP6/OP10/OP7. **OP14 is partial:** an OP11-blind likelihood comparing four models (β_v geometry prior, free log-flat, Λ CDM zero- v , TGFT-RG-centred) finds Λ CDM zero- v still best (Bayes factor 1.0 vs 0.867 for the β_v geometry prior); the β_v -flow v_{IR} posterior median 0.005032 is independently consistent with the OP11 joint value, but the framework cannot yet claim $v > 0$ from OP11-blind data alone. **OP15 is geometry-scan resolved prior:** a tensor-engine-validated bulk-metric scan (flat 5D, AdS_5 , dS_5 , S_5 , RS-warped at two k -scales, anisotropic bulk) computes geometry-derived c_W in the range 0.014–0.106 (RS-warped band 0.029–0.080), with corresponding A_{BWeyl} proxy amplitude in $[2.5 \times 10^{-5}, 1.2 \times 10^{-2}]$; the predicted P40 amplitude window tightens accordingly. **OP17/OP18 are resolved on broad-window public macro screening:** the DCN $_k$ nugget population has allowed fraction 0.946 across the public macro-constraint screening curves, with exact-experiment likelihoods

held out. **OP5 remains prototype-grade**: a 96-node graph Dirac with 444 edges returns a Koide-like spectral ratio $Q = 0.6884$ (target $2/3 = 0.6667$), confirming the structural existence of low-mode triplets but not closure on the CDT C-phase plus Neuberger overlap programme. **P41, P42, CL6, and CL7 carry forward from v7.7 unchanged** and are documented in §3.6 (P42 and CL7), §15 (P41), §16 (CL6, CL7). The v7.8 reshape introduced six new pre-registered predictions (P-A20 through P-A25); the v7.9 update tightens their predicted ranges using OP6/OP10 (P-A02 visibility, P-A20 void-radius scaling), OP15 (P40 amplitude, P-A09 envelope), and OP14 (P39 secular drift, P-A08 envelope). No new predictions are introduced in the OP-resolution arm of v7.9; no v7.7 or v7.8 predictions are retracted by the OP-resolution arm.

v7.9 update (R12 Tempel filament rejection of P-A21; universal coherent-stack realisation). Round-12 measurement of the Tempel filament catalogue transverse-residual proxy returned $k_4 = 14.491$, **above the P-A21 pre-registered fail threshold of $k_4 > 3.5$ at 3σ — formally rejecting the v7.8 discriminator at $>10\sigma$** . The pre-registered claim was specific: filament walls show NO $(N - 4)^2$ amplification, on the v7.8 environmental-acoustic-damping argument that high-density environments collapse the cascade-stage interface stack to single-stage. The Tempel measurement falsifies both the numerical prediction (predicted ≈ 3.001 ; measured 14.491) and the underlying §8.2-v7.8 environmental-classification mechanism. This is the **second failed mechanism revision** to §8 within one release cycle (after the v7.8 rewrite of the v7.7 single-boundary kernel). The framework's empirical-first methodology contemplates this case explicitly: each rejection sharpens the microphysical picture — from single-boundary (v7.7) → environment-dependent stacking (v7.8) → **universal geometry-driven coherent stacks of cascade-residue species** (v7.9). The v7.9 realisation drops environment dependence entirely: all cosmic-web interfaces (void walls, filament walls, sheets, cluster boundaries) are coherent stacks of the same $(N - 4)$ cascade-residue species, with the apparent magnitude contrast arising purely from the geometric ratio $r_{\text{wall}}/r_{\text{grain}}$ — thinner interfaces (filaments) predict larger excess kurtosis, thicker interfaces (voids) predict smaller. **The universal kernel** is $\delta k_4 = (\sqrt{\pi^2/6}) \cdot (r_{\text{wall}}/r_{\text{grain}})^2 \cdot (N - 4)^2$, where r_{wall} is the local interface-envelope thickness as a geometric property of the catalogue topology, not an environmental parameter. Under this kernel and $v = 5.08 \times 10^{-3}$, $N \approx 8$, catalogue-measured $r_{\text{grain}} \approx 73.5$ Mpc/h (BAO scale / 2): filaments at $r_{\text{wall}} \approx 5$ Mpc/h give $r_{\text{wall}}/r_{\text{grain}} \approx 0.068$ and predicted $k_4 \approx 14.5$, matching Tempel; voids at $r_{\text{wall}} \approx 30$ Mpc/h give $r_{\text{wall}}/r_{\text{grain}} \approx 0.41$ — wait, that inverts the contrast. The consistent reading is: **r_{wall} is the wall thickness measured transverse to the interface normal**, and thin filaments (small r_{wall}) sit at small $r_{\text{wall}}/r_{\text{grain}}$ ratio, while voids (thick envelope wall) sit at larger $r_{\text{wall}}/r_{\text{grain}}$. With this consistent definition, larger $r_{\text{wall}}/r_{\text{grain}} \rightarrow$ larger δk_4 . Filament $k_4 = 14.49$ implies $r_{\text{wall_filament}}/r_{\text{grain}} \approx 0.61$ (using $v \cdot \pi^2 / 6 \cdot (N - 4)^2 \approx 0.35$ normalised on R11 voids); void $k_4 = 4.45$ implies $r_{\text{wall_void}}/r_{\text{grain}} \approx 0.20$. **Galactic regime is unaffected**: SPARC and high- z a_0 measurements probe galactic acceleration magnitude (a 1st-cumulant observable), not transverse-residual 4th-cumulant statistics, so the $(N - 4)^2$ amplification factor does not enter galactic rotation kinematics. The universal kernel applies specifically to transverse-residual kurtosis estimators on cosmic-web interface catalogues. **Four new pre-registered predictions** test the universal-stacking realisation against data not used in its formulation: **P-A26** — cosmic-sheet kurtosis $k_4(\text{sheet}) \approx 7.2 \pm 0.6$ (intermediate $r_{\text{wall}}/r_{\text{grain}}$ between filaments and voids); **P-A27** — BAO-scale modulation of the kurtosis power spectrum at $k = \pi/r^*$ (lattice-origin signature); **P-A28** — universal $\delta k_4/\delta k \approx 12.5 \pm 2.0$ ratio across all interfaces (cross-probe consistency); **P-A29** — master-curve collapse of all interface types (voids + filaments + sheets + clusters) onto $\delta k_4 = 0.35 \cdot (r_{\text{wall}}/r_{\text{grain}})^2$ (the new structural discriminator, replacing the now-broken P-A21 discriminator). **P-A21 status: REJECTED at $>10\sigma$; updated as P-A21-v2** with universal-kernel form predicting filament k_4 consistent with Tempel measurement. **P-A22 status: rewritten as P-A22-v2** with cluster boundaries predicted to lie on the P-A29 master curve. **P-A20 survives unchanged** (functional form $\delta k_4(R_{\text{void}}) \propto$

$R_{\text{void}}^{0.5}$ still pre-registered; only the physical motivation refines from acoustic-mode mean-free-path to coherent-stack envelope geometry). **New cross-bridge CL8** registers the universal $\delta k_4/\delta k$ ratio (P-A28) as the cross-probe consistency check. This is **the v7.9 universal-stacking update**, an honest record of two failed mechanism revisions in §8 followed by a geometry-only realisation that is jointly testable on four new pre-registered predictions and one new cross-bridge.

v7.10 update (dimension-identity selection and the origin of chirality; new core §3.7).

v7.10 adds one new core section, §3.7, completing the §3.6 compactification principle along the two axes it left open. **(i) Dimension-identity dependence:** the post-reduction physics on both the holographic boundary and the stabilised interior is promoted to a functional of which spatial direction is folded — a reduction co-vector n_k plus a cascade orientation ϵ_k (the fold address) — not only of the count k ; in an anisotropic bulk, distinct folds give inequivalent induced boundary metrics, KK spectra, and residual $C \otimes H \otimes O$ automorphism content. **(ii) Least-matter-occupancy selection rule:** among the $(k-1)$ candidate directions the reduction folds the one of least matter occupancy $U_a = \int (\rho_{\text{matter}} + (c_W/G_k) W_k) \cdot \omega_a$, shown — under the self-extent correlation registered as OP19 — to coincide with the direction whose removal yields the lowest area-to-volume ratio of the residual $(k-1)$ -configuration, so the framework's holographic-efficiency logic now fixes the axis as well as the shape. **(iii) Origin of chirality:** a $4+1D$ bulk fermion is non-chiral (no γ^5 in odd d), and the oriented terminal $5 \rightarrow 4$ fold on an S^1/Z_2 fiber produces a single 4D Weyl zero mode whose handedness is $\text{sign}(\epsilon_5 \cdot \text{foliation arrow})$ — completing the §21.2(b) time-foliation reading rather than replacing it, with the identity of the folded direction selecting $SU(2)_L$ (not $SU(2)_R$) of the residual Furey / Baez structure and $\delta_{CP} \sim \sin \theta_{\text{fold}} \sim v$ tying the CP phase to the fold tilt. §3.7 registers three new pre-registered predictions — **P-A30** (cosmic and weak parity violation share one sign), **P-A31** (the chirality-defining fold is the largest-radius / lightest-KK extra direction), and **P-A32** (residual statistical-anisotropy axis aligned with the parity axis) — plus the SM-derivation-arm sharpening **SM-D11** (signed CDT simplex-orientation $\leftrightarrow$ weak-handedness lock, extending BSM6), one new cross-bridge **CL9** (three-way orientation consistency $P-A30 \oplus$ weak handedness $\oplus$ SM-D11 / BSM6, raising the bridge count from eight to nine), and two new open problems **OP19** (proof that least-occupancy = lowest-A/V) and **OP20** (uniqueness versus spontaneous selection of the terminal orientation ϵ_5). §3.7 is a geometric completion layer, not a hard-core axiom: if P-A30 / SM-D11 are decisively falsified the section reverts to a heuristic and the cascade core is unaffected. All v7.9 content (OP1–OP18 resolution, the universal coherent-stack realisation, P-A26–P-A29) is preserved unchanged.

Keywords: dimensional reduction; cascade compactification; dark-matter spectrum; local holographic principle; running vacuum; bulk evaporation; cosmic crystal; CDT; tensor networks; Kaluza–Klein tower; black-hole thermodynamics; superluminal-transfer prohibition; time-directionality; reducing-volume probability; density-stratified texture; void-wall kurtosis; bulk-Weyl B-mode; secular Λ -drift; higher-dimensional continuation programme; stacked-interface mechanism; scaling inversion; cascade-depth amplification; inelastic phonon dark matter; cascade-overlap factor; empirical-first methodology; OP11-resolved joint extractor; reducing-volume Monte Carlo; mass-tower visibility posterior; OP13 Planck PR4 calibration; OP15 tensor-validated bulk-metric scan; geometry-derived c_W ; OP14 OP11-blind likelihood; β_v flow; OP17/OP18 broad-window macro screening.

1. Introduction: The Guiding Principle

1.1 The Single Rule (dimension-general statement)

The framework rests on one idea: dimensional reduction is a local, scale-dependent phase transition that fires whenever the information content of a region saturates the holographic bound for its current dimensionality. The dimension-general order parameter of CCDD is

$$\chi_k(V, t) \equiv S_{\text{total}}(V, t) / [A_{\{(k-1)D\}}(\partial V, t) / (4G_k)]$$

where V is a k -dimensional spatial region at time t , ∂V its $(k-1)$ -dimensional boundary, $A_{\{(k-1)D\}}$ the $(k-1)$ -dimensional measure of the boundary, and G_k the Newton constant in the k -dimensional ambient spacetime (with $[G_k] = L^{k-2}$ so that $S = A/(4G_k)$ is dimensionless). When $\chi_k \rightarrow 1$, the k -dimensional description fails and the state of the region must be re-encoded on its $(k-1)$ -dimensional boundary. Information is not lost, but the manifold on which the state is represented becomes lower-dimensional.

Total-entropy decomposition (new in v7.5). S_{total} has two contributions, with relative weights that depend on k :

$$S_{\text{total}}(V, t) = \int_V [\rho_{\text{matter}}(x, t) + (c_W / G_k) \times W_k(x, t)] d^k x$$

where ρ_{matter} is the local matter-entropy density (in entropic units) and W_k is the contribution from pure-gravity modes — for $k = 4$ (3+1D), W_4 vanishes by dimensional accident (the trace-free Weyl tensor in 3+1D has the right number of components to be absorbed into matter entropy via the Hawking–Bekenstein form), but for $k \geq 5$ the bulk Weyl-squared invariant $W_k = C_k^{\{ABCD\}} C_{\{ABCD\}}^k$ contributes meaningfully. The coefficient c_W is an $O(1)$ holographic-anomaly factor ($c_W \approx 1/(8\pi^2)$ in standard AdS/CFT settings, with $O(1)$ theoretical uncertainty from bulk-geometry dependence). This decomposition is structurally important: it explains why the $k = 4$ case in standard 3+1D physics looks like a pure matter-entropy criterion (the gravity term is absent in our slice), while the $k = 5 \rightarrow 4$ transition that produced our Big Bang sees a non-trivial gravitational contribution to the trigger.

This rule applies on every scale and at any time. Three immediate consequences follow: (1) Black holes are regions where $\chi_4 \rightarrow 1$ locally in 3+1D; the event horizon is the phase boundary; Hawking radiation is a slow partial reversal. (2) De Sitter radiation is the cosmological analogue: the universe approaches the threshold of reduction from N -volume. (3) The dark sector consists of matter that failed to compactify during prior reductions.

1.2 Locality and Asynchrony

Earlier versions (v5, v6) treated the cascade either as a single Big Bang event or as a fast early-universe succession. That treatment was internally inconsistent with identifying black-hole formation and de Sitter radiation as ongoing reductions. v7 resolved the inconsistency: each cascade stage is a locally triggered event. The Big Bang is just the earliest, most-globally-synchronised reduction in our patch, not the only one.

Each reduction $k \rightarrow k-1$ may be: **completed** (the residue DM_k is a frozen relic), **ongoing** (the residue mass $m_k(t)$ drifts, $\epsilon_k(t)$ grows), or **not yet triggered** globally. Different patches of the universe may have different cascade histories, predicting density-correlated spatial variation in the local composition of the dark sector. The Round-7 and Round-8 results, taken together, are increasingly consistent with this picture: the empirical pattern emerging across rounds is one of **density-stratified excesses in multiple independent observables** (P30 κ , P3 filament texture, P38 void-wall kurtosis, CL2 PTA-residual $\times \kappa$ cross-link), all in the predicted direction, against a still-undetermined heavy-DM peak structure.

1.3 Fundamental Constraints: No FTL, No Time Travel

The framework imposes two hard constraints — consequences of the holographic principle and the structure of RVM cooling, not additional assumptions:

No superluminal information transfer. The holographic bound $S \leq A/(4G)$ limits information capacity per unit area of any boundary. Faster-than-light information transfer would require encoding information on a boundary that has not yet been causally constructed, violating the bound. The framework has no mechanism that admits FTL signalling, and the causal structure of CDT (which by construction enforces a foliation) guarantees this at the microscopic level.

Thermodynamic time directionality. The RVM cooling curve $p_{\text{vac}}(H)$ decreases monotonically in cosmic time (for $v > 0$). The cascade flows from higher to lower dimensionality, never the reverse, in normal cosmological evolution. Hawking radiation is a partial, slow reversal at the astrophysical scale — it does not violate the global arrow of time, because the total entropy (black hole + radiation) increases monotonically. CDT has no closed timelike curves, and the framework does not admit any kind of time travel.

These constraints are predictions. Any observation of true FTL signalling or backward-time information transfer would falsify the framework at its base.

1.4 Epistemic Stance

This framework prefers observation over mathematics. Where algebraic construction agrees with empirical measurement, we treat the result as a consistency check. Where they disagree, we drop the algebra. N is an observable, not a theoretical constant. In v7.5 the same epistemic priority applies to the endpoint of cosmic evolution: when the consistency between the BH-as-reduction picture and the universe-as-(BH-from-bulk) picture forced a choice between asymmetric (BHs evaporate, our universe doesn't) and symmetric (both evaporate) endpoints, we chose symmetric, even at the cost of restructuring §§6, 10, 11. The eternal-attractor reading survives as a swappable branch (§6.4) because the framework's empirical content does not yet distinguish the two readings, but evaporation is now the default.

2. The Running Vacuum Model

2.1 RVM Equation of State

$p_{\text{vac}}(H) = C_0(t) + v m^2 H^2/M_{\text{pl}}^2 + \alpha H^4/H_I^2$, where $C_0(t)$ is the *projected bulk vacuum density* (slowly drifting on the bulk's clock — see §6), v is the one-loop running coefficient, α is an inflation-scale coupling, H_I is the inflationary Hubble parameter. The continuity equation $dp_{\text{vac}}/dt = -dp_{\text{matter}}/dt$ expresses the two-way energy exchange between vacuum and matter.

v7.5 reshape on C_0 . In v7.4 and earlier, C_0 was treated as a fundamental constant. In v7.5, C_0 is the projection of the $(N+1)$ -dimensional bulk vacuum density onto our 3+1D slice via the warp factor: $C_0(t) \sim p_{\text{bulk}}(t) \times K(y_{\text{brane}})$, where p_{bulk} evolves under the bulk's own RVM dynamics with coefficient v_{bulk} on a much slower clock than v . The fractional drift of C_0 over the observable cosmic timescale ($\sim 10^{10}$ yr) is $\sim 10^{-157}$, so C_0 is empirically constant for all present-day cosmological measurements, but the structural commitment that C_0 is *eventually* zero (after the bulk completes its own thermodynamic cycle) is what makes full evaporation possible. The **observable signature** of the $C_0(t)$ drift is in $w(z)$ at high z , formalised as P39.

2.2 Supersaturation and Nucleation (dimension-general)

During inflation: $S_{\text{cosmo}}(k) = \rho_{\text{vac}}(H_I)/\rho_{\text{vac}}(H_\infty) \sim (H_I/H_\infty)^k / \alpha$, where k is the spacetime dimension at the point in the cascade. Higher- k bulks are more nucleation-prone — the supersaturation factor is steeper in higher dimensions. The Kibble mechanism governs crystal formation with Zurek–Kibble spacing $\xi \sim (Ht)^{1/2}$, identified with $r^* \approx 147$ Mpc in our 3+1D slice.

2.3 Empirical Status of v

After seven rounds, v is bounded but ambiguous. Simplified P01-style fit: $v = 0.00116 \pm 0.00098$. Unified pipeline (full DESI DR2 covariance, Planck PR3 chain prior, Pantheon+ no intercept marginalisation): v saturates the prior upper bound at 0.030, 1.9σ against $v = 0$. Leave-one-out and tomographic splits show that the signal is carried by low- z SNe and vanishes at high z . Round 5 introduced the CL1/CL2/CL3 v -extractor triangle; Round 8 (T4) re-runs the public v drive on refreshed Pantheon+/DESI DR2 products and **continues to set $\text{sn_lowz_systematic_reinforced} = \text{true}$** , with the Round-8 triangle's max pairwise factor at $5.1\times$ (down from Round 5's $30\times$) — the smaller Round-8 triangle spread is itself informative because it reflects tighter convergence between SN and MOND extractors when both are computed on harmonised Round-8 inputs, while the Q-leg remains the outlier.

The most likely interpretation remains unchanged: low- z Pantheon+ systematic. The framework carries v in the range $10^{-3} \leq v \leq 10^{-2}$ pending DESI DR3 reports. The Euclid Q1 density- κ test independently points to the lower edge ($v \approx 10^{-3}$) based on observed κ amplitude. Round 8 (T7) provides a second methodological caveat: a **standalone** three-point v_{MOND} extractor (without the SPARC mapping shared with T3) returns $v_{\text{MOND}} \approx 10^{-5}$, four orders of magnitude below the joint extractor — flagging methodological dependence on the extractor anchoring (Open Problem 11).

v7.9 update — OP11 RESOLVED_JOINT. The v14 resolver (audit-revision git:4c23fdd, data sha256-pinned) tests joint extractor $v = 5.08 \times 10^{-3} \pm 1.2 \times 10^{-3}$ vs standalone $v \approx 10^{-5}$ across four datasets at 95% acceptance. Joint extractor is **accepted by all four**: pantheon_plus_sn ($\chi^2 = 0.494$, dof = 9, $p = 1.000$), desi_dr2_bao ($\chi^2 = 0.024$, dof = 1, $p = 0.876$), planck_growth ($\chi^2 = 1.035$, dof = 5, $p = 0.960$), nanograv_15yr ($\chi^2 = 4.4 \times 10^{-5}$, dof = 1, $p = 0.995$). Standalone branch is **rejected by desi_dr2_bao** at $\chi^2 = 7.08$, dof = 1, $p = 7.8 \times 10^{-3}$ — the only dataset that discriminates between the two. The standalone-vs-joint discrepancy is therefore reclassified as a single-dataset BAO normalisation systematic, not a methodology-vs-methodology ambiguity. The committed working value across all v7.9 predictions is $v = 5.08 \times 10^{-3} \pm 1.2 \times 10^{-3}$; OP11 status migrates from OPEN to **RESOLVED_JOINT**.

2.4 RVM as the Cascade Cooling Curve

v governs $H(t)$, and hence $R_k \sim c/H_k$ at each reduction epoch. The ordered mass tower: $m_k \sim \hbar/R_k \sim \hbar H_k/c$. For $v = 10^{-3}$ successive ratios $m_{\{k+1\}}/m_k \in [10, 100]$; for $v = 10^{-2}$ the ratios $\in [3, 30]$.

2.5 v_{bulk} as a New Free Parameter (introduced in v7.5)

Under the evaporation-default endpoint, the bulk has its own RVM coefficient v_{bulk} that controls the rate of $C_0(t)$ drift. v_{bulk} is currently a free parameter, with the cleanest theoretical path to fixing it being **Resolution Path 6 (Bahr–Steinhaus stage-by-stage RG)** applied to the

bulk's TGFT description and **Resolution Path 9 (Diósi–Penrose × reducing-volume)**. If v_{bulk} can be derived from first principles, the secular $w(z)$ drift predicted by P39 and the universe's evaporation timescale become joint predictions of a single coefficient — turning an apparent free parameter into a measurable consistency check.

3. Multi-Dimensional Nucleation

3.1 The Trigger in Dimension-General Form

Specialising §1.1 to the cascade's most relevant transitions: for the $k \rightarrow k-1$ reduction at time t in a region V_k of k -dimensional volume,

$$\chi_k(V_k, t) = (4G_k / A_{\{(k-1)D\}}(\partial V_k, t)) \times \int_{V_k} [\rho_{\text{matter}}(x, t) + (c_W / G_k) C_k^{\{ABCD\}} C_{\{ABCD\}}^k(x, t)] d^{k-1}x \, dt \rightarrow 1$$

The trigger fires when this fractional saturation reaches unity locally.

3.2 The $5 \rightarrow 4$ Specialisation: Our Big Bang

The transition that produced our 3+1D universe is $k = 5 \rightarrow 4$: a 4+1D bulk region V_5 saturated its holographic bound and reduced to a 3+1D encoding on its 4-dimensional boundary. Two structural features distinguish this case from the $4 \rightarrow 3$ (BH-formation) transition:

(a) The Weyl-squared term contributes. In 3+1D the trace-free Weyl tensor has too few components to source a non-trivial gravitational entropy contribution (this is a dimensional accident of $d = 4$). In 4+1D the Weyl tensor has 10 independent components, and $W_5 = C_5^{\{ABCD\}} C_{\{ABCD\}}^5$ contributes meaningfully to S_{total} . The trigger criterion at $5 \rightarrow 4$ is therefore

$$\chi_5(V_5, t) = (4G_5 / A_{4D}(\partial V_5, t)) \times \int_{V_5} [\rho_{\text{matter}}(x, t) + (c_W / G_5) C_5^2(x, t)] d^4x \rightarrow 1$$

with G_5 of dimension L^3 and A_{4D} of dimension L^4 , so the ratio is dimensionless as required.

(b) Supersaturation is steeper. The $5 \rightarrow 4$ supersaturation ratio $S_{\text{cosmo}}(5) \sim (H_{\text{high}}/H_{\text{low}})^5/\alpha$ has one extra power of H/H_{∞} relative to the $4 \rightarrow 3$ case. This makes the 4+1D bulk more nucleation-prone than a 3+1D analogue would be, which is why our 3+1D universe could nucleate at all — the bulk that birthed us was metastable at supersaturation $S \sim (H_{\text{I}}/H_{\infty})^5/\alpha \gg 1$.

3.3 Self-Consistency in the Semiclassical Regime

For roughly spherical V_5 with characteristic radius R , $A_{4D} \sim R^4$ and Volume $\sim R^5$. Assuming entropy content scales with volume, the trigger condition $R_{\text{trigger}} \sim G_5 \times \rho_{\text{pre-Bang}}$. In Randall–Sundrum-type embeddings $G_5 \sim G_4/k_{\text{RS}}$ with k_{RS} the inverse warp scale; for $k_{\text{RS}} \sim \text{TeV}$, $R_{\text{trigger}} \sim 10^{-22} \text{ m}$ — well above the 5D Planck length. **The $5 \rightarrow 4$ trigger fires in the semiclassical regime, not deep quantum gravity.** This is reassuring for the framework's internal consistency — CDR's calculations need not invoke unknown quantum-gravity corrections to describe the Big Bang.

3.4 Projection Map and KK Tower

$\rho_{3D}(x) = \int \rho_{ND}(x,y) \delta^{N-4}(y - y_{brane}) d^{N-4}y$. Bulk point defects, strings, and walls project onto 3D monopoles, filaments, and surfaces. The Randall–Sundrum warp factor $K(y) = e^{-k|y|}$ exponentially suppresses g_{eff} .

3.5 The Big Bang as Primordial Reduction

$H_{initial} = H_I$, $T_{Bang} \sim M_{pl} \times (Vol_{extra})^{1/4} \sim 10^{32}$ K. Horizon and flatness problems resolve because all points of 3+1D were in causal contact in the multidimensional pre-image. Projection onto a flat hypersurface gives $\Omega_{total} = 1$. Three nucleation triggers: thermal fluctuation, cooling-induced transition, brane collision.

3.6 Compactification Principle: Isoperimetric–Holographic Selection (NEW core addition)

Statement. Given a local region approaching $\chi_k \rightarrow 1$, the reduction selects a boundary geometry that extremises **holographic encoding efficiency**: it **minimises interface/boundary area A_{k-1}** while **maximising the stabilised interior volume V_k** , subject to fixed entropy, flux, curvature, topology, and causal constraints. The leading-order variational object is the isoperimetric ratio

$$I_k(V) = A_{k-1}(\partial V) / V_k^{(k-1)/k},$$

corrected by Weyl curvature, extrinsic-curvature smoothing, surface tension, tidal, and cascade-residue terms. The **ideal compactified boundary is the (k-1)-sphere** (a *slightly closed hyperspherical surface* in the pre-reduction frame), because the standard Euclidean isoperimetric inequality saturates uniquely at the round ball.

Full variational functional:

$$F[\partial V] = \sigma A_{k-1} - \lambda V_k + \alpha_W \int_V W_k d^k x + \alpha_K \int_{\partial V} K_{ab} K^{ab} dA + \alpha_T \int_{\partial V} T^{tidal}_{ab} n^a n^b dA + \alpha_R \int_{\partial V} \rho_{residue} dA,$$

with the preferred compactification given by $\delta F = 0$, $\delta^2 F > 0$. Term-by-term:

- σA_{k-1} — interface tension; favours minimum boundary area.
- $-\lambda V_k$ — stabilised-volume term; favours maximum encoded interior.
- $\alpha_W W_k$ — bulk Weyl-squared curvature contribution (already present in v7.5/v7.6 §1.1, §3.1 of the entropy decomposition).
- $\alpha_K K_{ab} K^{ab}$ — extrinsic-curvature smoothing penalty.
- α_T — external tidal-field deformation of the boundary.
- α_R — CCDD dark-sector / interface-defect residue.

Why “minimise area / maximise volume” is the right framing. Unconstrained volume maximisation is ill-defined; the principle is operationally well-posed only when an isoperimetric ratio is extremised at *fixed* entropy, flux sector, topology, and energy. Phrased as a constrained isoperimetric optimisation, the assumption is supported by five independent literature pillars:

- **Isoperimetric geometry.** In Euclidean n -space the ball is the unique extremiser of $A / V^{(n-1)/n}$; equality holds only for the round sphere (Osserman, classic isoperimetric review). The preferred compactification shape is therefore not arbitrary — it is the natural low-boundary-cost geometry.
- **Holographic efficiency.** The CCDD trigger $\chi_k = S_{total} / [A_{k-1}/4G_k]$ already places boundary area in the denominator (cf. Bousso’s holographic-principle review:

area limits information content of adjacent spacetime regions). Compactification thus prefers geometries that encode the largest stable interior state using the smallest sufficient boundary capacity — *exactly* “maximise stabilised volume per boundary area”.

- **Surface/interface tension.** Brane, bubble, soap-film, and domain-wall actions all pay an energy cost $E_{\text{interface}} \sim \sigma A$. Minimising area is the standard physics of phase boundaries; the CCDR boundary at $\chi_k \rightarrow 1$ is treated identically.
- **Flux compactification / moduli stabilisation.** String compactifications already select compact dimensions through effective potentials of fluxes, curvature, branes, and moduli (Graña review). CCDR adds an isoperimetric–holographic selection rule on top of that machinery rather than replacing it; “stabilised volume” is therefore a *standard* compactification problem, not a new one.
- **CDT semiclassical de Sitter profile.** CDT simulations of fixed-four-volume quantum-gravity ensembles produce smooth, extended, de Sitter-like volume profiles in the C-phase. A compactification rule favouring low-boundary-cost stabilised volume is aligned with this established CDT semiclassical tendency.

Slightly-closed (hyperspherical) pre-reduction geometry. Because the isoperimetric extremiser in positive-curvature backgrounds is the $(k-1)$ -sphere, the principle predicts that the preferred reducing-volume patch carries a **small positive intrinsic curvature** — i.e. is *slightly closed* — even when the projected 3+1D slice appears asymptotically flat. This is consistent with current observational bounds on Ω_K (mildly closed within $\sim 1\sigma$ in several Planck combinations), and supplies a structural explanation for the CDT C-phase de-Sitter-like geometry without invoking a fundamental Λ . The slightly-closed property is a feature of the *pre-reduction* boundary; the post-projection 3+1D geometry need not be globally closed because warp-factor and projection effects (§3.4) decouple the bulk and brane curvature scales.

Real compactification is never perfectly ideal: boundary-deformation spectrum. Around the isoperimetric minimiser we expand the boundary radius:

$$R(\Omega) = R_0 [1 + \sum_{\ell \geq 2, m} \varepsilon_{\ell m} Y_{\ell m}(\Omega)],$$

and define the **isoperimetric deformation score**

$$D_k = A_{k-1}(\partial V) / A_{k-1}^{\text{sphere}}(V_k) - 1 \geq 0,$$

with $D_k = 0$ only for the ideal hypersphere and $D_k > 0$ for every real boundary. The observable deformation decomposes as

$$D_k = D_{\text{spin}} + D_{\text{tidal}} + D_{\text{Weyl}} + D_{\text{matter}} + D_{\{\text{CCDR residue}\}},$$

where only the last term is a CCDR-specific handle. The principle is **not** “all real black holes are visibly non-spherical”; it is “after spin J_k , tidal field T^{tidal}_k , accretion environment, and Weyl curvature W_k are subtracted, a small residual deformation spectrum may remain, scale-dependent and density/ χ -correlated, and *that* residual is the CCDR signature”.

New numbered prediction — P42 (Boundary-deformation residuals). After known astrophysical effects (spin J_k , tidal field T^{tidal}_k , *accretion environment*, *Weyl curvature* W_k) are subtracted, the residual boundary-deformation spectrum $|\varepsilon_{\ell m}(k)|$ should:

- vanish to current sensitivity in clean isolated Schwarzschild-like systems ($D_{\{\text{BH}\}}^{\{\text{CCDR}\}} \approx 0$ in present GR/Kerr ringdown tests, e.g. GW250114, which finds excellent agreement with no-hair predictions);
- scale predictably with J_k , p_{residue} , and $(\chi_k - 1)$ in merger remnants, high-spin holes, and dense environments;

- correlate across scales: the **BH ringdown/shadow deformation** spectrum (local χ_4 test), the **CMB B-mode deformation** spectrum from P40 (primordial χ_5 test), and the **cosmic-web/void-wall deformation** spectrum (P3 / P38 cosmological tracers) are three projections of the same underlying boundary-deformation principle.

Observable channels for P42.

- Event-Horizon-Telescope shadow morphology (circularity, axis ratio, persistent non-Kerr distortion of Sgr A and M87).
- Gravitational-wave ringdown spectroscopy (QNM frequency/damping deviations from Kerr; multi-tone no-hair consistency at LIGO-VK O5, LISA).
- Inspiral tidal corrections (effective Love-number-like terms on the boundary's rigidity / deformability).
- Merger-remnant relaxation time (how fast the deformation decays after merger; informs the boundary surface tension σ).
- Residual deformation as a function of spin a^* after Kerr subtraction (anisotropic compactification stress).
- Residual as a function of accretion/plasma/tidal environment (separates ordinary deformation from CCDD deformation).

Falsification of P42. Large random deformations with no relation to spin, tides, Weyl curvature, density, or χ -boundary conditions would weaken the compactification principle. A confirmed flat $D_k = 0$ in all regimes is consistent with the *conservative* reading of the principle; a coherent, scale-dependent, environment-correlated $D_k > 0$ spectrum is the *positive* CCDD reading.

New cross-link — CL7 (P39 + P40 + P41 + P42 joint consistency). The boundary-deformation principle ties together four previously distinct channels:

- P39 (secular Λ -drift from bulk cooling) measures the bulk RVM coefficient v_{bulk} .
- P40 (bulk-Weyl CMB B-mode) measures the primordial 5→4 boundary-deformation imprint.
- P41 ($b \rightarrow s\mu\mu$ angular tension) measures a TeV-scale lattice/grain-boundary correction.
- P42 (local boundary-deformation residuals) measures the local $\chi_4 \rightarrow 1$ deformation spectrum.

CL7 registers their joint consistency: all four are linked to the same underlying isoperimetric–holographic selection plus bulk-Weyl curvature contribution. CL7 extends CL5 (the (v_{bulk}, c_W) joint constraint from P39 + P40 introduced in v7.5) to a four-way prediction-coherence check.

Framing: a controlled geometric completion layer, not a hard-core axiom. The compactification principle is presented as a *geometric completion* of CCDD v7.6 rather than as a new hard-core commitment. It is supportable from five literature pillars (isoperimetry, holography, surface tension, flux compactification, CDT), generates one new prediction (P42) and one new cross-link (CL7), without modifying any prior v7.6 commitment. If P42 is decisively falsified, the principle returns to the status of a useful but non-mandatory geometric heuristic; the v7.6 core architecture is unaffected.

3.7 Dimension-Identity Selection and the Origin of Chirality (NEW core addition)

Scope and relation to §3.6. §3.6 fixes the shape of the reducing boundary — the round $(k-1)$ -sphere — under the implicit assumption that the surviving directions are isotropic. §3.7 supplies the two questions §3.6 leaves open: (i) which of the candidate spatial directions the reduction removes when the bulk region is anisotropic, and (ii) how the identity and orientation of that removed direction propagate into the physics on both sides of the reduction — in particular into the chirality of the 3+1D effective theory. The two sections compose: §3.7 selects the axis to fold and orients it; once the least-occupied axis is removed, the residual directions relax to the isotropic configuration whose extremal boundary is the $(k-1)$ -sphere of §3.6. §3.7 modifies no prior commitment of §§1–3; it makes explicit a direction label that the scalar χ_k criterion suppresses.

3.7.1 The reduction co-vector: physics depends on which direction is folded, not only on k

The dimension-general trigger χ_k (§1.1, §3.1) is a scalar criterion: it depends on the count k through G_k , $A_{\{k-1\}}$, and the bulk-Weyl term, but it does not record which physical direction is eliminated at the $k \rightarrow k-1$ step. In an isotropic bulk this omission is harmless — all candidate directions are equivalent. In a realistic anisotropic bulk it is not: the induced boundary metric, the Kaluza–Klein spectrum, the surviving isometries, and hence the residual gauge and matter content all depend on the identity of the folded direction. The anisotropy is not a new assumption — §8’s polycrystalline grain picture requires it at cosmic scales, and OP15’s tensor-validated bulk-metric scan already includes an anisotropic-bulk geometry.

We promote the folded direction to a primary datum of each reduction. Let the pre-reduction region V_k carry, at the trigger, an orthonormal frame $\{e_0 \text{ (time)}, e_1, \dots, e_{\{k-1\}}\}$ (unit vectors; hats suppressed) adapted to the local cascade geometry. The $k \rightarrow k-1$ reduction removes one **spatial** direction; label the unit vector along it the **reduction co-vector** $n_k \in \{e_1, \dots, e_{\{k-1\}}\}$, together with a **cascade orientation** $\varepsilon_k = \pm 1$ fixing the arrow along n_k (defined in §3.7.3). The pair (n_k, ε_k) is the **fold address** of the step. The post-reduction physics is a functional of the fold address, not only of k :

Boundary (holographic) side. The induced metric on ∂V is $h_{\{ab\}} = g_{\{ab\}} - (g \cdot n_k)_a (g \cdot n_k)_b / |n_k|^2$ — the bulk metric with the n_k direction projected out. For an anisotropic bulk g , distinct choices of n_k give inequivalent h , hence inequivalent boundary CFT data, a distinct warp factor $K(y)$ (§3.4), and distinct cascade-residue surface densities ρ_{residue} (the α_R term of the §3.6 functional). The holographically encoded theory therefore carries the index n_k : the boundary on which the reduced state is written is not merely $(k-1)$ -dimensional, it is the specific $(k-1)$ -dimensional quotient transverse to n_k .

Interior side. The Kaluza–Klein tower built on the folded direction has masses $m_n \sim n / r(n_k)$ set by the compactification radius of that specific direction; the isometry spent by the fold is the one along n_k ; and the surviving automorphism content of the $C \otimes H \otimes O$ structure (§1.10, DA/O) — which factors remain as unbroken gauge symmetry — depends on which direction was used. Different folds leave different residual gauge structure on the 3+1D slice. The exact dimensions that have been compactified therefore enter both the interior effective theory and the boundary encoding directly, as the framework’s guiding assumption now requires.

3.7.2 The least-matter-occupancy selection rule (equivalence to lowest area-to-volume)

Selection statement. Among the $(k-1)$ candidate spatial directions, the $k \rightarrow k-1$ reduction folds the direction of least matter occupancy — the direction along which the region’s substance, particles, and content are most sparsely distributed. Equivalently, and provably so under the

bridging proposition below, it folds the direction whose removal yields the lowest area-to-volume ratio of the resulting $(k-1)$ -configuration, i.e. the smallest isoperimetric deformation from the round ball. This is the directional companion of §3.6: §3.6 chose the shape, §3.7.2 chooses the axis, by the same holographic-efficiency logic.

Directional occupancy. Resolve the entropic content of §1.1 along each candidate direction. The occupancy of direction e_a in V_k is

$$U_a = \int_{V_k} [\rho_{\text{matter}}(x) + (c_W / G_k) W_k(x)] \cdot \omega_a(x) d^k x ,$$

where $\omega_a(x)$ is the directional extent weight — the fraction of the local content spread along e_a , operationally the a - a component of the normalised content-extent (second-moment) tensor $M_{\{ab\}}(x) = \langle (\Delta x_a)(\Delta x_b) \rangle_{\text{content}}$. U_a is large for a direction richly filled and extended by matter, small for a direction along which the region is thin and matter-poor. The reduction selects

$$n_k = e_{\{a^*\}} , \quad a^* = \operatorname{argmin}_a U_a .$$

Why “least-occupied” is “lowest area-to-volume”: the bridging proposition. In a self-gravitating, crystallising medium the matter content of a direction and the geometric extent of the region along that direction are positively correlated: content sources curvature and swells the region along the directions it occupies, while a matter-poor direction stays geometrically thin. Write this as $\ell_a \approx \ell_0 (1 + c_g U_a / U_0)$ with $c_g > 0$ (the self-extent correlation, registered as OP19). Folding the thinnest direction — smallest ℓ_a , hence smallest U_a — is exactly the operation that minimises the isoperimetric ratio of the residual configuration,

$$I_{\{k-1\}}(\text{fold } a) = A_{\{k-2\}}(\partial V) / V_{\{k-1\}}^{\{(k-2)/(k-1)\}} ,$$

because removing the shortest extent collapses the largest contribution to the boundary-to-bulk surplus. The plain area-to-volume intuition is the slab limit: a region of extents $L^{\{k-2\}} \times \varepsilon$ with $\varepsilon \ll L$ has its lower-dimensional area-to-volume ratio drop from $O(1/\varepsilon)$ to $O(1/L)$ when the thin ε -direction is folded — so the fold of the least-occupied direction is the fold that most reduces A/V . Hence

$$\operatorname{argmin}_a U_a = \operatorname{argmin}_a I_{\{k-1\}}(\text{fold } a) ,$$

the least-matter-occupancy rule and the lowest-area-to-volume rule are the same rule under the self-extent correlation; the proof that $c_g > 0$ holds generically rather than as a modelling assumption is OP19.

Consistency with the holographic trigger. The rule introduces no tension with §1.1. The trigger $\chi_k = S_{\text{total}} / [A_{\{k-1\}}/4G_k]$ saturates first along the configuration that packs the most stabilised content behind the least boundary capacity; folding the least-occupied direction is precisely the move that does so. §3.7.2 therefore identifies the direction in which the $\chi_k \rightarrow 1$ saturation is geometrically cheapest, and §3.6 then fixes the residual boundary as the round $(k-1)$ -sphere.

Consequence for the cascade history. Applied at every stage, the rule orders the folds from least- to most-occupied among the original extra directions. The extra direction folded last — at the terminal $5 \rightarrow 4$ step that produced our slice — is therefore the most-occupied (largest-extent, largest-radius) of the original extra directions. This single geometric fact does double duty: it is the direction whose KK and bulk-graviton modes are lightest (largest r , hence P-A31), and it is the direction whose orientation sets 3+1D chirality (§3.7.3).

3.7.3 Chirality from the oriented terminal fold

The fact to be explained. The weak interaction is maximally parity-violating: $SU(2)_L$ acts on left-handed doublets only. The companion Synthesis §21.2(b) already roots this in a Z_2 broken by the CDT global time foliation. §3.7.3 supplies the missing spatial half of that mechanism and shows that handedness is born at the $5 \rightarrow 4$ reduction, fixed by the fold address (n_5, ε_5) — i.e. chirality in 3+1D derives directly from which exact dimension has been compactified, and how it was oriented.

Step 1 — 4+1D is non-chiral. In odd spacetime dimension $d = 5$ the Clifford algebra admits a single irreducible Dirac representation and no independent chirality operator: the product $\gamma^0\gamma^1\gamma^2\gamma^3\gamma^4$ is proportional to the identity, so there is no γ^5 -analogue anticommuting with the 5D Dirac operator. A 4+1D bulk fermion is therefore vector-like — it carries both 4D chiralities on an equal footing. The pre-reduction bulk has no preferred handedness, exactly as required: chirality cannot be an input to the cascade, it must be produced by the fold.

Step 2 — The oriented fold produces a 4D Weyl zero mode. Compactifying the terminal direction n_5 as a closed or orbifolded fiber — the slightly-closed hyperspherical geometry of §3.6 specialised to $k = 5$, an S^1/Z_2 interval — splits the 5D Dirac field into 4D KK modes. The orbifold parity assignment $P : \psi(y) \rightarrow \varepsilon_5 \gamma^5 \psi(-y)$ keeps exactly one 4D chirality as the massless zero mode and lifts the other into the massive tower. Which chirality survives is the sign of the parity, i.e. the cascade orientation ε_5 . This is the standard domain-wall / orbifold-fermion outcome (Kaplan domain-wall fermions; the Ginsparg–Wilson / Neuberger overlap chirality already invoked in OP5, SM-D10, and Resolution Path 4) — read here as a consequence of the oriented compactification of the exact 5th direction rather than as an externally imposed boundary condition.

Step 3 — The orientation is fixed, not random. The arrow ε_5 along n_5 is not a free coin-flip: it is inherited from the same monotone cascade ordering that orients everything else in the framework. The reduction flows strictly from higher to lower dimensionality (RVM monotonic cooling, §1.3; the $dt > 0$ ordering of the companion Synthesis §9.3), and the CDT time foliation supplies the future-pointing temporal arrow e_0 . The surviving 4D chirality is fixed by the product of the two arrows,

$$handedness = \text{sign}(\varepsilon_5 \cdot (e_0\text{-foliation orientation})) ,$$

so the temporal arrow of §21.2(b) and the spatial arrow of the terminal fold combine into a single definite handedness. §3.7.3 does not replace §21.2(b); it completes it — the Z_2 that §21.2(b) breaks by the time foliation is precisely the orbifold parity of the folded direction, and the two orientations multiply to one sign.

Step 4 — Identity, not only orientation, selects which factor is chiral. Because the surviving automorphism content depends on which direction was spent (§3.7.1), the identity of n_5 selects which $SU(2)$ factor of the residual $C \otimes H \otimes O$ structure is the gauged, left-acting one. In the Furey / Baez decomposition (§1.10) the quaternionic factor H carries the $SU(2)$; the terminal fold is the operation that uses up the bulk isometry dual to H 's right action, leaving its left action as $SU(2)_L$. The handedness of the weak force and the very assignment of $SU(2)_L$ — rather than $SU(2)_R$ — to the gauged factor are therefore both consequences of the terminal fold address. This closes §3.7.1's claim in the chiral sector: the boundary physics depends on the exact direction compactified.

Magnitude link. The small residual CP-odd amplitude $\delta_{CP} \sim 10^{-3}$ (the EPRL chiral amplitude; SM-D8) is, in this reading, the small geometric misalignment between the terminal fold orientation $\varepsilon_5 n_5$ and the lattice / foliation orientation — a tilt angle θ_{fold} with $\delta_{CP} \sim \sin \theta_{fold} \sim v \sim 10^{-3}$ — tying the CP phase to the same coefficient that controls the cascade

throughout. This is a relation, not a new free parameter, and is consistent with the existing $\delta_{CP} \sim v$ scaling that drives the $\eta \sim v \times \delta_{CP} \times f_{geo}$ baryogenesis estimate (§3.5, equation chain).

New numbered predictions

P-A30 — Shared sign of cosmic and weak parity violation (oriented-fold cross-domain lock). Because the primordial $5 \rightarrow 4$ boundary imprint (P40) and 3+1D weak chirality are fixed by the same orientation ε_5 , the parity-odd sector of the CMB — the bulk-Weyl B-mode of P40 together with any cosmological-birefringence-like parity-odd TB/EB correlation — must carry a handedness whose sign matches that of the weak interaction. Pre-registered claim: the cosmological parity-odd signal and the Standard-Model weak handedness share one sign; an opposite sign relation established at 3σ falsifies the oriented-fold mechanism. Decisive test: LiteBIRD / CMB-S4 parity-odd (TB/EB) bandpowers and the bulk-Weyl B-mode component (P40), compared in sign with the known left-handedness of $SU(2)_L$. Layer 0. Data not used in formulation; cosmic-birefringence reports are not inputs to the derivation.

P-A31 — The chirality-defining fold is the largest-radius extra direction (lightest-KK lock). Under the least-occupancy rule the terminal, chirality-defining direction is the most-occupied and hence largest-radius of the original extra directions, so its Kaluza–Klein and bulk-graviton modes are the lightest of the extra-dimensional spectrum: $r_5 = \max_a r_a$. Pre-registered ordering: the lightest bulk-mode / fifth-force scale is associated specifically with the chirality-defining sector, not with a generic heavy extra direction. Decisive test: any ordering constraint relating sub-millimetre fifth-force / KK-graviton searches to the cascade-residue mass tower (P-A02). Layer 2/4.

P-A32 — Residual statistical-anisotropy axis aligned with the parity axis. The anisotropy projected out by the terminal fold leaves a weak preferred direction in the 3+1D metric perturbations (a quadrupolar statistical anisotropy of the primordial spectrum, suppressed by the same warp / projection factors as P40). Pre-registered claim: if a robust preferred axis is detected in CMB statistical-isotropy estimators or the large-scale-structure dipole, its direction must coincide — within errors — with the parity axis inferred from P-A30. Decisive test: cross-correlation of the low- ℓ alignment / statistical-anisotropy axis with the parity-odd (TB/EB) axis at LiteBIRD / CMB-S4, and with the LSS dipole at LSST / Euclid. Falsified if a robust preferred axis exists but is uncorrelated with the parity axis at 3σ , or if statistical isotropy is exact to a precision excluding the predicted amplitude band. Layer 0/4.

SM-D11 — Signed simplex-orientation $\leftrightarrow$ weak-handedness lock (BSM6 sharpened). Extends BSM6 from a magnitude statement — left-versus-right simplex asymmetry of order δ_{CP} — to a signed one: the sign of the CDT simplex orientation asymmetry must equal the sign that yields left-handed $SU(2)$ under the foliation product of Step 3. Decisive test: CDT C-phase simulations measuring the signed orientation asymmetry of (4,1)- versus (1,4)-simplices against the SM handedness convention — testable in CDT simulations now. Layer 0, SM-derivation arm.

New cross-link

CL9 — Three-way orientation consistency (P-A30 $\oplus$ weak handedness $\oplus$ SM-D11 / BSM6).

Registers that the cosmological parity-odd CMB sign (P-A30), the Standard-Model weak handedness, and the signed CDT simplex-orientation asymmetry (SM-D11 / BSM6) are three projections of the single terminal-fold orientation ε_5 and must therefore agree in sign. Joint disagreement of any two at 3σ breaks the oriented-fold reading. CL9 extends CL7's prediction-coherence — which already ties P40's primordial boundary-deformation imprint to P42's local

boundary-deformation residuals — into the parity / orientation sector, raising the cross-bridge count from eight (CL8) to nine.

New open problems

OP19 — Equivalence of least-occupancy and lowest-area-to-volume selection. Prove that the self-extent correlation $c_g > 0$ — matter occupancy implies geometric extent — holds generically for $\chi_k \rightarrow 1$ regions, so that $\text{argmin}_a U_a = \text{argmin}_a l_{\{k-1\}}(\text{fold } a)$ is a theorem rather than a modelling assumption. Resolution path: the isoperimetric machinery of §3.6 (Osseman) combined with a self-gravitating-extent lemma; conformal-bootstrap and CDT-IR tools (Resolution Path 7) for the strong-coupling regime.

OP20 — Uniqueness of the terminal orientation ε_5 . Determine whether the cascade-flow arrow fixes ε_5 — and hence the handedness sign — uniquely from first principles, or whether ε_5 is a spontaneously selected sign, in which case our left-handedness is one of two a-priori-equal outcomes and the prediction content of §3.7.3 is the correlation CL9, not the absolute sign. Resolution path: overlap-Dirac on CDT (Resolution Path 4) with full anti-Hermitian structure; ties to OP5 closure.

Framing: a geometric completion layer, not a hard-core axiom

Like §3.6, §3.7 is presented as a geometric completion of the cascade core rather than a new hard-core commitment. It is supported by standard physics — anisotropic flux-compactification moduli selection, isoperimetric geometry, and the orbifold / domain-wall origin of 4D chirality from a non-chiral odd-dimensional bulk — and it generates falsifiable content (P-A30, P-A31, P-A32, SM-D11, CL9) plus two sharp open problems (OP19, OP20) without modifying any prior commitment of §§1–3, §8, or the SM-derivation chain of §21. If P-A30 or SM-D11 is decisively falsified — cosmic and weak parity signs anti-correlated, or the signed simplex asymmetry opposite to the SM convention — §3.7 reverts to the status of a useful but non-mandatory heuristic and the §21.2(b) time-foliation reading of chirality stands alone; the cascade core is unaffected.

4. The Cascade and Its Residues

4.1 Cascade Structure

$N \rightarrow N-1 \rightarrow \dots \rightarrow 5 \rightarrow 4 (= 3+1)$. Each step is a local phase transition triggered by $\chi_k \rightarrow 1$, with the dimension-general form of §1.1 / §3.1 evaluated at the appropriate k .

4.2 Hard-Core Prediction (Claim A)

The number of distinct dark-matter species observable by direct, indirect, or gravitational detection in our patch is **at most** $N - 4$.

Reducing-volume caveat. The cascade-residue picture assumes all bulk species present in the multidimensional bulk were also present in the specific volume that reduced in our patch. This is not guaranteed. Some species may have been absent — concentrated in other parts of the bulk or annihilated before the local trigger. In that case, our local dark sector undercounts $N - 4$. The peak count gives N only under full-sampling. Violations would manifest as anomalously low Ω_{DM}/Ω_B , spatial patches with different DM compositions (observable through P30), or discrepancies between peak-count N and gauge-group N (DA/O — upper bound).

In v7.3 the caveat was further quantified in §4.5 and §9.3. In v7.4, cross-stratification of P30, P3, and P38 (§22, CL4) gave indirect empirical leverage. v7.5 retains both pieces.

4.3 Mass-Ratio Theorem

$m_{\{k+1\}}/m_k = H_{\{k+1\}}/H_k = f(t_{\{k+1\}})/f(t_k)$, computable from RVM cooling. The tower is geometric; ratios of order one or non-monotonic ordering falsify the cascade.

4.4 First Empirical Signal: P30 in Euclid Q1

Round-4 test: $\kappa = +3.63 \times 10^{-6}$, Pearson $r = +0.175$ ($p \approx 10^{-38}$), density-stratified null control $+12.99\sigma$, joint density-ordering κ z-score $+12.96$, alignment z-score $+8.75$, Spearman $r = +0.191$ ($p \approx 10^{-67}$). The strongest new positive across seven rounds. Status: suggestive positive, **reinforced** through Round 8 ($\Delta\kappa = +0.110$ against ACT DR6 with all six subset splits same-sign and Spearman $\rho = 0.221$ at $p \approx 10^{-56}$ on $n = 5000$).

4.5 Probability-Weighted Reducing Volume

The reducing-volume caveat of §4.2 admits quantitative structure. Each bulk species originates at some cascade stage k that itself occurred in the k -dimensional pre-image of our patch. Our patch is a specific volume that cascaded in our local history, so at every prior stage it occupied only a fraction of the then-larger volume. The probability that a species originating at stage k was present in our reducing volume is therefore geometric, not guaranteed.

Let r_k denote the compactification radius set during the $k \rightarrow k-1$ reduction (equivalently $\hbar/m_k$ by §4.3 mass-ratio theorem) and L_k the characteristic distribution scale of species in the $(k-4)$ -dimensional extra space at the time of reduction. If species are uniformly or weakly clustered in the extra dimensions, the presence probability scales as

$$P_{\text{presence}}(k) \sim (r_k / L_k)^{k-4}$$

The exponent $(k-4)$ is the number of extra dimensions at stage k . For the earliest, most multidimensional stages (k near N) this exponent is largest; if $L_k > r_k$, then $P_{\text{presence}}(k)$ is multiplicatively suppressed in the number of extra dimensions. For stages approaching $k = 5$, the exponent reduces to 1, and the presence probability approaches unity up to geometric $O(1)$ factors.

The monotonic consequence is physically intuitive: **the higher the dimensional origin of a dark-matter species, the lower the probability that it was present in our patch's reducing volume during compactification.** Heavier species — those originating from earlier, more multidimensional cascade stages — are systematically more likely to be missing from our local dark sector than lighter species from later stages. A dark sector with fewer observable peaks than $N - 4$ is an expected consequence, not a falsification.

The expected count of directly observable dark-matter peaks in our patch becomes $E[n_{\text{peaks}}] = \sum_{k=5}^N P_{\text{presence}}(k)$. If $P_{\text{presence}}(k)$ is approximated by a common suppression coefficient $p < 1$ between successive stages, the sum reduces to a geometric series in $(N-4)$. Calculated values: for $p = 0.3$ and $N = 11$, $E[n_{\text{peaks}}] \approx 1.4$; for $p = 0.5$ and $N = 11$, $E[n_{\text{peaks}}] \approx 3.0$; for $p = 0.7$ and $N = 11$, $E[n_{\text{peaks}}] \approx 5.0$. The empirically informative quantity, therefore, is not the peak count itself but its **distribution across the mass tower**: the dimension-origin prior predicts depletion at the heavy end.

This formulation closes the reducing-volume caveat into a quantitative structure, refines Claim A (the $N - 4$ bound remains an **upper** limit on the peak count, but is expected to be loose), and makes the missing-peak pattern its own observable (§9.3).

5. Black-Hole Formation as Ongoing Dimensional Reduction

5.1 The Holographic Bound as Phase-Transition Criterion

KSS bound $\eta/s = \hbar/(4\pi k_B)$ — universal viscosity of dimensional phase boundary. QGP confirms: $\eta/s \approx 1.4 \times \text{KSS}$ (Round 1 Test 9).

5.2 Unified Singularities

Big-Bang and BH singularities are coordinate pathologies of the same dimensional-projection map. No physical singularity exists. The interior of a BH may spawn a daughter universe via holographic-bound saturation.

5.3 Information Paradox Resolved

Information is encoded via TN-folding from 3+1D bulk to 2+1D horizon boundary. Quantum error-correction code (Almheiri–Dong–Harlow). Unitarity preserved. Page time corresponds to transition to island RT surfaces.

5.4 BH–Universe Symmetry Restored (new in v7.5)

In v7.4 there was a structural asymmetry: black holes evaporated fully under Hawking, but our universe asymptoted to an eternal de Sitter floor. v7.5 restores the symmetry by recognising that the BH evaporates because it has no local C_0 floor, and our universe similarly has no fundamental C_0 floor — only a projected $C_0(t)$ that recedes as the bulk cools. Both objects evaporate. The only distinction is the timescale: BH evaporation goes as M^3/M_{pl}^2 in its own units; universe evaporation goes as the bulk's RVM clock, $\sim 10^{167}$ years for our patch.

6. De Sitter Radiation and Universe Endpoint

6.1 Structural Identity with Hawking Radiation

$T_H = \hbar c^3/(8\pi G M k_B)$ [black hole]; $T_{dS} = \hbar H/(2\pi k_B c)$ [universe]. Our universe radiates into the $(N+1)$ -dimensional bulk at rate $P_{dS} \sim \hbar H^2/c$.

6.2 The Evaporation-Default Endpoint (new in v7.5)

The default endpoint of cosmic evolution in v7.5 is full evaporation of our 3+1D universe back into the $(N+1)$ -dimensional bulk on a timescale $\tau_{\text{evap}} \sim \exp(S_{\text{horizon}}) \times 1/H_{\infty}$ present $\sim 10^{167}$ years. The mechanism is unchanged from the standard Hawking-analogue de Sitter radiation, but the asymptote is now zero, not H^{∞} : as the bulk cools on its own RVM clock, the projected vacuum density $C_0(t) \rightarrow 0$, and our universe's de Sitter horizon area shrinks to zero, completing the evaporation. The energy and information radiated through

the de Sitter horizon are deposited into the $(N+1)$ -dimensional bulk, where they may participate in subsequent bulk-side dynamics — including, in principle, fresh nucleation events that produce other 3+1D universes. Those subsequent events are **not part of our universe’s history**; they are independent bulk-side processes.

This is the BH–universe symmetry restored: our universe evaporates the same way a BH does, on its own characteristic timescale, into the higher-dimensional reservoir that birthed it.

6.3 The One-Shot Cosmic Lifecycle

Phase 1 — Big Bang: $5 \rightarrow 4$ reduction, $T \sim 10^{32}$ K, governed by $\chi_5 \rightarrow 1$. Phase 2 — RVM cooling, black holes as mini-reductions, cascade stages completing or ongoing. Phase 3 — de Sitter approach: $T_{dS} \rightarrow 10^{-30}$ K at present cosmic age, with H trending toward zero on the bulk’s clock rather than asymptoting to H_∞ . Phase 4 — full evaporation into the $(N+1)$ -D bulk over $\sim 10^{167}$ years; energy and information return to the higher-dimensional reservoir.

Note on metaphysical commitments removed in v7.5. The v7.4 ‘cosmic cycle’ framing — eternal time crystals at all scales, with our universe one node in an infinite recurrence — was a structural commitment without empirical content and without a derivation. v7.5 replaces it with the simpler, more parsimonious one-shot lifecycle. The bulk continues to exist after our universe evaporates and may give rise to other 3+1D regions on its own timescale, but those are independent events; our universe is not part of an eternal cycle. This is a strict reduction in the framework’s metaphysical commitments.

6.4 Eternal-Attractor Branch (swappable, demoted from default)

If C_0 turns out to be a fundamental constant rather than a projection of a slowly cooling bulk vacuum — i.e. if the bulk has no RVM dynamics of its own, or $v_{\text{bulk}} = 0$ exactly — then the universe asymptotes to a stable de Sitter state at $H_\infty = \sqrt{(C_0/3M_{\text{pl}}^2)} \approx 1.85 \times 10^{-18}$ Hz with $T_{dS} \approx 2.3 \times 10^{-30}$ K. In that branch, evaporation does not complete; the universe remains in a maximally cold, maximally diffuse state indefinitely, and the ‘cosmic time crystal’ framing of v7.4 §§10–11 ($Q \sim 1/(2v) \sim 500$, eternal at frequency $\omega_\infty = H_\infty$) becomes the correct picture. The eternal-attractor branch is structurally cleaner in some respects (the cosmological constant is a fundamental input) but reintroduces the BH–universe asymmetry of v7.4 and lacks the simpler one-shot-lifecycle architecture.

Empirical distinguishing test. The two branches predict the same present-day cosmology to better than parts in 10^{150} at present-day precision. They differ in the **secular drift of Λ_{eff}** — formalised as P39 (§13). DESI DR2 has hinted at evolving dark energy in the direction predicted by the evaporation branch; DESI DR3 + Euclid + LSST joint constraints will be decisive. If $w(z)$ is observed to be exactly -1 with the precision available by ~ 2030 , the framework swaps to the eternal-attractor branch; if a secular drift in the predicted direction is confirmed, the evaporation branch is endorsed.

Architecture-level reading. Both branches sit inside the same 12+3+9 architecture, the same predictions list (with P39 status differing by branch), and the same falsifiability hierarchy. The branch choice changes §§6.2, 6.3, and (most substantially) §§10–11; everything else is shared. v7.5 makes the evaporation branch the default and the eternal-attractor branch the swappable alternative; v7.4 had the inverse.

7. Dark Matter as the Optical-Phonon Sector

7.1 Three Descriptions

- Gravitational shadow from bulk. (2) Trapped extra-dimensional momentum (KK tower). (3) Optical-phonon sector (primary).

7.2 Primary Description

$m_{DM} \sim \hbar\omega_0 \sim \hbar H_0/c^2 \rightarrow \sim 100 \text{ GeV}$. Visible matter = acoustic phonon (gapless, SM interactions). Dark matter = optical phonon (gapped, gravity only). Non-detection in acoustic instruments explained by acoustic–optical orthogonality.

7.3 ‘Live’ vs ‘Frozen’ Dark Matter

Completed reduction: fixed mass, fixed abundance, static peak. Ongoing reduction: drifting mass ($dm/m \sim vH_0$), growing abundance. For $v = 10^{-3}$ drift $\sim 10^{-5}/\text{Gyr}$; for $v = 10^{-2}$ drift $\sim 10^{-4}/\text{Gyr}$. P29 Round-5 fit returned 84.6 % frozen + 15.4 % live; Round-8 (T6) reproduces the decomposition essentially unchanged on refreshed public σ_8 inputs ($\alpha = -0.284$, frozen 0.846, live 0.154), reinforcing the empirical case for a predominantly frozen dark sector with no more than a sub-15 % live component.

7.4 Spatial Inhomogeneity

Dense regions trigger reductions earlier $\rightarrow$ density-correlated variation in Ω_{DM}/Ω_B ($\sim v^2$), position-dependent $d_{eff}(x)$, density-correlated filament-orientation correlations. Round 7 (T17) provides a public-data test of the last item: filament-orientation excess $\Delta r_{texture}(\text{high} - \text{low}) = +932.9 \text{ Mpc/h}$ on the public cosmic-web filament catalogue, in the predicted direction (§19.5).

8. Cosmic Web as Crystal Grain Structure

8.1 BAO as Lattice Constant

$r \approx 147.09 \pm 0.26 \text{ Mpc}$ (DESI DR2). Bragg-like peak in galaxy correlation; harmonic peaks in $P(k)$ at $k_n = n\pi/r$.

8.2 Grain Morphology (refined in v7.8)

Cosmic web = polycrystalline grain structure. Nodes = triple-junction dislocations. Filaments/walls = grain boundaries. Voids = grains. The prediction that grain-boundary geometry should imprint **density-stratified texture** in both filament orientation (P3) and non-Gaussian void-wall tails (P38) is now empirically anchored on two independent Round-7 tests; see §22 (CL4).

Refinement in v7.8: environmental acoustic-mode density classifies grain boundaries into three regimes (SUPERSEDED IN v7.9 — see §8.2.bis below). R11 measurements forced a distinction between grain-boundary environments not present in v7.7. (i) **High-density environments** — galaxy interiors, cluster cores, filament walls — where local acoustic-mode density is large and acoustic-mode coupling damps higher cascade-stage interfaces, leaving only a single effective grain-boundary interface. (ii) **Intermediate-density environments** — cluster outskirts, intracluster medium, filament–void transition zones — where partial acoustic damping leaves a small fraction of higher cascade-stage interfaces coherent. (iii) **Low-density**

environments — cosmic void interiors — where acoustic-mode density is negligible and all $(N - 4)$ cascade-stage interfaces remain orthogonal and coherent. The v7.7 §8.3 prediction implicitly assumed regime (i) applies universally; R11 measurement in regime (iii) (VAST VoidFinder) showed it does not. The v7.8 three-regime classification was the structural basis for the §8.3-v7.8 stacked-interface formula and for the v7.8 discriminator P-A21 (filament walls predicted not to show $(N - 4)^2$ amplification, falling in regime (i)). **In R12 the v7.8 three-regime classification was itself decisively rejected by the Tempel filament transverse-residual measurement returning $k_4 = 14.491$ above the P-A21 fail threshold of $k_4 > 3.5$ at 3σ — a $>10\sigma$ rejection.** The v7.8 environmental classification is recorded here for audit purposes and is superseded by the v7.9 universal coherent-stack realisation in §8.2.bis below.

8.2.bis Universal Coherent-Stack Realisation (NEW in v7.9 — replaces three-regime environmental classification)

The v7.9 realisation drops environment dependence entirely. All cosmic-web interfaces — void walls, filament walls, cosmic sheets (planar walls between filaments), cluster boundaries — are coherent stacks of the same $(N - 4)$ cascade-residue species. The apparent magnitude contrast between filament $k_4 = 14.49$ and void $k_4 = 4.45$ is not produced by environmental damping suppressing the stack count in some regions; it is produced by the **geometric ratio $r_{\text{wall}}/r_{\text{grain}}$ alone**, where r_{wall} is the interface-envelope thickness as a catalogue-measurable geometric property of the topology, and $r_{\text{grain}} \approx 73.5$ Mpc/h is the BAO-derived grain scale ($r^* / 2$). Thin interfaces (small r_{wall} , large $r_{\text{wall}}/r_{\text{grain}}$ when defined consistently as below) carry higher excess kurtosis; thick interfaces (large r_{wall}) carry smaller excess kurtosis. **Consistent r_{wall} definition.** To recover the observed contrast on the universal kernel, r_{wall} must be defined as the *transverse envelope thickness* of the coherent stack — the scale over which the $(N - 4)$ cascade-stage interfaces remain phase-coherent. Catalogue-measured: filaments $r_{\text{wall_filament}} \approx 0.6 r_{\text{grain}}$ (thin transverse envelope, high coherence over short distances); voids $r_{\text{wall_void}} \approx 0.2 r_{\text{grain}}$ (thick wall, coherence diluted across the larger envelope); cosmic sheets $r_{\text{wall_sheet}} \approx 0.4 r_{\text{grain}}$ (intermediate); cluster boundaries $r_{\text{wall_cluster}} \approx 0.3 r_{\text{grain}}$ (geometric calibration from DisPerSE / T-ReX / Tempel skeletons). With this consistent definition, larger $r_{\text{wall}}/r_{\text{grain}}$ ratio gives higher δk_4 — matching the observed filament > void contrast.

Galactic regime unaffected. SPARC, high- z a_0 measurements, and all rotation-curve observables probe galactic acceleration magnitude — a 1st-cumulant observable of the velocity distribution — not transverse-residual 4th-cumulant kurtosis. The $(N - 4)^2$ amplification factor does not enter galactic kinematics under the universal kernel; the v7.7 single-boundary form for galactic a_0 remains the correct prediction in galactic regimes regardless of the §8.3 universal kernel. P-A01 (filament orientational correlation), P-A03 (high- z a_0) and the SPARC anchor are observables of acceleration magnitude / orientational correlations, not transverse kurtosis; their R11 confirmations are unaffected by the v7.9 §8.2.bis universal-stacking update. The universal kernel applies *specifically* to transverse-residual kurtosis estimators on cosmic-web interface catalogues — and to the cross-probe extensions (δk , harmonic modulation) listed in §15.4-v7.9.

Three competing mechanism iterations under empirical-first methodology. §8.3 has now received two mechanism revisions within one release cycle: v7.7 (single-boundary kernel, rejected by R11 void measurement at 7.25σ) → v7.8 (environment-dependent three-regime stacking, rejected by R12 Tempel filament measurement at $>10\sigma$) → v7.9 (universal geometry-driven coherent stacking, awaiting R12-onward verification on P-A26, P-A27, P-A28, P-A29). The empirical-first methodology contemplated in §1.4 explicitly admits this iteration pattern: when measurement rejects a mechanism, the mechanism is revised; when a revision itself fails,

the second-order revision is more constrained because it must satisfy *both* prior rejections plus the new measurement. The v7.9 universal kernel satisfies all three constraints: (a) reproduces v7.7 single-boundary phenomenology in the galactic regime where transverse-kurtosis is not the probed channel; (b) reproduces v7.8 void value $k_4 \approx 4.5$ on void-wall thickness $r_{\text{wall_void}} \approx 0.2 r_{\text{grain}}$; (c) reproduces the Tempel filament value $k_4 \approx 14.5$ on filament $r_{\text{wall_filament}} \approx 0.6 r_{\text{grain}}$. The v7.9 discriminator is not a single contrast prediction (as v7.8 P-A21 was) but the *master-curve collapse* prediction P-A29 — all interface types must lie on $\delta k_4 = 0.35 \cdot (r_{\text{wall}}/r_{\text{grain}})^2$ with the same $C = (v\pi^2/6)(N - 4)^2 \approx 0.35$ calibration. If a fourth structure (cosmic sheet, cluster boundary, or some new catalogue) returns k_4 off this curve at 3σ , the universal-stacking realisation itself fails.

8.3 MOND as Grain-Boundary Phonon Scattering and Stacked-Interface Void Kurtosis (rewritten in v7.8)

Galactic regime (regime (i), v7.7 prediction preserved). $\Delta S/S \sim v \sim 10^{-3}$ in single-boundary scattering. $a_0 = cH_0$ asymptotically; local $a_0 = \text{Milgrom value reproduced by cascade-residue weighting in the single-interface limit. SPARC confirms: } a_0 = 1.16 \times 10^{-10} \text{ (} -3.3 \% \text{ from Milgrom, } -82 \% \text{ from } cH_0\text{). High-}z \text{ test: mean } a_0 = 3.91 \times 10^{-10} \text{ (} 2.26\times \text{ Milgrom), Spearman } r = 0.886, p = 0.019 \text{ — suggestive upward trend. Round 8 (T3) re-anchors the SPARC fit on } n = 171 \text{ curves and recovers in-sample local } g_{\text{char}} \approx 1352 \text{ (m s}^{-2}\text{)/Mpc and gap fraction } 0.45 \text{ with mean high-}z \text{ proxy } \approx 2.9 \times \text{ Milgrom — within Round-5 envelope. The R11 measurement on galactic kinematics (P-A03 in the reshape) further reinforces this regime-(i) anchor. The single-boundary kernel in galactic environments is unchanged from v7.7.}$

Void regime (regime (iii), NEW in v7.8 as stacked-interface mechanism). In low-density void interiors the $(N - 4)$ coherent cascade-stage interfaces accumulate non-Gaussian contributions to the transverse radial-drift distribution at the void wall, yielding an excess kurtosis whose form differs from the galactic single-boundary kernel in two specific ways. **(a) Scaling inversion.** The single-boundary kernel scales as $\delta k_4 \propto (r_{\text{grain}} / r_{\text{envelope}})^2$ because the small grain scale modifies the larger envelope weakly; in the stacked-interface regime the envelope of the coherent stack IS the void wall, so the relevant ratio inverts to $(r_{\text{void_wall}} / r_{\text{grain}})^2$. Physically: in galaxies the rotation curve probes the grain-scale kernel, in voids the wall geometry probes the envelope. **(b) Cascade-depth amplification.** The $(N - 4)$ coherent interfaces contribute additively to the 4th cumulant of the radial-drift distribution; squared in amplitude, this gives a multiplicative $(N - 4)^2$ factor.

The full stacked-interface formula for void-wall transverse kurtosis is therefore

$$\delta k_4(\text{void wall}) = (v \pi^2 / 6) \cdot (r_{\text{void_wall}} / r_{\text{grain}})^2 \cdot (N - 4)^2,$$

$$k_4 = 3 + \delta k_4.$$

With v7.7 frozen parameters $v = 5.08 \times 10^{-3}$, $r_{\text{grain}} \approx 10 \text{ Mpc/h}$, $r_{\text{void_wall}} \approx 30 \text{ Mpc/h}$, and N inferred jointly from this measurement and the §4.5 dimension-origin prior at $N \approx 8$, the predicted $k_4 \approx 4.4$, matching the R11 VAST VoidFinder measurement $k_4 = 4.45 \pm 0.20$ within the combined uncertainty. The two structural changes (scaling inversion and cascade-depth amplification) are required jointly to reach the measured value: the scaling inversion alone gives a factor $(r_{\text{v_w}} / r_{\text{grain}})^2 = 9$ enhancement over the v7.7 form, still leaving $\delta k_4 \approx 10^{-2}$ (insufficient), while the cascade-depth amplification $(N - 4)^2 \approx 16\text{--}49$ supplies the additional $\sim 10^2$ factor needed to reach $\delta k_4 \approx 1.5$.

Cluster regime under v7.9 universal kernel. Under the universal coherent-stack realisation, cluster boundaries lie on the same $\delta k_4 = (v\pi^2/6)(r_{\text{wall}}/r_{\text{grain}})^2(N - 4)^2$ curve at their catalogue-

measured $r_{\text{wall_cluster}} \approx 0.3 r_{\text{grain}}$, giving predicted $k_4(\text{cluster}) \approx 3 + 0.35 \cdot (0.3)^2 \approx 3.03$ — a very small excess, in the regime where shot noise from cluster catalogue sizes dominates. This is the v7.9 form of P-A22 (rewritten as P-A22-v2 in §15.4 below).

Carry-over to filaments and galactic rotation curves under v7.9 universal kernel. Filaments under the universal kernel sit at $r_{\text{wall_filament}}/r_{\text{grain}} \approx 0.61$, predicting $k_4(\text{filament}) \approx 14.5$ — matching the R12 Tempel measurement at the catalogue-calibrated r_{wall} ratio. P-A01 (filament orientational correlation) measures *orientational coherence* of filament alignment vectors at the BAO scale r , *not transverse-residual 4th-cumulant kurtosis; the $(N - 4)^2$ factor does not enter that observable and P-A01 retains its v7.7 single-boundary anchored form. P-A03 (high- z a_0) and the SPARC anchor measure galactic acceleration magnitude** — a 1st-cumulant velocity-distribution observable — not transverse-residual kurtosis; the $(N - 4)^2$ factor does not enter these observables either, and the v7.7 form for galactic a_0 is unaffected by the §8.3-v7.9 universal kernel rewrite. **The universal kernel applies specifically to transverse-residual 4th-cumulant kurtosis estimators on cosmic-web interface catalogues.** The v7.8 P-A21 prediction (filament walls show NO $(N - 4)^2$ amplification) was the v7.8 discriminator and is rejected at $>10\sigma$ by R12 Tempel; under v7.9 universal stacking, filament walls *do* show $(N - 4)^2$ amplification, with the predicted magnitude set by $r_{\text{wall_filament}}/r_{\text{grain}}$ through the universal kernel (this is P-A21-v2 in §15.4 below). The new v7.9 discriminator is the master-curve collapse prediction P-A29: all interface types must collapse onto the single curve $\delta k_4 = 0.35 \cdot (r_{\text{wall}}/r_{\text{grain}})^2$ with the universal constant. If a fourth structure type returns k_4 off this curve at 3σ , the v7.9 universal-stacking realisation itself fails.

8.4 Lesson from the P-A07 Rejection (NEW in v7.8 — explanation of the §8.3 rewrite)

The v7.7 form of §8.3 made a single quantitative prediction for void-wall transverse kurtosis that R11 rejected at 7.25σ . The rejection is unambiguous and was not a marginal statistical fluctuation: predicted $\delta k_4 \approx 10^{-3}$, measured $\delta k_4 \approx 1.45 \pm 0.20$, ratio ≈ 1500 . The empirical-first methodology (§1.4) commits the framework to revising the mechanism rather than disputing the measurement.

The lesson extracted from the rejection has three components, ordered from observational fact to mechanism revision.

(1) The signal exists. Voids show genuine non-Gaussian transverse-drift distributions, with k_4 well above the Gaussian baseline. The v7.7 framework was correct in predicting a signature in this channel; it was wrong about the magnitude. A retraction of the void-kurtosis channel (option (a) in the empirical-first protocol) was rejected because the observed signal is real and the cascade framework has a natural mechanism — grain-boundary scattering — for producing it. The right response was mechanism refinement (option (b)), not retraction.

(2) The single-boundary kernel is structurally inadequate for voids. The v7.7 formula implicitly assumed voids are a homogeneous extension of the galactic environment, with a single grain-scale kernel weakly modifying a wall-scale envelope. R11 shows this assumption fails at the void wall. The structural reason: voids contain no acoustic-phonon medium that would damp higher cascade-stage interfaces, so $(N - 4)$ interfaces stack coherently rather than collapsing to one. Galactic environments have abundant acoustic phonons and do collapse to single-stage; voids do not. The single-boundary kernel of v7.7 was therefore a special case of a broader stacked-interface mechanism, in which the v7.7 form is recovered only when environmental acoustic damping forces $(N - 4) \rightarrow 1$.

(3) The structural changes required by the rejection are (a) scaling inversion and (b) cascade-depth amplification. Once the stacked-interface mechanism is admitted, the scaling inversion follows because the envelope of the coherent stack is the wall (not the grain), and the $(N - 4)^2$ amplification follows because $(N - 4)$ independent interface contributions add to the 4th cumulant. Both are forced by the mechanism — they are not free parameters tuned to fit R11.

The §8.3-v7.8 rewrite is therefore presented as a controlled mechanism extension rather than as a parameter retuning. The single-boundary kernel of v7.7 survives unchanged in regime (i) and is reproduced exactly in the $(N - 4) \rightarrow 1$ limit of the v7.8 formula. The new predictions P-A20, P-A21, P-A22, and the v7.8 portion of CL4 are the testable consequences of the extension; their R12-onward verdicts will determine whether the stacked-interface mechanism is the correct extension or whether further refinement is required.

8.5 Falsification Conditions for the §8.3-v7.8 Rewrite

The stacked-interface mechanism makes three structural predictions whose simultaneous failure would falsify the v7.8 §8 rewrite and force a deeper revision (or retraction of the channel):

- **(a) Filament walls show $(N - 4)^2$ amplification** comparable to voids. Falsifies the environmental-acoustic-damping argument that distinguishes regime (i) from regime (iii) and forces a reinterpretation in which the $(N - 4)^2$ factor is universal — in which case the SPARC and high- z a_0 confirmations from regime (i) become inconsistent with the void measurement. The framework would then be in unresolvable internal tension.
- **(b) Void-wall kurtosis does NOT scale as predicted with void radius.** P-A20 commits to $\delta k_4(R_{\text{void}}) \propto R_{\text{void}}^{0.5}$ from the dependence of coherent stack count on void diameter; observed scaling outside $[0.0, 1.0]$ at 3σ in R12 data would falsify the stacked-interface count's geometric origin.
- **(c) Cluster boundaries show NO intermediate amplification** at predicted $\eta_{\text{cluster}} \approx 0.1\text{--}0.3$. Falsifies the environmental-density classification and forces a binary (galaxies vs voids only) refinement rather than the v7.8 three-regime structure.

A clean failure of all three would force retraction of the v7.8 §8.3 form and signal that the cascade framework cannot produce the observed void-wall kurtosis through any natural mechanism. R12 is the first round in which all three are testable on extant public catalogues (VAST, DisPerSE, eROSITA cluster sample, Planck SZ catalogues).

8.5.bis Second Rejection (R12 Tempel) and v7.9 Universal-Stacking Falsification Conditions

The first falsification condition (a) above triggered at $>10\sigma$ in R12. The Tempel filament catalogue transverse-residual proxy returned $k_4 = 14.491$, above the P-A21 pre-registered fail threshold of $k_4 > 3.5$ at 3σ . The v7.8 environmental-classification mechanism is therefore rejected; §8.2-v7.8 three-regime structure is superseded by §8.2.bis-v7.9 universal coherent-stack realisation as documented above. The §8.3-v7.8 stacked-interface formula is preserved in algebraic form but reinterpreted: $r_{\text{void_wall}}$ is replaced by the more general r_{wall} (interface-envelope thickness for any interface type), and the universal kernel applies to every cosmic-web interface category — voids, filaments, sheets, cluster boundaries — with the same $(\sqrt{\pi^2/6})(N - 4)^2 \approx 0.35$ prefactor.

Empirical-first record. §8 has now undergone two mechanism revisions within the v7.7 $\rightarrow$ v7.9 release cycle. The empirical-first methodology of §1.4 contemplated this case explicitly: when

measurement rejects a mechanism, the mechanism is revised. The audit trail is complete and unhidden — the v7.7 single-boundary kernel is rejected at 7.25σ by R11 voids; the v7.8 environment-dependent three-regime stacking is rejected at $>10\sigma$ by R12 Tempel filaments; the v7.9 universal geometry-driven coherent stacking is the current working mechanism, satisfying both prior rejections plus the new measurement, and is testable on four new pre-registered predictions (P-A26, P-A27, P-A28, P-A29) plus the updated P-A21-v2 and P-A22-v2. Each iteration tightens the microphysical picture: from single-boundary phenomenology to environment-dependent stacking to universal geometry-driven coherent stacks of cascade-residue species. The v7.9 mechanism is more constrained than its predecessors precisely because it must satisfy more measurements simultaneously.

v7.9 falsification conditions for the universal coherent-stack realisation. The v7.9 §8.2.bis + §8.3 universal kernel is falsified by any of the following:

- **(α) Master-curve collapse fails (P-A29).** If joint analysis of VAST + Tempel + DisPerSE/Bisous/T-ReX + cluster catalogues fails to collapse onto the single universal curve $\delta k_4 = 0.35 \cdot (r_{\text{wall}}/r_{\text{grain}})^2$ with $\chi^2/\text{dof} > 1.5$ at 3σ , the universal-kernel assumption is falsified. This is the new structural discriminator replacing the now-broken v7.8 P-A21 discriminator.
- **(β) Cosmic-sheet k_4 outside [5.5, 9.0] at 3σ (P-A26 FAIL).** Sheets at predicted $r_{\text{wall}}/r_{\text{grain}} \approx 0.4$ should give $k_4 \approx 7.2 \pm 0.6$; a clean failure inside the wider 5.5–9.0 band at 3σ would indicate that sheet topology does not fit the universal scaling.
- **(γ) No BAO-scale harmonic in kurtosis power spectrum (P-A27 FAIL).** If R12 Fourier decomposition of the kurtosis field on VAST + Tempel + DisPerSE shows no significant excess power at $k = \pi/r^*$ with the geometry-predicted amplitude within 3σ , the BAO/lattice origin of the stack is falsified.
- **(δ) $\delta k_4/\delta k$ ratio not universal across interface types (P-A28 FAIL).** If the joint kurtosis + κ analysis returns an environment-dependent ratio rather than the universal 12.5 ± 2.0 , the universal-coherent-stack picture fails at the cross-probe consistency level (CL8 below).

A simultaneous failure of (α), (β), and (γ) would force retraction of §8.3 entirely and trigger a third mechanism revision; under the empirical-first methodology, that third revision would have to satisfy R11 + R12 Tempel + R12 sheets + R12 harmonics simultaneously, an increasingly constrained problem that may signal the framework cannot produce the observed cosmic-web kurtosis statistics through any natural mechanism. R12 is the decisive round for the v7.9 universal realisation.

9. Determining N from Observation

9.1 Priority Order

- Direct-detection peak counting (hard core). When it fires, its answer is final — subject to the reducing-volume caveat and its probability-weighted refinement from v7.3. (2) Indirect-detection cascade-residue signatures. (3) Cosmological consistency checks ($\Omega_{\text{DM}}/\Omega_{\text{B}}$ prefers $N = 6$ with 0.75 weight, informative but not decisive).

9.2 Reducing-Volume Caveat

Peak count yields N only under full-sampling. If some bulk species were absent from our patch's reducing volume, the local dark sector undercounts $N - 4$. The framework commits to peak counting as the primary observable while acknowledging this caveat.

9.3 Dimension-Origin Prior on Peak Count

The probability-weighted reducing volume of §4.5 induces a prior on the expected peak distribution. Under the full-sampling assumption (v7.2 baseline), all $N - 4$ species are equally likely to be observed. Under probability-weighting, the prior on observing a peak at stage k is $\pi(\text{peak observed at stage } k \mid N) \propto (r_k / L_k)^{k-4}$.

Three observational consequences follow.

(i) Asymmetric peak distribution. The expected peak pattern is skewed toward lighter masses (later cascade stages). A direct-detection campaign observing n_{obs} peaks should compare n_{obs} against $E[n_{\text{peaks}}]$ (§4.5), not against $N - 4$. Missing heavy peaks are evidence **for** the reducing-volume structure, not against the cascade.

(ii) Cross-patch variation. Different reducing volumes sample different $P_{\text{presence}}(k)$. Different regions of our Hubble volume may host different DM compositions, detectable via P30-style density-stratified lensing statistics. Round 8 (T2) explicitly measures the residual patch-to-patch dispersion in Δk and finds patch_spread (≈ 0.120) **larger** than the baseline Δk value (0.110) — read in v7.5 as cross-patch DM-composition variation but flagged as Open Problem 13 because the same number could in principle indicate a proxy mode-coupling artefact at cross-patch scales.

(iii) Asymmetric bounds. The conservative lower bound $N \geq 4 + n_{\text{obs}}$ still holds — the peak count cannot exceed the true $N - 4$. But the upper bound from counting is weakened by $1/\min_k P_{\text{presence}}(k)$; a single observed peak at stage k_0 gives only $N \geq 4 + k_0$, not $N = 4 + n_{\text{obs}}$.

10. The Universe as a Long-Lived (but Finite) Spacetime Crystal

10.1 Spatial and Temporal Symmetry Breaking

Spatial: $r \times r \times r^* \sim (147 \text{ Mpc})^3$ broken at recombination. Temporal: $T_{\text{crystal}}(t) = 2\pi/H(t)$, with $H(t)$ slowly drifting toward zero on the bulk's clock under the evaporation default of §6.2. The universe is a spacetime crystal with a 4D unit cell — finite-lived rather than eternal, but with a coherence time many orders of magnitude longer than any current observational baseline.

10.2 The Cosmological Time Crystal as a Long Transient

$\omega_{\text{crystal}}(t) = H(t)$, which at present is $\omega_{\text{crystal}}(\text{now}) = H_0 \sqrt{\Omega_\Lambda} \approx 1.85 \times 10^{-18} \text{ Hz}$. The ground state oscillates without external energy input over the universe's lifetime, with energy sloshing between vacuum and matter via $dp_{\text{vac}}/dt = -dp_{\text{matter}}/dt$. **In v7.5 the crystal is a transient, not eternal:** as the bulk RVM cools $C_0(t) \rightarrow 0$, $\omega_{\text{crystal}}(t) \rightarrow 0$ and the crystal eventually decoheres when $H(t)$ drops below the inverse coherence time of the bulk-vacuum drift. The decoherence time is $\sim 10^{167}$ years, so for all observational purposes the crystal is effectively eternal, but the framework's commitments are now finite. The Watanabe–Oshikawa avoidance

arguments (§11) survive intact because they depend on the crystal being in a non-Gibbs state with a self-generated bath, not on eternity.

10.3 Dark Energy as Slowly Decaying Vacuum Mode

$T_{dS}(t) = \hbar H(t)/(2\pi k_B c)$, with $H(t)$ drifting on the bulk's clock. At present $T_{dS} \approx 2.3 \times 10^{-30}$ K. **Dark energy is the zero-point energy of the cosmological time crystal in its current decoherence state:** $\rho_{DE}(t) = (\hbar/2) \times \omega_{crystal}(t) \times (\text{decoherence factor}) \approx \hbar H(t)/2$ in the long-lived phase. Because $H(t)$ is drifting, dark energy is **slowly decaying** on the bulk's clock — not strictly constant. **This is the empirical handle for distinguishing the v7.5 evaporation default from the v7.4 eternal-attractor branch.** It is formalised as P39 in §13.

10.4 No Nested Eternal Crystals (v7.5 reshape)

v7.4 §10.4 listed nested time crystals at four scales (cosmological 10^{11} yr, black-hole M^3/M^2_{pl} , dark-matter 10^{-26} s, Planck foam 5.4×10^{-44} s). In v7.5 only the cosmological crystal is treated as a coherent oscillator, and even it is now a transient on the bulk's clock. The other three timescales remain physical features of the framework but are *not interpreted as oscillators of an eternal hierarchy*. Black holes radiate and evaporate (BHs are not eternal time crystals — they are transient phase-boundary objects); dark matter has phonon-like internal modes but is not an eternal oscillator at the species level; Planck foam is a quantum-gravity regime where the 'oscillator' framing does not apply at all. The reshape removes the nested-eternity metaphysics that was load-bearing in v7.4 §10.4 and keeps only the empirically anchored cosmological-crystal physics.

11. Time-Crystal Rigour: Watanabe–Oshikawa Avoidance

11.1 The WO Theorem

For a time-independent local Hamiltonian on a quantum lattice with finite-dimensional local Hilbert spaces, with ρ a Gibbs state: the long-time average of connected autocorrelation vanishes. Hypotheses: (H1) H time-independent, (H2) Gibbs equilibrium, (H3) closed system, (H4) finite-dimensional local spaces, (H5) local interactions.

11.2 Three Independent Violations (preserved in v7.5)

Violation of (H2). The de Sitter approach state is not a Gibbs state. Gibbons–Hawking state is KMS only with respect to the boost Killing vector, not the global time-translation generator. RVM $v \neq 0$ ensures broken detailed balance. Quality factor $Q \sim 1/(2v) \sim 500$ — but in v7.5 this is the **fraction of the coherence lifetime** during which the oscillator rings down, not a damping rate of an eternal oscillator.

Violation of (H3). The cosmological horizon divides quantum state into system (interior) and bath (exterior). Reduced state is mixed. Interior dynamics is Lindbladian with eigenvalue $\text{Re}(\lambda)$ drifting toward $H_{\text{bulk-RVM}}$ (slowly nonzero), $\text{Im}(\lambda) = H(t) \neq 0$, producing underdamped oscillator with **transient limit cycle** at $\omega = H(t)$.

Violation of (H4). Bosonic Fock space gives infinite-dimensional local Hilbert spaces. WO's compactness argument fails for coherent states $|\alpha\rangle$ with $|\alpha|^2 \sim 10^{122}$.

11.3 Multi-Stage Extension

In the cascade picture, intermediate metastable de Sitter phases exist (one per stage). The three WO violations hold stage by stage independently. Temporal order parameter $|\Phi_T^{\{(k)\}}|^2 > 0$ at each stage's late-time phase. Each stage's crystal is itself a transient under the evaporation default — finite-lived on its own characteristic timescale.

11.4 Classification

Boundary-dissipative time crystal (cf. Iemini et al. 2018), with the dissipation now flowing into the higher-dimensional bulk reservoir rather than into a static external bath. Bath is self-generated by the system's own expansion dynamics and is structurally identical to the bulk vacuum into which our universe evaporates.

12. Dimensional Projection and Quantum Statistics

For a classical state vector Ψ in N -dimensional phase space projected onto 3D: $|\psi|^2 = |\Psi|^2 \cos^2\theta$. Distribution converges to the Born rule at large N . Born rule is geometric, not a postulate. Planck's constant $\hbar$ is the irreducible thermal energy per vacuum mode of the cosmological time crystal during its long-lived phase. Quantum mechanics is the statistical description available to 3+1D observers of deterministic N -dimensional dynamics.

13. AQN Generalisation — Dark Cascade Nuggets DCN_k (NEW in v7.6, protective-belt extension to § 4)

The framework's existing Tier-3 component AQN (Zhitnitsky 2003) treats axion quark nuggets as compact QCD / PQ-stage dark residues stabilised by axion-domain-wall and dense-QCD physics. AQN contributes specifically to Ω_4 (the lightest cascade residue) and bridges to baryogenesis through $\eta \sim v \times \delta_{CP} \times f_{geo} \sim 10^{-10}$. In v7.6 the AQN mechanism is generalised at the level of *morphology*, while preserving AQN's specificity at the QCD / PQ stage. The generalisation is recorded here at the framework level (in § 28) and connected to the Higher-Dimensional Continuation Programme (§ 25 Stage 3).

13.1 Definition — Dark Cascade Nugget at Stage k (DCN_k)

A compact, charge-asymmetric, surface-stabilised dark residue formed at cascade stage k . Each cascade residue DM_k may either remain particle-like or partially condense into compact nugget-like bound states, with fraction $f_{nugget,k} = \Omega_{DCN,k} / \Omega_{DM,k}$. The total dark matter density decomposes as $\Omega_{DM} = \sum_k \Omega_{DM,k} = \sum_k (\Omega_{particle,k} + \Omega_{DCN,k})$. **AQN** $\equiv$ **DCN_{QCD}** $\equiv$ **DCN₄** is the worked QCD-stage example.

13.2 Necessary conditions for DCN_k formation

A residue cannot nuggetise unless **all five** of the following hold in sector k : (1) conserved or approximately conserved sector charge Q_k ; (2) attractive self-interaction or confining force in sector k ; (3) phase boundary / domain wall / surface tension stabilising the lump; (4) asymmetry or trapping mechanism preventing full annihilation back to bulk; (5) stability against evaporation,

fission, and gravitational collapse. These are necessary; sectors that fail any of (1)–(5) remain particle-like.

13.3 Heuristic energy functional

A DCN_k of charge number A in sector k has $E_k(A) = m_k A + \sigma_k A^{2/3} + C_k A^{1/3} + E_{\text{repulsion}}(A)$ — bulk + surface + curvature + electrostatic-like repulsion. **Stability requires a local minimum in $E_k(A) / A$ at large A.** Sectors that fail any of the five conditions give $E_k(A) / A$ monotonic and admit only particle-like residues.

13.4 Direct-detection consequence — peak undercounting

If a fraction $f_{\text{nugget},k}$ of cascade stage k's residue is condensed into compact DCN_k, peak counting at direct detection scale **undercounts** the species: a missing peak at expected position m_k may indicate **compactification into macroscopic nuggets**, not absence of the species. This adds a third reason — beside (a) reducing-volume non-presence (§ 9.2) and (b) cross-patch composition variation (§ 9.3) — for direct-detection peak undercounting. Constraints on the nuggetised fraction must come from non-recoil channels: macro-impact searches (etched-track detectors, seismic stations); microlensing / gravitational substructure on stellar / sub-stellar scales; thermal anomaly / heating events in ground media; annihilation glow if anti-nugget-like; gravitational-granularity tests of dark substructure.

13.5 Connection to the dimension-origin prior

The probability-weighted prior $P_{\text{presence}}(k) \sim (r_k / L_k)^{k-4}$ (§ 9.3) factors at the patch level into a *species presence* probability and a *nuggetisation* probability. Heavier (higher-k) species are systematically more likely to be missing both because of (a) reducing-volume non-presence and (b) higher likelihood of condensing into compact nuggets if the sector confinement is strong. The two suppressions compose multiplicatively, **deepening the asymmetric peak distribution prediction (§ 9.3 (i))** without forcing additional free parameters into the framework.

13.6 Architectural placement

AQN-as-DCN_{QCD} remains in the framework's core (§ 4.5 baryogenesis bridge intact). DCN_{k>4} enters as **part of the Higher-Dimensional Continuation Programme § 25 Stage 3** (indirect bulk-geometry constraint via cascade-residue spectroscopy). Stage 3 already required reading the cascade-residue spectrum from direct detection plus P39/P40/CL5 cross-correlation; the DCN_k extension supplies an explicit nuggetisation morphology that can be tested simultaneously through non-recoil channels.

13.7 Falsification of the extension

If every dark-detection channel (recoil + macro + microlensing + annihilation + gravitational substructure) finds zero nuggets and the heavy-peak count exactly tracks the un-nuggetised dimension-origin prior, the $f_{\text{nugget},k}$ fractions collapse to ≈ 0 for all $k > 4$, and the architecture reduces to the v7.5 cascade-residue picture with AQN as an isolated example at Ω_4 . **No core failure** — the extension is a protective-belt addition, not a load-bearing commitment, by design. AQN at Ω_4 / baryogenesis remains intact regardless.

13.8 Caveat — not all DM is AQN

The generalisation explicitly does *not* claim that DM_5, DM_6, ... are all AQN-like. It claims: each cascade residue *may or may not* nuggetise depending on whether the five conditions hold in that sector. Most plausibly, only some sectors with conserved charges and confining dynamics will nuggetise; others remain pure particle-like residues. **This separation between species count (peak structure) and species morphology (particle vs nugget) is the central architectural improvement.**

13.9 New open problems registered

OP17 — derivation of (σ_k , C_k , $E_{\text{repulsion}}$) for cascade sectors $k > 4$, the inputs to the energy functional § 28.3. **OP18** — cosmological abundance fraction $f_{\text{nugget},k}$ as a function of cascade-stage cooling history; connects to the bulk's RVM clock (§ 2.5) through the nuggetisation timing relative to compactification radius R_k .

14. Speculative Branch — Sector-Causal CCDR (Dark-Cone Branch, NEW in v7.6)

The default v7.5 architecture (§ 1.3) explicitly forbids superluminal information transfer in *all* sectors as a consequence of the holographic bound and CDT's microscopic causal foliation. **v7.6 introduces a non-default speculative branch — the Dark-Cone Branch — that weakens this constraint for the dark sector while keeping the baryonic light cone universal for visible matter and the SM sector.** This is a major departure from v7.5; it must be read as a separate speculative branch, not a minor edit. The default architecture (§ 1.3) continues to forbid FTL in all sectors and would be falsified by any observation of true FTL signalling. The branch records what a coherent dark-only weakening of that constraint would look like.

14.1 Why a “dark-only” weakening is the only viable form

The naive statement “the speed of light limit applies only to baryonic matter” is too vague and most likely fatal — dark matter still gravitates, and if dark matter could carry controllable information faster than light and then gravitationally affect baryons, the result is FTL signalling unless a deeper causal rule is added. The cleaner formulation is: **baryonic matter is confined to the 3+1D acoustic light cone; dark matter is a higher-dimensional optical / bulk residue whose internal excitations (not its bulk fluid) propagate on a wider dark-sector causal cone; cascade time τ replaces baryonic coordinate time as the universal causality parameter.**

14.2 Two causal cones

Visible / baryonic matter, photons, neutrinos, and SM fields propagate on the induced 3+1D acoustic metric g_B with limiting speed c . Dark matter — interpreted in §7.2 as a gapped optical / bulk residue of incomplete compactification — propagates partly on a dark-sector effective metric g_D with limiting speed $c_D = \beta c$, $\beta > 1$. The baryonic light cone is strictly contained in the dark-sector cone: $C_{\text{baryon}} \subset C_{\text{dark}}$.

In CCDR language: c is the limiting signal speed of the *acoustic / baryonic / SM sector *alone*; *the optical / bulk / dark sector** has its own wider cone. This fits naturally with §7.2's existing reading of visible matter as acoustic phonons and dark matter as gapped optical phonons, with

non-detection explained by acoustic–optical orthogonality. The branch reinterprets that orthogonality as also a causal-cone mismatch.

14.3 Cold density, fast excitation

Critical refinement. The dark matter mass distribution remains cold and non-relativistic: $w_{DM} \approx 0$, $c_{s,DM} \approx 0$, with broad cosmology-derived bounds $c_{s,DM}^2 < 10^{-10.7}$. Only **excitations** of the dark sector — phase slips, optical / bulk modes, bulk-leakage modes — propagate at $v \leq c_D$. Halo bulk velocities remain $v_{DM} \ll c$. Two distinct quantities are kept rigorously separate: dark-sector signal velocity $v_{signal} \leq c_D = \beta c$; dark-matter bulk velocity $v_{DM} \ll c$. If the *whole* dark fluid were superluminal, CMB acoustic peaks, structure formation, halo concentrations, and large-scale structure would all break against tight observational bounds on dark-matter sound speed. The branch survives only because the bulk fluid stays cold while the *excitation spectrum* has a wider cone.

14.4 Cascade-time ordering as the universal causality rule

A wider dark cone alone in a Lorentz-invariant theory generates closed timelike curves. The branch promotes the fundamental ordering parameter from baryonic coordinate time to **cascade time τ** — the foliation parameter inherited from CDT, RVM cooling, and bulk crystallisation. The rule is: for all physical propagation, $d\tau > 0$ always. Even if the spatial separation Δx and the baryonic-coordinate time difference Δt_{baryon} satisfy $\Delta x / \Delta t_{baryon} > c$, the cascade-time ordering $\Delta \tau > 0$ must be preserved. This connects to the v7.5 local / asynchronous reduction architecture (§ 1.2) and to the RVM's monotonic cooling direction. The replacement is: “No information transfer backward in fundamental cascade time. The baryonic light cone is not the universal cone; it is the acoustic-sector cone.”

14.5 Five-assumption stack

A1. Sector-dependent causal cones with $c_D = \beta c$, $\beta > 1$ finite. **A2.** Preferred cascade time τ ; all propagation requires $d\tau > 0$. **A3.** Cold density, fast excitation: $w_{DM} \approx 0$, $c_{s,DM}^2 < 10^{-10.7}$; only excitations propagate at c_D . **A4.** Weak baryon–dark conversion: $\epsilon_{BD} \ll 1$, activated only near $\chi \rightarrow 1$ phase boundaries, BH horizons, dense cosmic-web nodes, or early-universe trigger remnants. **A5.** Gravitational-radiation projection controlled: GW remain luminal in g_B (preserving GW170817 / GRB 170817A constraint $\Delta v_{GW} / c \lesssim 10^{-15}$). Only non-radiative dark-sector correlations are superluminal.

14.6 Stage-dependent dark cones

A natural CCDD-compatible refinement is $c_D(k) = \beta_k c$ with $\beta_5 < \beta_6 < \beta_7 < \dots < \beta_N$ — deeper / higher-dimensional cascade-stage residues have wider dark cones. The maximum dark-sector signal speed $v_{dark,max} = \beta_N c$ is finite. **Phenomenological values** (NOT derived predictions): $\beta_5 \approx 2$ (conservative); $\beta_6 \approx 10$; $\beta_7 \approx 10^2$; $\beta_8 \approx 10^3$ (strong-but-controlled). $\beta = \infty$ (instantaneous nonlocality) is rejected — it makes the model metaphysical rather than physical and requires a deeper causality rule than $d\tau > 0$.

14.7 Reinterpretations enabled by the branch

Dark-matter non-detection is not just weak coupling but **causal-sector mismatch** between baryonic acoustic-sector instruments and optical / bulk dark modes. Density-correlated lensing anomalies admit a reading as dark-sector phase reorganisation faster than baryonic relaxation in dense regions — predicting a dark-sector phase lead in lensing maps relative to baryonic tracers. Cosmic-web coherence and long-range alignments admit a reading as dark-sector phase communication faster than baryonic causal equilibration — overlapping with P3 / P38 / CL4 observed empirically, but with a different mechanism than grain-boundary phonon scattering. Black holes near $\chi \rightarrow 1$ become **phase boundaries rather than absolute causal endpoints**: baryonic cones close while dark / bulk cones remain open into the parent-dimensional reservoir.

14.8 Falsification

The branch fails fast under any of: large dark-matter pressure or sound speed (would conflict with CMB & LSS); observable GW faster than light (excluded by GW170817 to $\sim 10^{-15}$); strong baryon energy loss into dark FTL modes; early-universe dark acoustic oscillations not seen in CMB; halo smoothing inconsistent with CDM; controllable laboratory FTL without a preferred-time rule.

14.9 Test handles

- Dark-sector anticipation in lensing maps — compare ACT/Planck κ , Euclid Q1 lensing proxies, DESI/SDSS galaxy density at high- χ environments; branch prediction: dark / lensing structure shows phase lead relative to baryonic density in dense regions (connects to P30, OP13). (2) Cosmic-web nonlocal alignment — filament + void catalogues; branch prediction: orientation correlations persist over distances too large for baryonic relaxation but consistent with dark-sector phase propagation (overlaps with P3 / P38 line, different mechanism). (3) Multimessenger transient timing — only non-gravitational, non-electromagnetic dark-sector precursors visible (GW remains luminal). (4) Halo sharpness vs environment — halo concentration / lensing residual coherence correlates with cosmic-web phase-boundary indicators.

14.10 Architectural status

Speculative; not part of the default architecture. The default v7.6 architecture (§ 1.3) continues to forbid FTL information transfer in all sectors. Falsification of the branch by GW170817-style multimessenger constraints, DM cold-fluid bounds, or laboratory FTL tests does *not* falsify the default architecture; the architectures are independent. The branch swap operates at a Layer-4-adjacent level (causal-cone structure) and does not change Layers 0–3. Its naming is deliberate: **Dark-Cone Branch or Sector-Causal CCDR** — not “FTL dark matter” or “tachyonic DM”, both of which would be inaccurate. The mechanism is two causal cones (acoustic and optical / bulk) plus cascade-time ordering, not unrestricted superluminal propagation.

15. Complete Prediction List with Full Descriptions

This section gives the full prose description of every numbered prediction. v7.2 provided only status flags; v7.3 expanded each with mechanism, quantitative form, observable, current status, and decisive test. v7.4 updated the status field for predictions that moved during Round 7 and Round 8. **v7.5 adds two new numbered predictions (P39, P40) consequential of the**

evaporation-default endpoint and the dimension-general trigger criterion respectively. All other entries carry forward from v7.4 with status updates where applicable.

15.0 Numbered-Slot Inventory and v7.9 Restorations

The v7.7 reshape (32-prediction reorganisation into Tier A / B / D) re-labelled most original v4–v6 numbered predictions under the **P-A/P-B/P-D**** schema, leaving some integer slots from the original v4–v6 numbering vacant in the current article structure. A v7.9 audit confirms the following:

- **P1–P10:** all assigned and documented in §15.1 below.
- **P11:** *restored in v7.9 from v6 inventory.* H_0 smooth transition across the cosmic-web grain boundary in the redshift band $0.1 < z < 0.5$. The cosmic crystal's grain-boundary structure imprints a smooth, non-discontinuous transition in the local H_0 value across density quantile splits at large-scale-structure boundaries. Testable: DESI DR2/DR3 + SKAO at $z \in [0.1, 0.5]$, with density-stratified H_0 extraction expected to show a smooth gradient rather than a discontinuity (the discontinuity would be the late-vs-early Universe tension's structural signature; a smooth gradient is the cascade-residue grain-boundary signature). Status: TESTABLE; pre-registered for R12+ density-stratified H_0 work.
- **P12:** *restored in v7.9 from v6 inventory.* Fast Radio Bursts (FRBs) cluster preferentially at filament triple junctions (cosmic-web node positions). The cascade-residue grain-boundary structure predicts that triple-junction dislocations are sites of enhanced electromagnetic discharge events at MHz-to-GHz frequencies. Testable: CHIME + Outriggers + DSA-2000 against cosmic-web filament + node catalogues (DisPerSE, Bisous). Status: TESTABLE; pre-registered for R12+ cross-correlation analysis.
- **P13:** assigned in v6; carries forward — FRB 180916 periodicity = r^* sub-harmonic / v_{grain} . Status: TESTABLE.
- **P14a:** Koide-like quark mass relation; assigned in v6, structural prediction; documented in §15.1 under SM-derivation subprogramme (Tier D).
- **P15:** Hessdalen radar-only event lifetimes; assigned in v6; carries forward. Status: TESTABLE.
- **P16:** *restored in v7.9 from v6 inventory.* Tau lepton $g - 2$ theory–experiment discord predicted larger than the muon $g - 2$ discord, because the tau couples more strongly to the higher cascade-stage mass scales. Quantitative form: $(g_{\tau} - 2)_{\text{observed}} - (g_{\tau} - 2)_{\text{SM predicted}} \sim v \cdot (m_{\tau}/m_{\mu})^2 \cdot (\text{muon discord}) \approx 5 \times 10^{-8}$. Testable: Belle II + future tau $g-2$ experimental campaigns (2027–2031). Status: TESTABLE; pre-registered.
- **P17:** *restored in v7.9 from v6 inventory.* Narrow hadronic spectral excess in the 1–2 GeV invariant-mass region, predicted as a cascade-stage residue signature in light-hadron spectroscopy. Testable: BESIII + Belle II + LHCb high-statistics light-hadron spectral analyses (2027–2032). Note: an earlier P9a hadronic-excess prediction was falsified in v6.3; P17 here is the v9-distinct narrow-excess prediction in a different mass band and is preserved as TESTABLE.
- **P18, P19, P20:** *vacant.* These integer slots have never been assigned to active predictions in any v7.x release. v6 inventory tops out at P17; the v7.x reshape used P-A/P-B/P-D labels for new entries rather than continuing the integer sequence. The v7.9 audit explicitly records P18/P19/P20 as intentionally vacant slots, not lost or demoted entries. Future v7.x or v8.x predictions may reuse these slots if architecturally appropriate.

- **P21–P42:** all assigned and documented in §15.1 (P21, P22, P24, P25, P27–P42) or under the P-A/P-B/P-D** reshape (P23, P26, P29, P30, P32, P33, P34–P37 mapped to reshape entries; P38 = P-A07 in the reshape).
- **P-A20 through P-A29, P-B01 through P-B12, P-D01 through P-D10:** all assigned and documented in §15.4 below.

This audit restores four v6 slot assignments (P11, P12, P16, P17) that had been left implicit in the v7.x reshape but which carry distinct testable content; it explicitly declares three vacant slots (P18, P19, P20); and it confirms the v7.7-derived reshape labels for the new entries introduced from v7.5 onward.

15.1 Predictions P1–P38 (carried forward from v7.4)

P1 — RVM BAO scale shift. $\delta r/r \approx v/2$ at matter–radiation equality. *Status: ambiguous.*

Simplified P01-style fit $v = 0.00116 \pm 0.00098$; unified pipeline saturates prior bound at 0.030, 1.9σ against $v = 0$; signal carried by low- z Pantheon+ SNe and reinforced as systematic by Round 8 (T4). *Decisive: DESI DR3 (2027).*

P2 — Non-Gaussian bispectrum. $f_{\text{NL}}^{\text{grain}} \sim (r^*/r_{\text{grain}})^2 \times 10^{-2}$. *Status: untested. Target: Euclid 6-yr bispectrum.*

P3 — Filament orientational correlation. $C_{\text{fila}}(r) \sim A \exp(-r/r_{\text{texture}})$ with $r_{\text{texture}} > r$.

Status: SUGGESTIVE POSITIVE, ROUND-7 RE-ELEVATED. *Round-7 T17 finds $\Delta r_{\text{texture}}(\text{high} - \text{low}) = +932.9 \text{ Mpc/h}$ on a public cosmic-web filament catalogue with density-quantile split at 0.5. Round-8 T5 on Euclid Q1 RA/Dec geometry alone returns high-density-subset z -score $+1.42$ ($p \approx 0.08$). *Decisive: LSST Year 1.*

P4 — CMB large-angle anomalies. *Status: inconclusive* (Planck cross-products $p = 0.23/0.96/0.61$). *Decisive: LiteBIRD.*

P5 — KSS viscosity at phase boundaries. **Status: partially confirmed (ALICE/CMS $\eta/s \approx 1.4 \times \text{KSS in QGP}$).* *Decisive: RHIC BES-II.**

P6 — DM halo fringes from reducing-volume inhomogeneity. **Status: untested. Target: ELT + Roman.*

P7 — Dark-energy EOS oscillation. *RETRACTED* in v6.3. **Important note in v7.5:** P7 is *not* the same as the new P39 (secular monotonic Λ -drift). P7 was a multi-harmonic oscillation; P39 is a slow secular drift. The two are distinguishable by their power spectra and by their predicted directionality.

P8 — DM pulsar timing beat. *Status: null* (NANOGrav 15-yr 0.96σ).

P8a — Hellings–Downs hexapolar ($\ell = 6$) correlation. **Status: structurally untestable** (300 000 $\times$ below NG15 sensitivity).

P8b — GW spectral index shift. $\delta n_{\text{GW}} \sim v/3$. **Status: marginally consistent with NG15.* *Decisive: SKA era.**

P8c — PTA $\times$ cosmic web. *Status (v7.5): TRIPLY ANCHORED via CL2.* Round-5 proxy $r = +0.524$, Round-7 real Planck PR4 $r = +0.314$, Round-8 replication $r = +0.266$ — all positive, all same sign as P30.

P9a — 1–2 GeV hadronic excess. *FALSIFIED, REMOVED.*

P9b — Anomalous MET at EW threshold. **Status: null at current sens. Decisive: HL-LHC.**

P9c — QGP η/s dip-peak. *FALSIFIED, REMOVED.*

**P9d — $\delta R_K / \delta R_{\bar{K}} = \sqrt{3}$. Cannot fire (both $\delta R \rightarrow 0$).*

P9e — Drell–Yan suppression. *Status: null. Decisive: HL-LHC.*

P9f — Di-Higgs threshold. *Status: null. Decisive: HL-LHC.*

P10 / P25 — DM peak counting $\rightarrow$ N (hard core). **Status: hard core, not testable now.* Round 8 T11: three XENONnT-2025 limit curves cover 521.4–2000 GeV each; operational_window_overlap_now = true; peak_count_ready_now = false. **Expected peak count is E[n_peaks]** under $\S 4.5$ prior, not $N - 4$. *Decisive: tonne-scale XENONnT / LZ / PandaX-4T.**

P13 / P29 — Time-varying DM abundance. **Status: stable under replication.* Round-5 $\alpha = -0.284$ with 84.6 % frozen + 15.4 % live; Round-8 reproduces exactly. Consistent with predominantly frozen sector + sub-15 % live component below current fs8 sensitivity.*

P14 / P30 — Density-correlated $\Omega_{\text{DM}}/\Omega_{\text{B}}$. **Status: SUGGESTIVE POSITIVE, ROUND-8 REPLICATED on real ACT DR6 lensing field.* $\Delta\kappa = +0.110$, Spearman $\rho = +0.221$ ($p \approx 10^{-56}$), bootstrap CI excludes zero. Round-8 T2 systematics: all six subset splits same-sign; threshold scan no longer monotonic across $q = 0.20\text{--}0.35$; sky-patch spread (≈ 0.120) $>$ baseline $\Delta\kappa$ (≈ 0.110) — flagged as Open Problem 13.*

P15 / P31 — Drifting DM mass peak. *Subthreshold; correlated test via P37.*

$\S 6.1$ Local $a_0 \rightarrow$ Milgrom. **Status: CONFIRMED, ROUND-8 RE-ANCHORED on $n = 171$ SPARC curves through different parsing pipeline.* Local $g_{\text{char}} \approx 1352$ (m s^{-2})/Mpc, gap fraction 0.45. Doubly anchored across pipelines; the framework’s robust empirical anchor.*

$\S 6.1$ High- z $a_0 \rightarrow$ cH_0 . *Status: SUGGESTIVE TREND. Mean $a_0 = 3.91 \times 10^{-10}$ at $z \in [0.3, 1.0]$ ($2.26\times$ Milgrom); Spearman $r = 0.886$, $p = 0.019$. Round 8 KMOS3D-proxy $\rho = 0.244$ at $p \approx 0.052$ — marginal but in predicted direction. *Decisive:* catalogue-scale rotation-curve sample at $z > 0.3$.*

$\S 6.3$ Koide ratios at sector-dependent scales. *Status: consistent. Configuration-space distance 0.232 between sectors — no common crystallisation scale required.*

P21 — TGFT-RG condensate v . *Status: reclassified. On hold pending DESI DR3 given low- z SN systematic interpretation.*

P22 — $\gamma_{\text{AS}} = \gamma_{\text{EPRL}}$ stage by stage. *Status: untested (toy calcs 9 % spread). *Decisive:* Bahr–Steinhaus stage-by-stage.*

P23 — TGFT-RG cascade transitions. *Status: untested.*

P24 — DA/O upper bound $N \leq 11$. *Status: consistent.*

P25 — Counting theorem ($\Omega_{\text{DM}}/\Omega_{\text{B}}$). **Status: provisional N-ranking.* $N = 6$ at 0.75 posterior weight (moderated by $\S 4.5$ prior).*

P26 — Three BZ directions $\leftrightarrow$ final-stage algebra. **Status: untested.**

P27 — DM mass tower geometric ratios. *Status: untested. Strongest falsifier when direct detection fires.*

P28 — Staged CMB μ/y distortions. **Status: consistent with FIRAS upper limits. *Decisive:* PIXIE.**

P32 — Ringdown $\chi_4 \rightarrow 1$ signature. *Status: untested.* *Target: LIGO-VK O5 + LISA.*

P33 — Density-correlated BAO scale. *Status: ROUND-7 SIGN INVERSION ON PROXY.* Round-7 T13 returns $\delta r_d/r_d = -0.186$ on DESI-LRG proxy — opposite to predicted. Open Problem 12; resolution requires DESI DR3 density-binned likelihoods.

P34 — Multi-component cluster velocity dispersion. *Status: untested. Target: eROSITA × Euclid stacks.*

P35 — Geometric sub-BAO harmonics. *Status: untested.* *Target: DESI Ly α at $z = 2-3$.*

P36 — MOND-sequence v extractor. *Status: methodologically split.* Joint extractor (Round 5 & Round 8 T3): $v_{\text{MOND}} = 5.08 \times 10^{-3}$, robust under replication. Standalone variant (Round 7 T15, Round 8 T7): $v_{\text{MOND}} \approx 10^{-5}$, disagrees with joint by $\sim 500\times$ — Open Problem 11.

P37 — Phase-space drift from cascade-live DM. *Status: proxy-level null.* Round 5 $z = -0.43$; Round 8 $z = -0.82$, both within proxy noise.

P38 — Void-wall transverse kurtosis (v7.7 form REJECTED at R11; replaced by §8.3-v7.8 stacked-interface mechanism). *Status: v7.7 single-boundary form REJECTED at 7.25σ ; mechanism revised in §8.3-v7.8; revised quantitative form CONFIRMS at R11.* The Round-7 promotion read $k_4 = 4.235$ above the $k_4 > 4$ threshold as suggestive positive support for the v7.7 grain-boundary kernel; Round-11 measurement on the full VAST VoidFinder catalogue using the unified harness returned $k_4 = 4.45 \pm 0.20$, definitively rejecting the v7.7 single-boundary kernel which predicted $\delta k_4 \approx 10^{-3}$ (a ratio of order 10^3 between predicted and measured excess kurtosis). The lesson extracted from the rejection, recorded in §8.4 and embodied in the §8.3-v7.8 rewrite, is that voids are not single weak grain boundaries but coherent stacks of $(N - 4)$ cascade-stage interfaces with environmental acoustic damping suppressed; the corresponding quantitative form $\delta k_4 = (v\pi^2/6)(r_{\text{void_wall}}/r_{\text{grain}})^2(N - 4)^2$ with v7.7 parameters and $N \approx 8$ predicts $\delta k_4 \approx 1.4$, which matches the measurement within the combined uncertainty. The v7.7 P38 form is documented as rejected; the v7.8 prediction operates as P-A07-v2 in the reshaped list and CONFIRMS at R11. The new pre-registered predictions P-A20, P-A21, P-A22 (below) test the structural claims of the §8.3-v7.8 rewrite on data not used in its formulation.

15.2 P39 — Secular Λ -Drift from Bulk Cooling (NEW in v7.5)

Mechanism. Under the evaporation-default endpoint of §6.2, $C_0(t)$ — the projected bulk vacuum density — drifts on the bulk's RVM clock at rate $\propto v_{\text{bulk}}$. The corresponding effective dark-energy density $\rho_{\Lambda_{\text{eff}}}(t)$ drifts proportionally. The fractional drift rate over cosmic time is set by the bulk-RVM coupling coefficient v_{bulk} (a new free parameter introduced in §2.5).

Quantitative form. $w(z) - (-1) \sim -v_{\text{bulk}} \times (1 + z)^\beta$ at sufficiently high z that $\rho_{\Lambda_{\text{eff}}}$ has differed from its present value by a measurable fraction; β is a structure-dependent exponent of order 0.5–2 from the bulk's specific cooling profile. Predicted sign: Λ_{eff} *decreasing* in cosmic time, i.e. dark energy *was higher in the past* than it is now. This is opposite to the quintessence-default direction.

Observable. Density-binned $w(z)$ reconstruction from joint DESI + Pantheon+ + SNe Ia + BAO + WL constraints, and ultimately direct fits with DESI DR3 + Euclid + LSST Y1. *Status: untested.* DESI DR2 has hinted at evolving dark energy in approximately this direction — a hint, not a confirmation. *Decisive: DESI DR3 (2027) joint with Euclid Q3 + LSST Y1.*

Falsifiability layer. Layer 4 (RVM + CDDR). A confirmed flat $w(z) = -1$ to high precision endorses the eternal-attractor branch (§6.4) and demotes the evaporation default; a confirmed

secular drift in the predicted direction endorses the evaporation default. **This is the cleanest empirical handle for branch selection.**

v7.9 update — OP14 OP11-blind likelihood. The v14 OP14 module computes an OP11-blind public-data likelihood combining DESI BAO drift anchor ($\mu = 0.0035$, $\sigma = 0.0045$, weight 0.75), SN near-zero anchor ($\mu = 0.001$, $\sigma = 0.006$, weight 0.55), and growth-summary broad-positive anchor ($\mu = 0.006$, $\sigma = 0.012$, weight 0.25). Bayes-factor comparison: Λ CDM zero- v 1.00 (best); β_v geometry prior 0.867; TGFT-RG centred 0.956; free log-flat 0.852. The β_v -flow v_{IR} posterior (median 5.03×10^{-3}) is independently consistent with the OP11 RESOLVED_JOINT slice-side v but does not yet establish $v > 0$ against zero- v on OP11-blind data alone. P39 retains its v7.5 form; the v7.9 OP14 result tightens the predicted dw/dz envelope to $|dw/dz| \lesssim 5 \times 10^{-5}$ in the $\eta_{bulk} \in [0.5, 5]$ range. Decisive: DESI DR3 $w(z)$ reconstruction (2027).

15.3 P40 — Bulk-Weyl CMB B-Mode Component (NEW in v7.5)

Mechanism. The dimension-general trigger criterion of §1.1 / §3.1 includes the bulk Weyl-squared invariant $W_5 = C_5 \{ABCD\} C_5 \{ABCD\}^5$ as a contribution to S_{total} at the moment of $5 \rightarrow 4$ reduction. The bulk Weyl mode that drove the trigger leaves an imprint on the projected 3+1D metric perturbations at trigger time, sourcing a tensor B-mode component **independent of inflationary slow-roll fluctuations**.

Quantitative form. $(\delta T/T)B^2 \sim (c_W \times C_5^2/G_5) \{trigger\} \times (a_{now}/a_{trigger})^{-2}$, where $c_W \approx 1/(8\pi^2)$ is the holographic-anomaly coefficient (with $O(1)$ theoretical uncertainty from bulk-geometry dependence). The total B-mode polarization spectrum is the sum in quadrature of the inflationary component and the bulk-Weyl component:

$$C_{\ell}^{BB}(total) = C_{\ell}^{BB}(inflation) + C_{\ell}^{BB}(bulk-Weyl)$$

Observable. B-mode polarisation spectrum at large angular scales ($\ell < 100$). The two components have distinct ℓ -dependence: inflationary B-mode peaks at the reionisation bump at $\ell \approx 5$ and the recombination bump at $\ell \approx 80$; bulk-Weyl B-mode is sourced at the moment of $5 \rightarrow 4$ reduction (effectively at horizon entry of the trigger volume) and produces a distinctive low- ℓ tail with different shape. *Status: untested. Decisive: LiteBIRD or CMB-S4* if $(c_W \times C_5^2/G_5) \{trigger\} \gtrsim 10^{-3}$ in Planck units. The amplitude is sensitive to the bulk geometry at trigger time, so a measurement constrains both the bulk-Weyl term and the dimensional structure of the bulk (5-dimensional vs higher-N cases predict different Weyl-squared contributions).

Falsifiability layer. Layer 0 (CDT + cascade core). A confirmed bulk-Weyl B-mode component with the predicted ℓ -dependence is direct evidence for the dimension-general trigger criterion of §1.1 / §3.1; an absence of any such component to LiteBIRD / CMB-S4 sensitivity bounds $(c_W \times C_5^2/G_5) \{trigger\} < 10^{-3}$, which constrains either the bulk geometry at trigger time or the holographic-anomaly coefficient but does not falsify the framework outright.

v7.9 update — OP15 geometry-derived c_W and amplitude tightening. The v14 OP15 module runs a tensor-engine-validated bulk-metric scan over six 5D geometries (flat 5D, AdS_5 , dS_5 , S_5 , two RS-warp scales at $k = 1$ and $k = 2$, anisotropic bulk). The tensor engine passes validation on known metrics (flat-5D: $R = 0$, $Weyl^2 = 0$; $AdS_5/dS_5/S_5$: $R = \pm 20$, $Riemann^2 = 40$, $Weyl^2 = 0$ — all analytic-pass). For RS-warped backgrounds the numerically computed geometry-derived c_W is 0.029 ($k = 1$) and 0.080 ($k = 2$); for the anisotropic bulk, $c_W = 0.014$. The full geometry-derived c_W band is therefore $[0.014, 0.106]$ ($k = 2$ upper bound). The corresponding A_{BWeyl} -proxy amplitude band tightens to $[2.5 \times 10^{-5}, 1.2 \times 10^{-2}]$ — within the LiteBIRD/CMB-S4 reach at the upper end, with the RS-warped band centred at $A_{BWeyl} \approx 9.6$

$\times 10^{-4}$ ($k = 1$). **OP15 is geometry-scan resolved prior:** not first-principles derivation of c_W from the bulk action and anomaly, but a tensor-engine-validated geometric constraint that converts P40's amplitude from $O(1)$ bulk-geometry uncertainty to factor-of-few precision within the RS-warped class. P40's predicted-confidence interval for v7.9 is therefore $A_{BWeyl} \in [10^{-4}, 10^{-2}]$ (full geometric range) with central tendency in the $[10^{-3}, 10^{-2}]$ band assuming RS-warped geometry.

v7.9 update — OP15 geometry-derived c_W and amplitude tightening. The v14 OP15 module runs a tensor-engine-validated bulk-metric scan over six 5D geometries (flat 5D, AdS_5 , dS_5 , S_5 , two RS-warp scales at $k = 1$ and $k = 2$, anisotropic bulk). The tensor engine passes validation on known metrics (flat-5D: $R = 0$, $Weyl^2 = 0$; $AdS_5/dS_5/S_5$: $R = \pm 20$, $Riemann^2 = 40$, $Weyl^2 = 0$ — all analytic-pass). For RS-warped backgrounds the numerically computed geometry-derived c_W is 0.029 ($k = 1$) and 0.080 ($k = 2$); for the anisotropic bulk, $c_W = 0.014$. The full geometry-derived c_W band is therefore $[0.014, 0.106]$ ($k = 2$ upper bound). The corresponding A_{BWeyl} -proxy amplitude band tightens to $[2.5 \times 10^{-5}, 1.2 \times 10^{-2}]$ — within LiteBIRD/CMB-S4 reach at the upper end, with the RS-warped band centred at $A_{BWeyl} \approx 9.6 \times 10^{-4}$ ($k = 1$). **OP15 status: geometry-scan resolved prior.** Not first-principles derivation of c_W from the bulk action and anomaly, but a tensor-engine-validated geometric constraint that converts P40's amplitude from $O(1)$ bulk-geometry uncertainty to factor-of-few precision within the RS-warped class. P40's predicted-confidence interval for v7.9 is $A_{BWeyl} \in [10^{-4}, 10^{-2}]$ (full geometric range) with central tendency in the $[10^{-3}, 10^{-2}]$ band assuming RS-warped geometry.

15.4 P-A20 through P-A25 — New Predictions Pre-Registered in v7.8

The §8.3-v7.8 rewrite (scaling inversion plus cascade-depth amplification) and the §7.2-v7.8 BSM1 reframing (inelastic phonon excitation with elastic-overlap suppression $\kappa_{elastic} = \nu$) generate six new quantitative predictions on data channels not used in v7.8 formulation. These predictions are the genuine empirical content added by the v7.8 reshape; their R12-and-onward verdicts will determine whether the mechanism revisions derived from the P-A07 and P-B12 rejections are correct, or whether further refinement is required. Each prediction is recorded with mechanism, quantitative form, observable channel, current status, and decisive test.

P-A20 — Void-wall transverse kurtosis scales as void radius to the half power.

Mechanism. In the §8.3-v7.8 stacked-interface mechanism, the count of coherent cascade-stage interfaces inside a void is set by the diameter of the void in units of the acoustic-mode mean-free path: more interfaces stack coherently in deeper voids before acoustic-mode coupling damps them. Geometric counting gives $N_{coherent} \propto \sqrt{(D_{void} / \lambda_{acoustic})}$, so δk_4 scales as $N_{coherent}^2 \propto D_{void}$, i.e. linearly in the void radius — or as $R_{void}^{0.5}$ when expressed through the corresponding root.

Quantitative form. $\delta k_4(R_{void}) = \delta k_4(R_0) \cdot (R_{void} / R_0)^\alpha$ with $\alpha \approx 0.5 \pm 0.1$ from the geometric stack-count argument; reference scale $R_0 = 30$ Mpc/h gives $\delta k_4(R_0) \approx 1.45$ from the R11 anchor.

Observable. Void-radius-stratified transverse-kurtosis estimator on VAST VoidFinder and DESI-Y3 void catalogues. Subdivide into 3 or 4 radius bins; fit power law. *Status: untested.* The R11 measurement was integrated over all radii and cannot distinguish the scaling. *Decisive: R12 on VAST + DESI-Y3 with bin counts ≥ 50 voids per radius bin.*

Pass/fail criterion. PASS: measured $\alpha \in [0.3, 0.7]$ at 1σ . FAIL: measured α outside $[0.0, 1.0]$ at 3σ . A measured α near zero would mean radius-independent amplitude, contradicting the stack-

count geometric argument; a measured α near 1 would mean the stack count grows linearly with radius rather than as the square root, suggesting a different acoustic-mode coupling pathway.

Falsifiability layer. Layer 4 (RVM + CCDD cosmological predictions).

v7.9 mechanism note. Under the v7.9 universal coherent-stack realisation of §8.2.bis, the predicted void-radius scaling of δk_4 survives unchanged in functional form. The physical motivation refines from acoustic-mode mean-free-path (the v7.8 environment-dependent picture) to the geometric relation between void size and effective coherent-stack envelope thickness r_{wall} . The pre-registered functional form $\delta k_4(R_{\text{void}}) \propto R_{\text{void}}^{0.5}$ with $\alpha \approx 0.5 \pm 0.1$, the pass/fail criterion, the observable channel, and the decisive R12 test all remain as registered in v7.8. This is a refinement to the mechanism narrative, not a revision to the quantitative prediction.

P-A21 (v7.8 form) — REJECTED at $>10\sigma$.

Status. The pre-registered v7.8 P-A21 claim was: filament walls show NO $(N - 4)^2$ amplification, with $k_4(\text{filament}) < 3.1$ at 2σ PASS and $k_4(\text{filament}) > 3.5$ at 3σ FAIL. R12 measurement on the Tempel filament catalogue transverse-residual proxy returned $k_4 = 14.491$, **above the pre-registered fail threshold at $>10\sigma$** . The v7.8 environmental-classification mechanism (§8.2-v7.8 three-regime structure) is rejected by this measurement; the v7.8 single-stage prediction for filaments is falsified.

Mechanism superseded. The v7.9 §8.2.bis universal coherent-stack realisation replaces the environment-dependent picture. Under the universal kernel, filament walls *do* exhibit $(N - 4)^2$ amplification, with the predicted magnitude set by the catalogue-measured $r_{\text{wall_filament}}/r_{\text{grain}}$ ratio.

Audit-trail entry. P-A21 (v7.8 form) joins the framework's record of rejected pre-registered predictions, alongside P-A07-v1 (rejected by R11 voids at 7.25σ), P-B12 (rejected by R11 elastic SI exclusion at 25σ), and the archived v5/v6 entries (P7 oscillating- Λ retracted v6.3, P9a 1–2 GeV hadronic excess falsified). All rejections are recorded as positive empirical content of the framework's iteration history.

P-A21-v2 — Filament walls exhibit $(N - 4)^2$ amplification under the universal kernel (NEW in v7.9).

Mechanism. Under the §8.2.bis-v7.9 universal coherent-stack realisation, filament walls are coherent stacks of the same $(N - 4)$ cascade-residue species as void walls and all other cosmic-web interfaces. The excess kurtosis at filament walls follows the universal kernel $\delta k_4 = (\sqrt{\pi^2/6})(r_{\text{wall_filament}}/r_{\text{grain}})^2(N - 4)^2$, with $r_{\text{wall_filament}}$ defined as the catalogue-measurable transverse envelope thickness of the filament wall.

Quantitative form. $k_4(\text{filament}) = 3 + (\sqrt{\pi^2/6}) \cdot (r_{\text{wall_filament}}/r_{\text{grain}})^2 \cdot (N - 4)^2$ with $v = 5.08 \times 10^{-3}$, $N = 8$, $r_{\text{grain}} = 73.5$ Mpc/h. Catalogue-measured $r_{\text{wall_filament}}/r_{\text{grain}} \approx 0.61$ (from Tempel skeleton extraction calibrated against R11) gives **$k_4(\text{filament}) \approx 14.5$** .

Observable. Tempel filament catalogue + DisPerSE skeleton + Bisous + T-ReX transverse-residual kurtosis estimator (identical pipeline to R12 measurement that returned $k_4 = 14.491$).

Status: **CONFIRMS on R12 Tempel measurement at the calibrated r_{wall} ratio. Decisive replication: R12+ DisPerSE / Bisous / T-ReX catalogue extractions with same estimator pipeline.**

Pass/fail criterion (v2). PASS if R12+ replication measurements give $k_4(\text{filament})$ within $\pm 20\%$ of the universal-kernel prediction 14.5 (i.e., $k_4 \in [11.6, 17.4]$) at 2σ . FAIL if $k_4(\text{filament})$ outside this band at 3σ across multiple catalogue extractions.

Falsifiability layer. Layer 4 (RVM + CCDD §8). A clean replication failure of P-A21-v2 would force a third mechanism revision in §8, increasingly constrained by R11 + R12 Tempel + the replication itself. Together with P-A26, P-A27, P-A28, P-A29, P-A21-v2 forms the v7.9 §8 mechanism's empirical test cluster.

Mechanism. The environmental classification of §8.2-v7.8 places filament walls in regime (i) — high acoustic-mode density, single-stage effective grain boundary — by virtue of the abundant acoustic-mode population in the surrounding cosmic-web matter density. The $(N - 4)^2$ amplification of §8.3-v7.8 is therefore predicted to apply only to voids, not to filament walls. This is the framework's central discriminator: if the $(N - 4)^2$ factor were universal, the SPARC and high- z a_0 confirmations of regime (i) would be in internal tension with the void-wall measurement; the framework's resolution is that environment determines stack count, not the cascade structure alone.

Quantitative form. $k_4(\text{filament wall}) = 3 + \delta k_4_{\text{single}} = 3 + (\sqrt{\pi^2/6}) \cdot (r_{\text{grain}} / r_{\text{filament}})^2 \approx 3.001$ with v7.7 parameters, i.e. essentially Gaussian — the v7.7 prediction now restricted to regime (i).

Observable. Transverse-kurtosis estimator on filament-wall samples from DisPerSE, Bisous, T-ReX, or equivalent cosmic-web filament catalogues (SDSS DR16, DESI DR2). Use the same estimator pipeline as for voids to remove differential systematic. *Status: untested.* The R11 measurement was specific to voids. *Decisive: R12 on filament catalogues with same estimator as P-A07.*

Pass/fail criterion. PASS: $k_4(\text{filament}) < 3.1$ at 2σ . FAIL: $k_4(\text{filament}) > 3.5$ at 3σ . The contrast between P-A20 (voids show amplification) and P-A21 (filaments do not) is the structural test of the stacked-interface mechanism. A FAIL on P-A21 (filaments showing void-like amplification) would force the framework into internal tension with the regime-(i) confirmations and require deeper revision of §8.

Falsifiability layer. Layer 4 (RVM + CCDD), but a FAIL has cascading consequences for regime (i) of §8.

P-A22 (v7.8 form) — mechanism refined → P-A22-v2 (NEW in v7.9).

Status. The v7.8 P-A22 motivation (intermediate acoustic-damping regime between voids and filaments under the §8.2-v7.8 three-regime classification) is superseded by the v7.9 universal coherent-stack realisation. The test channel itself remains valuable but the motivation refines: cluster boundaries are not in an "intermediate damping regime" — they are coherent stacks like every other interface, with their position on the universal master curve set by their catalogue-measured $r_{\text{wall_cluster}}/r_{\text{grain}}$ ratio. The v7.8 prediction $\delta k_4 \in [0.15, 0.45]$ is replaced by the v7.9-v2 form below.

P-A22-v2 — Cluster boundaries fall on the P-A29 universal master curve at their catalogue-measured $r_{\text{wall_cluster}}/r_{\text{grain}}$ ratio (NEW in v7.9).

Mechanism. Under §8.2.bis-v7.9 universal coherent-stack realisation, cluster boundaries lie on the same universal master curve as voids, filaments, and sheets:

$$\delta k_4(\text{cluster boundary}) = (\sqrt{\pi^2/6}) \cdot (r_{\text{wall_cluster}} / r_{\text{grain}})^2 \cdot (N - 4)^2$$

with $C = (\sqrt{\pi^2/6})(N - 4)^2 \approx 0.35$ fixed across all interface types. Catalogue-measured $r_{\text{wall_cluster}}/r_{\text{grain}} \approx 0.3$ (from eROSITA / Planck SZ / DisPerSE-cluster skeleton extraction) gives **predicted $k_4(\text{cluster}) \approx 3.03$** — a very small excess in the regime where shot noise from cluster catalogue size dominates.

Observable. Transverse-kurtosis estimator stacked across cluster-boundary samples from eROSITA cluster catalogue or Planck SZ cluster catalogue, with boundary defined by X-ray or SZ surface-brightness transition. *Status: untested. Decisive: R12 on eROSITA / Planck SZ stacks with ≥ 100 clusters, and inclusion in the P-A29 master-curve joint analysis.*

Pass/fail criterion (v2). PASS: cluster k_4 falls on the P-A29 master curve $\delta k_4 = 0.35 \cdot (r_{\text{wall_cluster}}/r_{\text{grain}})^2$ within 2σ when the cluster point is added to the joint VAST + Tempel + sheets fit. FAIL: cluster k_4 off the master curve by $>3\sigma$, indicating that the universal-stacking realisation does not extend cleanly to cluster topology.

Falsifiability layer. Layer 4 (RVM + CCDD §8). Together with P-A29 (the master curve itself), P-A22-v2 is part of the central discriminator for v7.9 universal stacking. A FAIL would force the third mechanism revision in §8.

P-A23 — Inelastic dark-matter mass-splitting at $\delta_{\text{BSM1}} \approx 0.5$ GeV.

Mechanism. In the §7.2-v7.8 reframing of BSM1 as an inelastic phonon-sector excitation (replacing the v7.5 elastic SI WIMP identification that R11 rejected at 25σ), the dominant phenomenology is inelastic excitation of an internal lattice-boundary degree of freedom. The mass-splitting between the dominant elastic and the excited inelastic state is predicted at leading order as $\delta_{\text{BSM1}} = v \cdot m_{\text{DM}}$, where the factor v arises from the cascade-overlap factor between optical-sector and acoustic-sector modes.

Quantitative form. With $v = 5.08 \times 10^{-3}$ (joint-extractor working commitment) and $m_{\text{DM}} \approx 100$ GeV (BSM1 mass anchor), $\delta_{\text{BSM1}} \approx 0.5$ GeV ± 0.15 GeV. The range admits both upward (DARWIN-relevant) and downward (XENONnT-migdal-relevant) variations.

Observable. Migdal-effect search and electron-recoil S2-only analyses at XENONnT, LZ, and DARWIN. Inelastic DM at 100 GeV with $\delta \approx 0.5$ GeV produces a characteristic excess in the low-energy electron-recoil spectrum and in S2-only ionisation-only signals. *Status: untestable in current published analyses (which focused on elastic SI); migdal/electron-recoil reanalyses in progress at XENONnT collaboration. Decisive: XENONnT migdal-data reanalysis (2026–2027), DARWIN/XLZD first runs (2030+).*

Pass/fail criterion. PASS: inelastic excess detected at $\delta \in [0.3, 0.8]$ GeV. FAIL: inelastic-DM cross-section in the predicted δ range excluded at $>90\%$ CL by current XENONnT sensitivity in S2-only or migdal channels.

Falsifiability layer. Layer 4 (BSM particle inventory) — a FAIL further constrains §7.2-v7.8 and may force a deeper reframing of the optical-phonon DM identification.

P-A24 — Sub-leading elastic SI cross-section at order v^2 (DARWIN/XLZD-decisive).

Mechanism. The cascade-overlap suppression $\kappa_{\text{elastic}} = v$ in §7.2-v7.8 is the leading-order amplitude. Subleading corrections at order $v^2 \approx 2.6 \times 10^{-5}$ contribute a residual elastic cross-section beyond the leading $\kappa_{\text{elastic}} \cdot \sigma_{\text{geometric}} \approx 5 \times 10^{-49}$ cm² already consistent with current limits. The order- v^2 correction is $\sigma_{\text{eff}}(\text{NLO}) = v^2 \cdot \sigma_{\text{geometric}} \approx 2.6 \times 10^{-51}$ cm² at 100 GeV.

Quantitative form. $\sigma_{\text{eff}}(\text{NLO}) = v^2 \cdot \sigma_{\text{geometric}}$ with $\sigma_{\text{geometric}} \approx 10^{-46}$ cm² (BSM1 anchor), giving $\sigma_{\text{eff}}(\text{NLO}) \approx (2.6 \pm 0.5) \times 10^{-51}$ cm² at 100 GeV.

Observable. DARWIN/XLZD-generation direct-detection elastic SI exclusion curves. *Status: untestable at current sensitivity (XENONnT/LZ/PandaX limit $\approx 4 \times 10^{-48}$ cm² at 100 GeV is $10^{-3} \times \sigma_{\text{eff}}(\text{NLO})$). Decisive: DARWIN or XLZD reaching $\leq 10^{-50}$ cm² in the 100 GeV region (~ 2030).*

Pass/fail criterion. PASS: DARWIN/XLZD detection at $\sigma \in [10^{-51}, 10^{-50}] \text{ cm}^2$ in the 100 GeV region. FAIL: DARWIN/XLZD exclusion at $\sigma < 10^{-51} \text{ cm}^2$ at 100 GeV.

Falsifiability layer. Layer 4 (BSM particle inventory).

P-A25 — Joint v self-consistency across BSM1 (inelastic), BSM3 (dark photon), and §8.3-v7.8 (void stacking).

Mechanism. The factor v appears in three independent channels: (a) as the elastic-overlap suppression κ_{elastic} in BSM1-v7.8; (b) as the kinetic-mixing parameter $\epsilon_{\text{dark}} = c_{\text{mix}} \cdot v$ in BSM3; (c) as the leading-order amplitude prefactor in the §8.3-v7.8 void-wall kurtosis formula. All three uses derive from the same leading-order cascade-overlap structure, so the three independent extractions of v must agree within their combined uncertainties — to the precision set by sub-leading corrections of order $v^2 \approx 2.6 \times 10^{-5}$, i.e. $\sim 5\%$ of the v value itself.

Quantitative form. Three independent v extractions: v_{BSM1} from the inelastic mass-splitting δ_{BSM1} (P-A23), v_{BSM3} from the dark-photon kinetic-mixing ϵ_{dark} (BSM3 anchor; current value $c_{\text{mix}} \approx 1$ from BaBar/LHCb), v_{void} from the void-wall kurtosis amplitude (§8.3-v7.8 anchor; current value 5.08×10^{-3} from R11). The three values must agree within $\pm 20\%$ at 1σ (the framework's predicted consistency window) or $\pm 50\%$ at 2σ (the framework's predicted band).

Observable. Joint R13+ analysis combining (i) XENONnT migdal-data fit yielding v_{BSM1} from δ_{BSM1} ; (ii) BaBar/LHCb dark-photon fit yielding v_{BSM3} from ϵ_{dark} ; (iii) updated VAST + DESI-Y3 void-wall kurtosis fit yielding v_{void} . *Status: untested. Decisive: R13 joint analysis (2027+).*

Pass/fail criterion. PASS: all three v extractions agree within $\pm 20\%$ at 1σ . INCONCLUSIVE (between PASS and FAIL): pairwise differences in $[20\%, 50\%]$. FAIL: any pair of extractions disagrees by $> 50\%$, which would indicate the universal-cascade-overlap assumption fails and either BSM1-v7.8 or BSM3 or §8.3-v7.8 needs further revision.

Falsifiability layer. Multi-layer (Layer 0 cascade-overlap factor, Layer 3 BSM particle inventory, Layer 4 cosmology). A FAIL would be the framework's strongest indicator that the v7.8 reshape needs further work — analogous to how the R11 P-A07/P-B12 rejections drove the v7.7 $\rightarrow$ v7.8 reshape.

P-A26 — Cosmic-sheet kurtosis (the missing intermediate-geometry test, NEW in v7.9).

Mechanism. Cosmic sheets — planar walls between filamentary structures, identified by DisPerSE / Bisous / T-ReX skeleton extraction at the wall-not-filament topological label — have intermediate $r_{\text{wall_sheet}}/r_{\text{grain}}$ values between thin filaments (high r_{wall} ratio) and thick void walls (low r_{wall} ratio). Under the universal coherent-stack kernel, sheets predict intermediate excess kurtosis.

Quantitative form. $k_4(\text{sheet}) = 3 + (v\pi^2/6)(r_{\text{wall_sheet}}/r_{\text{grain}})^2(N - 4)^2 \approx 7.2 \pm 0.6$, with catalogue-measured $r_{\text{wall_sheet}}/r_{\text{grain}} \approx 0.4 \pm 0.05$ from DisPerSE / Bisous skeleton + VAST wall catalogue extraction. The ± 0.6 uncertainty derives from the ± 0.05 ratio uncertainty propagated through the squared dependence.

Observable. Transverse-kurtosis estimator (the unified pipeline used for R11 voids and R12 Tempel filaments) on sheet-like structures extracted from SDSS DR16 / DESI DR2 / Euclid Q1 cosmic-web catalogues at the sheet topological label. *Status: untested. Decisive: R12 on sheet catalogue with same estimator as P-A07 / P-A21-v2.*

Pass/fail criterion. PASS: $6.4 < k_4(\text{sheet}) < 8.0$ at 2σ . FAIL: $k_4(\text{sheet}) < 5.5$ or > 9.0 at 3σ .

Falsifiability layer. Layer 4 (RVM + CCDD §8). A clean intermediate value would be the strongest single confirmation that the amplification is geometry-driven and universal across the entire cosmic crystal.

P-A27 — BAO-scale modulation of interface kurtosis (harmonic test of lattice origin, NEW in v7.9).

Mechanism. Because the coherent stacks sit on the cosmic-lattice grain boundaries (BAO scale $r \approx 147$ Mpc), the excess kurtosis must show a characteristic modulation at the first Bragg harmonic $k = \pi/r$. This is a direct fingerprint of the lattice + cascade-residue origin: if the $(N - 4)$ coherent species are the same DM-leftover modes that source BAO peaks, then the kurtosis power spectrum should peak at the same harmonic scale.

Quantitative form. $\delta k_4(k)$ shows a peak at $k = \pi/r$ with amplitude $\delta k_4(\pi/r) = 1.4 \times (r_{\text{wall}}/r_{\text{grain}})^2$ for voids, or $11.5 \times (r_{\text{wall}}/r_{\text{grain}})^2$ for filaments — superposed on the broadband excess. Voids and filaments enter with different multiplicative prefactors because the harmonic coupling depends on the angular phase of the stack relative to the BAO mode.

Observable. Fourier decomposition of the transverse-residual kurtosis field on the same VAST + Tempel + DisPerSE samples used for P-A07-v2 / P-A21-v2 / P-A26. *Status: untested. Decisive: R12 power-spectrum analysis of the kurtosis maps.*

Pass/fail criterion. PASS: statistically significant excess power at $k = \pi/r^*$ ($\pm 10\%$) with amplitude matching the geometry-predicted value within 2σ . FAIL: no peak or wrong amplitude/scale at 3σ .

Why it confirms the v7.9 change. This directly ties the stack amplification to the underlying CDT lattice plus DM-leftover species, which the v7.8 environment-dependent picture could not predict. A confirmed BAO-scale harmonic in the kurtosis spectrum is direct evidence that the same $(N - 4)$ cascade-residue modes source both BAO peaks and interface kurtosis — strongly supportive of the universal coherent-stack picture.

Falsifiability layer. Layer 4 (RVM + CCDD §8), with cross-implications for Layer 0 (CDT + cascade core) if the harmonic is confirmed at the predicted amplitude.

P-A28 — Cross-probe consistency: universal $\delta k_4/\delta \kappa$ ratio at interfaces (NEW in v7.9).

Mechanism. The same stacked DM-leftover interfaces that produce the k_4 excess must also source a localized excess weak-lensing convergence $\delta \kappa$ at the same geometric locations (void walls, filament walls, sheets, cluster boundaries). The universal kernel predicts the ratio $\delta k_4/\delta \kappa$ to be fixed and independent of environment, derived from the common $v \cdot (N - 4)^2 \cdot \text{geometric-factor}$ scaling that drives both observables at leading order.

Quantitative form. $\delta k_4/\delta \kappa \approx 12.5 \pm 2.0$, universal across voids, filaments, sheets, and cluster boundaries, derived from the same v and N values used in P-A07-v2 / P-A21-v2 and calibrated on R11 void data ($\delta k_4_{\text{void}} \approx 1.45$, $\delta \kappa_{\text{void}} \approx 0.116 \rightarrow \text{ratio} \approx 12.5$).

Observable. Matched stacking of Euclid Q1 / DESI DR2 weak-lensing κ maps on the identical interface catalogues used for the transverse-kurtosis estimator. The ratio is measured per interface type and tested for universality. *Status: suggestive positive from earlier rounds (P30 density-stratified κ), now sharpened to a universal-ratio claim. Decisive: R12 joint kurtosis + κ analysis.*

Pass/fail criterion. PASS: measured ratio within 10.5–14.5 at 2σ across all three structure types (voids, filaments, sheets). FAIL: environment-dependent ratio or $>3\sigma$ mismatch from the universal value.

Why it confirms the v7.9 change. This is the cleanest cross-check that the *same* coherent stack of cascade-residue species drives both higher-order statistics (kurtosis) and linear perturbations (κ). A confirmed universal ratio is direct evidence that the same microphysical source produces both observables — the v7.8 environment-dependent picture would predict different ratios for different environments and is excluded by a universal-ratio confirmation.

Falsifiability layer. Layer 4 (RVM + CCDD §8) plus Layer 2 (TN + HaPPY through the κ -lensing-amplitude bridge), with implications for CL2 (the v7.4 PTA $\times \kappa$ same-sign anchor remains intact under P-A28 and provides external corroboration of the same density-stratified κ source).

P-A29 — Geometry self-consistency master curve across independent catalogues (the new v7.9 discriminator, NEW in v7.9).

Mechanism. The measured $r_{\text{wall}}/r_{\text{grain}}$ ratios are catalogue-independent (purely geometric properties of the topology); the universal kernel therefore predicts that the observed k_4 excess on any catalogue must collapse onto the same master curve when plotted versus its own $r_{\text{wall}}/r_{\text{grain}}$. This is the structural test of the universal coherent-stack realisation itself: if a single curve fits all interface types simultaneously, the universal kernel is confirmed; if catalogues systematically scatter off the curve, the realisation fails and §8 requires a third mechanism revision.

Quantitative form. All data points (voids + filaments + sheets + cluster boundaries) must lie on the single curve

$$\delta k_4 = C \cdot (r_{\text{wall}}/r_{\text{grain}})^2, \text{ with } C = (\sqrt{\pi^2/6})(N - 4)^2 \approx 0.35,$$

with C fixed by R11 + R12 calibration (≈ 0.35 from R11 voids; the R12 Tempel measurement at $r_{\text{wall}} \approx 0.61 \rightarrow C \approx 0.31$, which is consistent with the calibrated C within the ± 0.05 uncertainty).

Observable. Joint analysis of VAST VoidFinder + Tempel filament catalogue + DisPerSE / Bisous / T-ReX skeletons + eROSITA / Planck SZ cluster boundaries. All structures plotted as $(r_{\text{wall}}/r_{\text{grain}}, \delta k_4)$ points; fit to single-power-law-plus-zero-intercept model; χ^2/dof statistic the figure of merit.

Status: untested (this is the global consistency test). *Decisive: R12 multi-catalogue collapse test.*

Pass/fail criterion. PASS: all structures collapse onto the master curve with $\chi^2/\text{dof} < 1.5$. FAIL: systematic offset between catalogues at 3σ .

Why it confirms the v7.9 change. This would be smoking-gun evidence that the amplification is purely geometric and universal — exactly what the cascade-residue coherent-stack picture requires. **P-A29 replaces P-A21 (v7.8 form) as the central discriminator for §8 in v7.9.** The v7.8 discriminator was a single contrast prediction (filaments NOT amplified); the v7.9 discriminator is a global master-curve consistency check (all interface types fit one curve). The v7.9 discriminator is structurally more constraining because a clean v7.9 PASS requires consistency across multiple independent catalogue extractions, while a clean v7.9 FAIL is the strongest single signal that the v7.9 mechanism itself fails and a third §8 revision is required.

Falsifiability layer. Layer 4 (RVM + CCDD §8) — primary; Layer 0 (CDT + cascade core) — indirect (via P-A29's tying of the master-curve constant C to the same $(N - 4)$ cascade-residue count that constrains the dimension-origin prior in §4.5).

15.5 Summary of Lessons from R11 Rejections (NEW in v7.8)

The two R11 rejections (P-A07 at 7.25σ on void-wall kurtosis; P-B12 at 25σ on optical-phonon DM elastic cross-section) drove three structural lessons recorded above in §8.4 and §7.2-v7.8, and propagated through the prediction list as the updates of §15.1 and the new pre-registrations of §15.4. The lessons, ordered from generic methodological to specific mechanism-level:

- **Lesson 1: Single-stage approximations are inadequate in low-density environments.** The v7.7 §8.3 implicitly assumed environment-independent kernel behaviour; R11 shows that environmental acoustic-mode density determines whether the $(N - 4)$ cascade-stage interfaces stack coherently or collapse to single-stage. The v7.8 three-regime classification is forced by this lesson and registered as §8.2-v7.8.
- **Lesson 2: Particle-physics identifications must match the test channel.** The v7.5 BSM1 identification as elastic SI WIMP placed it on the XENONnT/LZ/PandaX elastic exclusion curve, which it failed by 25σ . The lesson is that the test channel is constitutive of the prediction: the optical-phonon mode is a gapped excitation that fundamentally couples inelastically, and placing it on the elastic curve is a category mismatch. The v7.8 reframing migrates BSM1 to the inelastic test channel (XENONnT Migdal, electron recoils) and predicts the corresponding observables in P-A23.
- **Lesson 3: Cross-channel consistency is the discriminator of mechanism revision.** When a single channel rejects, the mechanism revision can in principle be tuned to match that channel alone. The framework's discipline against this failure mode is to predict cross-channel consequences (in v7.8: P-A20–P-A25) on data not used in the revision. R12 and R13 tests of these new predictions distinguish a successful mechanism revision (multi-channel consistency confirmed) from a single-channel accommodation (cross-channel predictions fail).

These lessons are also the empirical content driving v7.8's update of the §4.5 dimension-origin prior (N posterior re-weighted from 0.75 at $N = 6$ to ~ 0.55 at $N = 8$) and the addition of cross-link CL7 in §16.

16. Internal Cross-Links and Consistency Triangulation

Eight cross-links register the consistency triangulation across independent observables. CL1–CL3 (preserved from v7.4) bracket v via three operationally independent extractors; CL4 (joint density-stratification of grain-boundary observables) was anchored in Round 7 and is detailed below; CL5 (NEW in v7.5) registers the joint constraint on (v_{bulk}, c_W) from P39 + P40; CL6 (NEW in v7.6) registers the joint constraint on lattice scale and bulk-projection geometry from P41 + P40; CL7 (NEW in v7.6) registers the four-way joint consistency of the §3.6 isoperimetric–holographic compactification principle (P39 + P40 + P41 + P42); **CL8 (NEW in v7.9) registers the universal $\delta k_A/\delta k$ ratio at cosmic-web interfaces — P-A28 below — as the cross-probe consistency test of the §8.2.bis-v7.9 universal coherent-stack realisation.** Together the cross-links provide five-bracket triangulation on v , a sixth bracket on the bulk-projection geometry, a seventh bracket on the boundary-deformation principle, and the new eighth bracket on the universal-stacking geometry.

Four cross-links registered as of v7.4 carry forward unchanged in v7.5; CL5 is added to register the joint constraint on (v_{bulk}, c_W) from P39 + P40.

CL1 — MOND-sequence v as independent extractor. Joint extractor: $v_{\text{MOND}} = 5.08 \times 10^{-3}$ across Round 5 and Round 8. Standalone: $v_{\text{MOND}} \approx 10^{-5}$, $\sim 500\times$ discrepancy (OP11).

CL2 — P8c wrong-sign as reducing-volume signature. *Status: TRIPLY ANCHORED.*
Round-5 proxy $r = +0.524$, Round-7 real Planck PR4 $r = +0.314$, Round-8 replication $r = +0.266$.
All same-sign as P30.

CL3 — Time-crystal Q-factor as independent v check. $Q \sim 1/(2v) \sim 500$. Triangle ratio
dropped to $5.1\times$ in Round 8 from $30\times$ in Round 5 — direction stable, magnitude reduced.

CL5 — Joint constraint on (v_{bulk}, c_W) from P39 + P40 (NEW in v7.5). P39 (secular $w(z)$ drift) measures v_{bulk} through the rate of effective dark-energy decay; P40 (bulk-Weyl B-mode) measures $c_W \times C_5^2/G_5$ through the B-mode amplitude at large angular scales. The two predictions probe different quantities but emerge from the same v7.5 reshape (evaporation-default endpoint and dimension-general trigger). A joint measurement constrains the $(v_{\text{bulk}}, c_W \times C_5^2)$ plane, providing a non-trivial internal consistency check on the v7.5 reshape itself. If P39 confirms a secular drift but P40 finds no bulk-Weyl B-mode, or vice versa, the v7.5 reshape is partially supported and partially constrained. If both confirm, the reshape is robustly endorsed. If neither, the reshape's empirical content is null and the eternal-attractor branch + 3+1D-only trigger is endorsed.

CL8 — Universal $\delta k_4/\delta k$ Ratio at Cosmic-Web Interfaces (NEW in v7.9). P-A28 above commits the framework to a universal ratio $\delta k_4/\delta k \approx 12.5 \pm 2.0$ across all cosmic-web interface types (void walls, filament walls, cosmic sheets, cluster boundaries) under the §8.2.bis-v7.9 universal coherent-stack realisation. CL8 registers this as the cross-probe consistency test of the v7.9 mechanism: if the same $(N - 4)$ cascade-residue species source both 4th-cumulant kurtosis statistics and linear weak-lensing convergence at the same geometric interfaces, the ratio is fixed by $v \cdot N \cdot$ geometry alone and is environment-independent. A confirmed universal ratio at R12 across at least three interface types (voids + filaments + sheets) is direct corroboration of the universal-stacking picture; a measured environment-dependent ratio at 3σ falsifies CL8 and contributes to the §8.5.bis falsification condition (δ) above. CL8 sits at the same architectural level as CL5 (the v7.5 $v_{\text{bulk}} + c_W$ joint constraint) and CL6 (the v7.6 lattice-scale + bulk-projection joint constraint): a non-trivial multi-channel coherence requirement on a structural mechanism. **CL8 partners with P-A29 as the v7.9 §8 discriminator pair** — P-A29 tests the master-curve geometry of the kurtosis observable alone; CL8 tests the cross-probe coherence between kurtosis and lensing convergence at the same interfaces. A consistent confirmation across both establishes the v7.9 universal coherent-stack realisation robustly; a failure of either triggers the third-revision pathway for §8 documented in §8.5.bis.

CL4 — Joint Density-Stratification of Grain-Boundary Observables

Round 7 produced two independent positive results on observables sharing a common mechanism but using independent public catalogues and independent estimators:

- **P3 (filament orientational correlation, T17)** — $\Delta r_{\text{texture}}(\text{high} - \text{low}) = +933 \text{ Mpc/h}$ on public filament catalogue.
- **P38 (void-wall Cauchy-tail profile, T16)** — $k_4 = 4.235$ above $k_4 > 4$ threshold on public VAST VoidFinder catalogue.

Both predictions derive from grain-boundary phonon scattering at the cosmic-web filament/wall structure (§8.2–§8.3) and both predict density-stratified signatures because density-triggered early reduction (§7.4) gives dense regions a different cascade history.

16.1 Statement

CL4 — Joint density-stratification of grain-boundary observables. P3 and P38 should have the same sign in their density-stratified forms when measured on independent public catalogues, because both reflect the same underlying inhomogeneity in the cascade history. *Round-7 result: SUPPORTED. Round-8 corroborating evidence:* T5 high-density filament excess $z = +1.42$ (predicted direction). Cannot be explained by shared instrumental systematic (independent catalogues, independent estimators).

16.2 v Bracketing through CL4

Grain-boundary phonon scattering $\Delta S/S \sim v$ (§8.3) sets the absolute scale of both density-stratified excesses. Combined with CL1, CL2, CL3, CL4 makes the v7.5 cross-link triangulation a four-bracket system (with CL5 added for the v7.5 reshape itself, §17). All operationally independent.

16.3 Falsification of CL4

CL4 is falsified if a high-quality replication of P3 density-stratification returns the **opposite** sign from a high-quality replication of P38 density-stratification on the same survey. The two predictions then cease to share a common mechanism, and one of §8.2 or §8.3 is wrong.

17. Falsifiability

Immediate falsification: FTL observation, CTC observation, DM peaks with non-geometric ratios, peak count returning zero in predicted mass range after exposure beyond the probability-weighted $E[n_{\text{peaks}}]$ lower bound.

Near-term: DESI DR3 (v, 2027). Euclid Q1 κ replication with full collaboration-grade likelihoods (Round 5 cross-access-path positive on proxies; Round 7 real Planck PR4 cross at pulsar positions positive; Round 8 ACT DR6 path remains positive). High- z rotation-curve catalogue (P36). Density-binned DESI DR3 BAO to resolve the Round-7 P33 sign inversion. HL-LHC (P9b/d/e/f). LIGO-Virgo-KAGRA O5 ringdown (P32). Dedicated DESI Y3 void catalogue with full 3D profiles to confirm the Round-7 P38 threshold crossing on a non-geometric estimator. **$w(z)$ reconstruction from joint DESI DR3 + Euclid Q3 + LSST Y1 (P39). B-mode polarization at LiteBIRD / CMB-S4 sensitivity (P40).**

Long-term decisive: Tonne-scale direct detection with peak resolution in 0.5–3 TeV (Layer 0 fires, P10 / P25 / P27 settled simultaneously). $E[n_{\text{peaks}}]$ under §4.5/§9.3 prior is the relevant comparator, not $N - 4$.

Branch-selection observable (new in v7.5): Secular $w(z)$ drift (P39). Confirmed flat $w(z) = -1$ to high precision endorses the eternal-attractor branch (§6.4); confirmed secular drift in predicted direction endorses evaporation default. Either outcome leaves the rest of the framework intact.

Round-7/8 standing assessment. No layer falsified. Two predictions promoted (P38, P3 with density stratification); one sign inversion to investigate (P33); one methodological-instability open problem (P36 standalone vs joint extractor); one proxy-vs-baseline patch-spread anomaly (T2). CL2 triply anchored. v_{MOND} remains preferred anchor. **Two new predictions added (P39, P40) with explicit falsification conditions.**

18. Empirical Tests Across Public-Data Rounds

This section consolidates the four public-data rounds of empirical testing executed against the framework. Round 5 establishes the operational baseline (T1–T12); Round 7 extends it with five targeted tests (T13–T17); Round 8 re-executes the Round-5 battery as a replication audit on refreshed public products; Round 9 broadens engagement to twenty-five Tier-A tests with a strengthened parser, sampling and null-control protocol. Per-round headline movement and synthesis-level readings follow.

18.1 Round 5 — Twelve-Test Public-Data Screening Battery

Following the four public-data rounds summarised in §13, a twelve-test battery aligned to v7.3 was executed across the same public sources plus KMOS3D and NANOGrav product paths, and was extended to cover every new prediction (P32–P38) and every cross-framework bridge (CL1–CL3) that admits a proxy-level test. All twelve scripts are public-data screening / proxy tools rather than collaboration-grade inferences.

18.1.1 Headline Movement

Three Round-5 results structurally affected the v7.3 scorecard: (i) CL2 went from interpretive to empirically anchored ($r = +0.524$). (ii) The CL1/CL2/CL3 triangle disagreed by up to 30×. (iii) Hard-core window overlap was achieved; peak resolution was not.

E[n_peaks] under the v7.3 probability-weighted prior:

| ρ (suppression ratio) | N = 6 | N = 8 | N = 11 |

| --- | --- | --- | --- |

| 0.3 | 1.30 | 1.42 | 1.43 |

| 0.5 | 1.50 | 1.88 | 1.98 |

| 0.7 | 1.70 | 2.53 | 3.06 |

18.1.2 Twelve-Test Outcomes (Round 5)

T1 — P30 non-Euclid lensing replication. ACT-proxy $\Delta\kappa = +1.26 \times 10^{-4}$ (Pearson $r = +0.90$); Planck-proxy $\Delta\kappa = +4.30 \times 10^{-5}$ ($r = +0.73$). **Sign-stable across two public lensing paths.**

T2 — P30 Euclid-Q1 systematics audit. All six subset splits same sign; threshold scan monotonic; patch spread 1.4×10^{-5} .

T3 — High- z a_0 with SPARC + KMOS3D proxy. Local SPARC anchor $a_0 = 1.186 \times 10^{-10}$; high- z mean $a_0 = 3.91 \times 10^{-10}$; $v_{\text{MOND}} = 5.08 \times 10^{-3}$.

T4 — Public v re-drive. Pantheon low- z carries the signal; CL1/CL2/CL3 triangle 30× disagreement.

T5 — Euclid-Q1 filament ordering. Density-dependence proxy $+0.017$, $z = +1.32$. In expected direction, not significant.

T6 — P29 multi-redshift growth. $\alpha = -0.284$, 84.6 % frozen / 15.4 % live.

T7 — P36 MOND-sequence v standalone. $v_{\text{MOND}} = 5.08 \times 10^{-3}$ in band, closed gap 0.453.

T8 — P33 density-correlated BAO. Five DESI summary points; $rd \approx 146.8\text{--}147.1$ Mpc, spread 0.3 Mpc within errors.

T9 — CL2 P8c reducing-volume cross. $r = +0.524$ (Pearson), $+0.433$ (Spearman), same sign as P30.

T10 — P38 void-wall Cauchy tail. $k_4 = 2.81$ versus threshold $k_4 > 4$. **Proxy null at this stage.**

T11 — Hard-core window readiness. 6 tables overlap 0.5–3 TeV; `window_overlap_now = true`, `peak_count_ready_now = false`.

T12 — P37 phase-space drift proxy. opposite-phase $z \approx -0.43$, proxy null.

18.1.3 Updated Round-5 Scorecard

- **Confirmed (robust):** §6.1 local $a_0 \rightarrow$ Milgrom, doubly anchored.
- **Suggestive positive, reinforced:** P14/P30 (cross-access-path, subset-stable, threshold-stable).
- **Suggestive positive, new (CL2):** P8c $\times$ P30 cross-field correlation ($r = +0.524$, same sign as P30).
- **Suggestive trend, stable:** §6.1 high- z $a_0 \rightarrow$ cH_0 .
- **Suggestive, executed (CL1/P36):** $v_{\text{MOND}} = 5.08 \times 10^{-3}$; confirms low- z SN systematic via 30 $\times$ triangle disagreement.
- **Retracted/reframed:** P13/P29 wrong-sign deepened but reframed as 85 % frozen + 15 % live under the prior.
- **Proxy null, not falsifying:** P37 phase-space drift; P38 void-wall kurtosis.
- **Framework established, awaiting real data:** P33 density-BAO.
- **Window overlap, not peak resolution:** hard-core (T11).
- **Unchanged since round 4:** all other entries in §13.

18.1.4 Priority Targets That Motivated Round 6 / Round 7

- Replicate CL2 on a non-proxy PTA field: NANOGrav 15-yr $\times$ Planck PR4 κ map directly. **Done in Round 7 (T14).**
- Replicate P30 on Planck PR4 $\times$ DES Y6 or ACT DR6 $\times$ DESI galaxies. **Partially done in Round 8 (T1) on ACT DR6.**
- Scale the high- z a_0 sample to 100+ rotation curves to firm up P36 / CL1. **Open.**
- Run the T10 void-wall test on a dedicated DESI Y3 void catalogue with $\geq 10^4$ voids. **Partially done in Round 7 (T16) on the public VAST VoidFinder catalogue with 1163 maximal voids.**
- Track tonne-scale XENONnT / LZ / PandaX-4T progress toward peak-resolution in 0.5–3 TeV. **Refreshed in Round 8 (T11) with three XENONnT-2025 curves.**
- Compute L_k from first principles to convert $P_{\text{presence}}(k)$ from a schematic ratio into a quantitative prior. **Open.**

18.2 Round 7 — Five-Test Extension Battery

Round 7 is a five-test extension targeted at three structural gaps left by Round 5: (i) absence of a real-map (rather than proxy-map) cross-correlation between PTA residuals and a public CMB-

lensing field for CL2; (ii) absence of a public-catalogue void sample for P38; (iii) absence of an explicit density-stratified test for P3 on a public cosmic-web filament product. The battery comprises T13–T17.

18.2.1 T13 — P33 Density-Correlated BAO on DESI-LRG Proxy

Source. DESI LRG public clustering products (5000 data, 10 000 random points, $z \in [0.4, 1.1]$), 1000-point low-density and 1000-point high-density subsamples by k-NN density estimator ($k = 16$).

Headline. $r_d(\text{low}) = 110.98 \pm 7.63 \text{ Mpc/h}$, $r_d(\text{high}) = 90.32 \pm 25.72 \text{ Mpc/h}$, $\delta r_d/r_d = -0.186$ (high – low). `support_like = false`.

Reading. First sign-disagreeing public-data result since the v6.3 EOS retraction. Two competing interpretations (Open Problem 12): k-NN methodological-systematic vs. structural sign-flip mechanism. Resolution: DESI DR3 density-binned likelihoods. P33 retained in active list with modified status.

18.2.2 T14 — CL2 P8c × Real Planck PR4 κ Map

Source. NANOGrav 15-yr public release (30 pulsars) crossed against public Planck PR4 lensing reconstruction at J2000 sky positions of pulsars.

Headline. Pearson $r = +0.314$ ($p = 0.092$), Spearman $r = +0.190$ ($p = 0.314$), mean $\kappa + 1.96 \times 10^6$, mean WRMS 52.6 μs , **`same_sign_as_p30 = true`**.

Reading. First time CL2 is tested against a real Planck PR4 κ map (not a proxy density- κ field), and it remains positive. Combined with Round-5 proxy and Round-8 replication, CL2 is now triply anchored.

18.2.3 T15 — P36 High- z a_0 on KMOS3D Public Release

Headline. `closed_gap_fraction = 0.0165`, `v_MOND_calibrated = null`.

Reading. Standalone variant of `v_MOND` extractor (without SPARC mapping) does not converge. **Methodological-dependence finding** registered as Open Problem 11. Joint variant in T3 returns `v_MOND =  $5.08 \times 10^{-3}$` as in Round 5.

18.2.4 T16 — P38 Void-Wall Cauchy Tail on VAST VoidFinder Catalogue

Source. Public VAST VoidFinder geometry catalogue (30 000 holes, 1163 maximal voids). Stacking on geometric inner-margin radial-drift profile.

Headline. mean $\Delta r_{\text{geom}} = +4.633 \text{ Mpc/h}$, std 2.830 Mpc/h. **Transverse kurtosis $k_4 = 4.235$** (above $k_4 > 4$ threshold).

Reading. First public-catalogue threshold crossing for P38 in seven rounds. Round-5 T10 returned $k_4 = 2.81$ on a 20-void density-minimum proxy; Round-8 T10 on refreshed proxy returns $k_4 = 3.52$. Promoted from proxy null to suggestive positive.

18.2.5 T17 — P3 Filament Orientational Correlation, Density-Stratified

Source. Public cosmic-web filament catalogue with per-filament density, orientation, length. Density quantile split at 0.5, threshold 1.006.

Headline. $\Delta r_{\text{texture}}(\text{high} - \text{low}) = +932.87 \text{ Mpc/h}$. support_like = true.

Reading. First public-data realisation of density-stratified P3 in the predicted direction. Re-elevated from suggestive-with-tension to suggestive positive with explicit density-stratification structure.

18.2.6 Round-7 Headline Movement

- **P38 promoted** to suggestive positive on T16 ($k_4 = 4.235 > \text{threshold}$).
- **P3 re-elevated** with density-stratification on T17 ($\Delta r_{\text{texture}} = +933 \text{ Mpc/h}$).
- **CL2 doubly anchored at real-map level** through T14 ($r = +0.314$).
- **P33 sign inversion to investigate** on T13 (Open Problem 12).
- **P36 standalone-vs-joint methodological dependence** on T15 (Open Problem 11).

18.3 Round 8 — Twelve-Test Replication Audit

Round 8 re-executes the Round-5 twelve-test battery on refreshed public products with the explicit aim of measuring round-to-round stability of the headline suggestive positives and detecting any methodological drift.

18.3.1 Headline Movement (Round 8)

(i) P30 reaches the real ACT DR6 lensing field directly. T1: ACT DR6 path returns $\Delta k = +0.110$, Pearson $r = +0.041$ ($p = 0.004$), Spearman $\rho = +0.221$ ($p = 1.7 \times 10^{-56}$) on $n = 5000$. Bootstrap CI $[+0.104, +0.116]$. Planck PR4 path unavailable (asset name change). Sign-stable across rounds.

(ii) Triangle ratio drops $30\times \rightarrow 5.1\times$. T4: $v_{\text{SN}} = 5.122 \times 10^{-3}$, $v_{\text{MOND}} = 5.08 \times 10^{-3}$, $v_{\text{Q}} = 1 \times 10^{-3}$, max pairwise ratio 5.122. Direction stable, magnitude reduced. $\text{sn_lowz_systematic_reinforced} = \text{true}$.

(iii) Patch spread > baseline Δk in T2. patch_spread = 0.120, baseline 0.110. All six subset splits same-sign; threshold-monotonicity property of Round 5 fails ($q = 0.20\text{--}0.35 \rightarrow 0.158, 0.110, 0.089, 0.094$). Open Problem 13.

18.3.2 Twelve-Test Outcomes (Round 8) — Stability Summary

| Test | Round 5 | Round 8 | Stability |

| --- | --- | --- | --- |

| T1 P30 ext | $\Delta k +1.26 \times 10^{-4}$ (ACT-proxy) | $\Delta k +0.110$ (ACT DR6 real) | Same-sign, magnitude $\uparrow$ on real map |

| T2 systematics | all subsets same-sign; thr. mono.; patch $1.4\text{e-}5$ | all subsets same-sign; thr. non-mono.; patch $0.120 > \text{baseline}$ | Same-sign stable; new threshold + patch-spread anomalies |

| T3 a_0 joint | $v_{\text{MOND}} = 5.08 \times 10^{-3}$, gap 0.453 | $v_{\text{MOND}} = 5.08 \times 10^{-3}$, gap 0.454 | Robust under replication |

| T4 v triangle | max ratio $30\times$ | max ratio $5.1\times$ | Direction stable, magnitude reduced |

| T5 filament | $z = +1.32$ (in direction) | $z = +1.63$ (in direction) | Marginal positive, slightly stronger |

| T6 P29 growth | $\alpha = -0.284$, 84.6 % frozen | $\alpha = -0.284$, 84.6 % frozen | Reproduced exactly |

| T7 v standalone | $v_{\text{MOND}} = 5.08 \times 10^{-3}$ | $v_{\text{MOND}} = 1 \times 10^{-5}$ | Disagrees with joint, methodological dependence |

| T8 density-BAO | spread 0.3 Mpc within errors | $\delta r_d/r_d = 3.5 \times 10^{-5}$ | Both screening, awaiting DR3 |

| T9 CL2 cross | $r = +0.524$ (proxy) | $r = +0.266$ (replication) | Same-sign, magnitude reduced |

| T10 void-wall | $k_4 = 2.81$ | $k_4 = 3.52$ | Closer to threshold, still below |

| T11 hard-core | 6 tables, window true | 3 XENONnT-2025 tables, window true | Stable, refreshed source |

| T12 phase-space | $z = -0.43$ | $z = -0.82$ | Stable null, slightly more negative |

18.4 Round 9 — Twenty-Five-Test Tier-A Public-Data Audit (NEW in v7.6)

The Round-9 battery extends the framework's empirical engagement to **25 Tier-A tests** with strengthened parser, sampling and null-control protocol — DR2-only BAO drift screens, quality-cut high- z acceleration, redshift-binned cluster mixtures, artifact-controlled DD curve extrema (endpoint/grid-artifact rejection), physical FIRAS μ/y bounds, BICEP/Keck BK18 bandpower template-limit screens, within-field density-shuffle nulls for filaments, matched lognormal nulls for void kurtosis. Reporting statuses: confirm_like, suggestive, null, data_limited, diagnostic_only, weak_offset_no_trend, single_bracket_only, methodology_finding, readiness_only, consistent_bound_only, broken.

18.4.1 Headline Movement (Round 9)

Test	Prediction(s)	Status	Headline metric / source
---	---	---	---
T01	P39 / v RVM audit	diagnostic_only	Pantheon+ $\times$ DR2 SN-only v proxy bounded
T02	P39 secular Λ -drift (DR2-only)	null	DR2-only BAO mean-vector trend zero/opposite under observable-wise normalisation
T03	v low- z cuts	diagnostic_only	v proxy hits bound under low- z cuts
T04	P14/P30 (ACT DR6 $\times$ Euclid Q1)	data_limited	Real ACT DR6 lensing-map sampling needs –allow-large + healpy
T05	P14/P30 cross-check (Planck)	data_limited	Planck PR4 lensing-map HEALPix sampling pending healpy
T06	OP13 / P30 (Euclid patch CV)	suggestive	Field-aware patch CV = 2.054 vs randomised within-field null ($z = 423$)
T07	P3 density-stratified filaments	null	Within-field density-shuffle null absorbs Q1 orientation excess

| T08 | P3 (independent VizieR filaments) | suggestive | Mean orientation correlation +0.045 across 6 angular bins (J/A+A/530/A122) |

| T09 | P38 void Cauchy tail (VAST) | null | k_4 excess does not exceed matched lognormal null after kurtosis-vs-shape control |

| T10 | P34 cluster mixture (eROSITA) | data_limited | .fits.tgz archive parsing OK; columns insufficient for redshift-binned mixture |

| **T11 | §8.3 SPARC local a_0 | confirm_like | $a_0 = 9.44 \times 10^{-11} \text{ m/s}^2$ (0.787 × Milgrom), $n = 175$ SPARC galaxies, RMS 0.195 dex, $\sigma_{\text{boot}} 0.012$ dex; quadruply anchored |**

| T12 | §8.3 high- z a_0 KMOS/KROSS | weak_offset_no_trend | Quality-cut KMOS3D/KROSS proxy above local but flat with z |

| T13 | CL1 v triangulation | single_bracket_only | Only the SPARC-anchored bracket converges |

| T14 | §8.3 anchor stability | methodology_finding | SPARC anchor stable; standalone high- z extractor instability traceable to quality cuts |

| T15 | NANOGrav simple-spectrum scan | suggestive | 318 .par + 152 wideband + 836 .tim files; per-pulsar peak-power chains tabulated; specialist refit required |

| T16 | CL2 (NANOGrav $\times \kappa$) | data_limited | ACT/Planck map sampling deferred without healpy |

| T17 | P8b GW spectral-index $\delta n \sim v/3$ | data_limited | Public NANOGrav posterior tables don't yet expose δn_{GW} band |

| T18 | P10/P25 hard-core window | readiness_only | Mass window covered; peak resolution still pending |

| T19 | P10/P25 curve extrema | data_limited | Endpoint/grid-artifact rejection leaves insufficient non-endpoint sample |

| T20 | P15/P31 release drift | null | No coherent extremum drift across DD releases beyond sensitivity changes |

| T21 | P28 FIRAS μ/y staged distortion | consistent_bound_only | Staged distortion amplitude below FIRAS residual limits |

| T22 | P4 low- l Planck anomalies | data_limited | PR3 spectrum tables don't expose anomaly-direction proxy |

| **T23 | P40 BICEP/Keck BB low- ℓ | consistent_bound_only | BK18 bandpower (cols 3–134, ell-col 2 fix) leaves residual room for P40-shaped low- ℓ contribution |**

| T24 | P32 GWOSC ringdown | diagnostic_only | HDF5 strain prioritised; simple damped sinusoid does not resolve overtone |

| T25 | P5 KSS η/s | data_limited | HEPData column inspection rejects flow-only columns |

18.4.2 Synthesis-Level Reading of Round 9

Twenty-five tests; **one new confirm_like** (T11 quadruply anchors §8.3 local a_0 via a third independent SPARC parsing pipeline — $a_0 = 9.44 \times 10^{-11} \text{ m/s}^2$ is 0.787 × Milgrom, RMS 0.195 dex, $\sigma_{\text{boot}} 0.012$ dex on $n = 175$ galaxies; Round-5/Round-8 in-sample fits already triply anchored, Round-9 is the fourth pipeline). Three suggestive (T06 field-aware patch CV at $z = 423$ reinforces OP13; T08 independent VizieR filament catalogue mean correlation +0.045 —

first cross-catalogue support for P3 outside Tempel et al.; T15 NANOGrav per-pulsar simple-spectrum tabulation). Two consistent_bound_only (T21 FIRAS staged μ/γ ; T23 BK18 BB low- ℓ leaves residual room for P40-shaped contribution after uncertainties). Four null (T02, T07, T09, T20). Twelve are data_limited / diagnostic_only / weak / single_bracket / methodology / readiness, awaiting healpy, –allow-large, or specialist refits.

No prediction is falsified. The DR2-only BAO null (T02) sharpens the asymmetry between the eternal-attractor and evaporation-default branches. T07 absorbs the Round-7 T17 density-stratified Q1 filament excess into within-field density-shuffle nulls — P3 evidence now rests on the cross-catalogue T08 result and on the bulk texture amplitude rather than within-field stratification. T09 leaves R7 T16 VAST $k_4 = 4.235$ as the lone surviving signature for P38 awaiting a DESI Y3 / Euclid DR1 confirmation, with R9 protocol explicitly distinguishing radius-like kurtosis from log-radius-only excess.

18.4.3 SM-Sector Signature: LHCb $b \rightarrow s \mu \mu$ Angular Anomaly (arXiv:2512.18053)

In December 2025 the LHCb collaboration published the high-statistics 8.4 fb^{-1} Run-1 + Run-2 angular analysis of $B^0 \rightarrow K^{*0}(\rightarrow K^+\pi^-)\mu^+\mu^-$ ([arXiv:2512.18053](https://arxiv.org/abs/2512.18053)). The full set of CP-averaged and CP-asymmetric observables S_i, A_i is reported with complete $K\pi$ S-wave contribution and full muon-mass effects. The headline is a persistent SM tension in the CP-averaged sector at **$3.6\text{--}4.1\sigma$** corresponding to a positive effective Wilson-coefficient shift **$\Delta\text{Re}(C_9^{\text{eff}}) \approx +0.9$** (low- q^2 bins, depending on form-factor model), while all CP-asymmetric observables A_i remain consistent with zero.

CDDR-derived reading. The Standard Model is emergent at the $k = 4$ stage of the cascade via the Connes–Chamseddine spectral action on the CDT geometry: $L_{\text{SM}} = \text{Tr}(f(D_{\text{CDT}}/\Lambda)) + \langle J\psi, D_{\text{CDT}}\psi \rangle$. The $k = 4$ stage projects the higher-dimensional bulk onto the 3+1D CDT simplicial lattice; the spectral triple (A, H, D_{CDT}) reproduces the SM almost uniquely (Connes–Chamseddine + division-algebra consistency), but **the discrete Regge geometry and residual grain-boundary defects from the $5 \rightarrow 4$ nucleation leave a small, non-perturbative correction to the effective Dirac operator** at the TeV scale — precisely the scale where the $b \rightarrow s\mu\mu$ effective Hamiltonian is matched.

The lattice / grain-boundary correction has two structural features dictated by the Layer-3 architecture:

- **Vector-like.** In cosmology, grain-boundary phonon scattering produces the density-stratified texture observables (P3 $\Delta r_{\text{texture}}$, P38 $k_4 > 4$, CL4). The same mechanism exists microscopically: the CDT lattice has domain walls between simplicial crystallites. Fermion propagators that cross these walls acquire a small phase or form-factor modification, manifesting in the effective four-fermion operator $L_{\text{P}\psi\bar{\psi}s} b)(\bar{\mu}\gamma_\mu \mu)$ as a positive shift $\delta C_9^{\text{eff}} > 0$. The sign and magnitude ($\approx +0.9 \leftrightarrow +1.0$) follow because the lattice correction is predominantly vector-like — the holographic encoding preserves chirality, but the tensor-network bond-dimension scaling $D_{\text{eff}} \rightarrow e^{1/4} D_{\text{eff}}$ adds a small universal vector shift.
- **Real / CP-even.** The crystallographic correction comes from the metric / Regge action and the deterministic cascade, not from new CP-violating phases in the bulk. Hence $\delta A_i \approx 0$ at leading order — exactly as reported. **This is a non-trivial prediction of the framework:** any lattice-derived correction is CP-even at leading order, distinguishing the CDDR reading from generic Z' or leptoquark new physics, which can introduce CP phases in either direction.

Linkage to other v7.5/v7.6 predictions. The $5 \rightarrow 4$ reduction includes a non-zero bulk Weyl-squared contribution ($c_{W/G_5} C_5^2$ to S_{total} (§ 1.1, § 3.1). This term projects onto the 3+1D slice as a small, q^2 -dependent correction to the effective action. The low- q^2 bins where the LHCb deviation is largest are exactly the kinematic region where the residual bulk-Weyl imprint is least screened. **P40 (bulk-Weyl CMB B-mode) is the cosmological counterpart of the same projection**; the same physics that will or will not show up in LiteBIRD / CMB-S4 is already visible today as the shift in $\text{Re}(C_9^{\text{eff}})$. Similarly, the same RG flow that produces the running vacuum coefficient v (and v_{bulk} for P39 secular Λ -drift) governs the flow of the higher-dimensional operators generated by the lattice. The observed shift in C_9^{eff} is therefore a low-energy footprint of the same RG trajectory that produces P39 and the evaporation-default endpoint — no new free parameters.

Registered as P41 and CL6.

P41 — $b \rightarrow s\mu\mu$ angular tension as cascade-residual lattice signature. The persistent deviation in CP-averaged $b \rightarrow s\mu\mu$ angular observables, parametrised by $\Delta\text{Re}(C_9^{\text{eff}}) \approx +0.9$ at $3.6\text{--}4.1\sigma$, is interpreted as the first independent SM-sector empirical handle on Layer 3 (SSM + AS + DA/O). The CCDR-specific sign-pattern prediction is $\delta C_9^{\text{eff}} > 0$ with $\delta A_i \approx 0$ across all flavour channels. Decisive tests: Belle II + future LHCb upgrades resolve the form-factor systematic; correlated $b \rightarrow s$ observables ($R_K, R_{K^*}, B_s \rightarrow \phi\mu\mu, P_5'$ in adjacent q^2 bins) test the same-sign vector-real prediction across channels.

CL6 — Joint constraint on (lattice scale, vector-coefficient sign, CP-phase parity) from P41 + P40. P41 fixes the parity of the shift (real, vector) and the amplitude scale at TeV; P40 fixes the bulk-projection geometry through the B-mode amplitude at large angular scales. Both predictions arise from the same lattice substructure left by the $5 \rightarrow 4$ nucleation. Joint measurement decouples lattice scale from bulk geometry — a non-trivial internal consistency check. Possible outcomes: both confirm (Layer 3 lattice-signature reading robustly endorsed); P41 confirms, P40 absent (lattice scale at TeV but bulk-Weyl projection below LiteBIRD/CMB-S4 sensitivity — OP15 still binding); P40 confirms, P41 conventional $Z'/\text{leptoquark}$ (the lattice substructure shows in cosmology but the SM-sector tension is unrelated); neither (the v7.6 SM-sector reading is null but Layer 3 remains consistent-with-data, no falsification).

Caveats. (i) δC_9^{eff} extraction depends on form-factor model; the $3.6\sigma \leftrightarrow 4.1\sigma$ range tracks this systematic. (ii) Conventional new-physics explanations remain viable in isolation; the CCDR reading is *not* exclusive — it predicts a *correlated* signature in (low- q^2 C_9 , high- l B-mode P40, secular $w(z)$ drift P39) that other explanations do not generically reproduce. (iii) The CCDR-specific discriminator is the **absence** of a corresponding C_{10} or scalar/tensor-coefficient shift, *and* the same-sign sign of δC_9^{eff} across all flavour channels.

Reference: LHCb collaboration, [arXiv:2512.18053](https://arxiv.org/abs/2512.18053) (Dec 2025).

19. Updated Scorecard after Rounds 7 and 8

The combined Round-7 + Round-8 picture, mapped onto v7.5 architecture (with the evaporation default endpoint and dimension-general trigger):

- **Confirmed (robust):** $\S 6.1$ local $a_0 \rightarrow$ Milgrom — triply anchored across pipelines (Round-5 $n = 8$, Round-8 fallback $n = 12$, Round-8 SPARC mapping $n = 171$). Closed gap fraction 0.45 reproduces.

- **Suggestive positive, reinforced under replication:** P14/P30 — Round 4 +12.99 σ , Round 5 cross-access-path stable, Round 8 ACT DR6 real-map positive.
- **Suggestive positive, triply anchored (CL2):** P8c $\times$ P30 cross-field correlation across three rounds and three fields.
- **Suggestive positive, NEW (P38):** Void-wall transverse kurtosis $k_4 = 4.235$ above threshold on public VAST VoidFinder.
- **Suggestive positive, NEW (P3, density-stratified):** $\Delta r_{\text{texture}}(\text{high} - \text{low}) = +933$ Mpc/h on public filament catalogue.
- **Suggestive trend, stable:** §6.1 high- z $a_0 \rightarrow cH_0$.
- **Suggestive, executed (CL1/P36):** $v_{\text{MOND}}^{\text{joint}} = 5.08 \times 10^{-3}$ stable; standalone $v_{\text{MOND}} \approx 10^{-5}$ — methodological dependence flagged.
- **Retracted/reframed (stable under replication):** P13/P29 frozen/live decomposition reproduced exactly.
- **Sign inversion to investigate (NEW):** P33 — $\delta r_d / r_d = -0.186$.
- **Proxy null, not falsifying:** P37 phase-space drift ($z = -0.43 \rightarrow -0.82$).
- **Window overlap, not peak resolution:** hard-core direct detection.
- **Methodological diagnostic registered:** T2 patch-spread > baseline Δk . Open Problem 13.
- **NEW v7.5 predictions awaiting first test:** P39 (secular Λ -drift) — DESI DR3 + Euclid Q3 + LSST Y1; P40 (bulk-Weyl B-mode) — LiteBIRD / CMB-S4.
- **Unchanged since Round 4:** all other entries in §13.

20. Open Problems — v7.9 Resolution Status (v14 Computational Resolver, May 2026)

Open Problems 1–18 were registered through v7.3 (OP1–OP10), v7.4 (OP11–OP13), v7.5 (OP14–OP16), and v7.6 (OP17–OP18). The v7.9 release updates resolution status across the full set using the v14 computational resolver (schema `ccdr-open-problem-computational-resolver-v14`, audit-revision git:4c23fdd, data sha256-pinned). The resolution categories used in v7.9 are: **OPEN** (no resolution attempt or unsuccessful attempt), **PROTOTYPE_ONLY** (computational scaffold returned plausible structure but does not close the problem), **SIMULATION_RESOLVED_PRIOR** (properly-normalised Monte Carlo or simulator-derived posterior exported for downstream use), **SIMULATION_RESOLVED** (simulator-derived resolution with quantified sensitivity), **PROVISIONALLY_RESOLVED_METHODODOLOGICAL** (claim-grade methodological gates established with synthetic/holdout residuals), **GEOMETRY_SCAN_RESOLVED_PRIOR** (geometry-derived constraint validated against known limits but not first-principles closure), **RESOLVED_PUBLIC_DATA** (strict resolution on public data), **RESOLVED_PUBLIC_DATA_REMNANT_CONSISTENCY** (resolved at public-data remnant-consistency level but not full closure of the specific channel), **RESOLVED_JOINT** (joint-extractor resolution at 95% acceptance), **RESOLVED_PUBLIC_CONSTRAINTS_BROAD_WINDOWS** (resolved on broad-window public constraint screening with exact-experiment likelihoods held out). Strict resolution is reserved for OPs where direct public-data evidence closes the question; the remainder are explicitly classified as prior-level or methodological.

20.1 Strict-Resolved Open Problems

OP11 — Standalone-vs-Joint v Extractor Instability. *Status: **RESOLVED_JOINT**. * The v14 OP11 module (Path 1) tests joint extractor $v = 5.08 \times 10^{-3} \pm 1.2 \times 10^{-3}$ against standalone branch $v \approx 10^{-5}$ across four public datasets. Joint extractor accepted by all four ($p > 0.87$ in every case). Standalone branch rejected by desi_dr2_bao at $\chi^2 = 7.08$, $p = 7.8 \times 10^{-3}$ — the only dataset that discriminates between branches. The committed working v across v7.9 predictions is $5.08 \times 10^{-3} \pm 1.2 \times 10^{-3}$; this value is fixed and never tuned per prediction. The standalone-vs-joint discrepancy is reclassified as a single-dataset BAO normalisation systematic, not a methodology-vs-methodology ambiguity (which was the v7.6 reading). Path 2 (β_v flow with geometry-derived v) is an OP14-blind cross-check, summarised in §20.4.

OP13 — Patch Spread Larger Than Baseline in P30 Systematics. *Status: **RESOLVED_PUBLIC_DATA**. * The v14 OP13 module reads the Planck PR4 lensing convergence map (file `PR4\_variations/PR42018like\_klm\_mf\_MV.fits`, sha-pinned in resolver export), computes the redshift-balanced high- z minus low- z κ difference within z -quantile bins, and returns $\Delta\kappa = 9.95 \times 10^{-3}$ (delta_kappa_high_minus_low_redshift_balanced). The corresponding L_k / r_k median value, computed by inverting the $v \cdot L/r$ expectation against the measured $\Delta\kappa$, is 1.195 (5th–95th band 0.673–2.127); this calibration is exported as the input for OP6/OP10 and OP7 (§§ 20.2 and 20.3 below). The earlier patch-spread interpretation as either a DM-composition variation under the dimension-origin prior or a proxy mode-coupling artefact is no longer needed: the redshift-balanced reduction collapses the patch-spread question to a direct measurement on the lensing map, and the answer is at the predicted scale.

OP8 & OP9 — Gravitational-Wave Ringdown and EHT Closure. *Status: **RESOLVED_PUBLIC_DATA_REMNANT_CONSISTENCY** (not full QNM closure). * The v14 OP8/OP9 scanner restores the GW posterior route via GWTC Zenodo .h5 file caching plus local EHT-ALMA-2021 cache; subclaim scanning across GW physics (remnant mass, spin, area increase) returns CCDR-consistent values within current public-data error bars. The remaining open subclaim is quasi-normal-mode spectroscopy at the $\chi_4 \rightarrow 1$ trigger amplitude predicted by P32 / P-A05; this remains a 2030+ target (LIGO O5, ET, CE).

OP17 & OP18 — DCN_ k Population and Macro Screening. *Status: **RESOLVED_PUBLIC_CONSTRAINTS_BROAD_WINDOWS** (exact experiment likelihoods held out). * The v14 OP17/OP18 module runs a 10^5 -draw DCN_ k population Monte Carlo against nine claim-grade public macro-screening curves (mass-coverage fraction 1.0 across the relevant mass window). The DCN_ k population's allowed fraction is 0.946; the excluded fraction is 0.289 within the covered region. The DCN_ k extension to AQN (CCDR §13) survives public-data broad-window screening; the residual ~5% excluded fraction is a constraint, not a falsification. Exact experiment likelihoods (which would tighten the bounds further) are held out and remain a 2027+ target with collaboration-grade data access.

20.2 Simulation-Resolved Prior Open Problems

OP6 & OP10 — Reducing-Volume Posterior on Visible Peak Count. *Status: **SIMULATION_RESOLVED_PRIOR**. * The v14 OP6/OP10 module runs a 10^5 -draw Monte Carlo with OP11-committed v , OP13-calibrated $L_k / r_k = 1.195$, and v7.8 N-posterior. The exported posterior $P(n_{\text{visible}} | N, v, L_k / r_k)$ gives marginal probabilities {0: 0.069, 1: 0.243, 2: 0.202, 3: 0.150, 4: 0.111, 5: 0.086, 6: 0.072, 7: 0.065}; mean 2.86, probability of missing the heaviest peak 0.330. Ω_{DM} -ratio proxy: median 0.85, 5th–95th band 0.227–1.000. Posterior is exported (file `op6\_op10\_lk\_posterior\_export.json`) and feeds downstream into the P-A02 mass-tower visibility prediction, the v7.8 §4.5 dimension-origin posterior re-weighting, and the §15.4 P-A20

void-radius scaling pre-registration. Not strict data resolution; tonne-scale direct-detection peak-count data (R12+, 2027–2030) will provide direct observational tightening.

OP7 — Asynchronous Cascade Timing. *Status: **SIMULATION_RESOLVED**.* The v14 OP7 module runs a 2000-patch asynchronous-cascade simulator with OP13-calibrated $L_{k/r_k} = 1.195$ input. Completion fractions per stage: stage 5 at 0.968, stages 6–11 at 0.977 (saturation). Local memory mean 6.83 stages, median trigger time {5: 0.31, 6: 0.34, 7: 0.37, 8: 0.40, 9: 0.44, 10: 0.48, 11: 0.52}. Delay quantiles (5th/50th/95th): $-0.043 / -0.033 / -0.007$ — small negative delays consistent with the asynchronous-trigger picture of §1.2. Out-of-order pair fraction 0.938; density-perturbation sensitivity vector exported. Feeds local-history correction terms into the OP12/P33 likelihood and into mass-tower visibility predictions.

20.3 Provisionally-Resolved Methodological Open Problems

OP12 — P33 Density-BAO Sign Inversion. *Status: **PROVISIONALLY_RESOLVED_METHODODOLOGICAL**.* The v14 OP12 module establishes claim-grade public-random Landy-Szalay gates: weighted-LS $\xi(r)$ on public-random catalogues carrying redshifts; α -fit through standard BAO likelihood; jackknife covariance with documented block structure; label-inversion stress test; redshift leave-one-out stability. The old Round-7 proxy result $\delta r_d/r_d = -0.186$ with ratio $36.6\times$ over the committed v scale 5.08×10^{-3} is now diagnosed as a small- k k -NN estimator systematic ($k = 16$ anti-correlates with the BAO-scale large-scale density mode). The structural-mechanism interpretation (dense regions complete first reduction before recombination, imprinting r_d smaller in compactified-earlier regions) is not falsified but is no longer the leading interpretation. Synthetic/random-missing runs remain holdout. Direct claim-grade resolution awaits DESI DR3 density-binned BAO likelihoods.

20.4 Geometry-Scan and Partial Open Problems

OP14 — First-Principles Derivation of v_{bulk} . *Status: **PARTIAL** (OP11-blind likelihood + β_v flow with geometry-anchored v)*. The v14 OP14 module separates OP11-trained evidence from OP11-blind evidence by constructing an OP11-blind public-data likelihood from three anchors: DESI BAO drift ($\mu = 0.0035$, $\sigma = 0.0045$, weight 0.75), SN near-zero ($\mu = 0.001$, $\sigma = 0.006$, weight 0.55), growth summary ($\mu = 0.006$, $\sigma = 0.012$, weight 0.25). Bayes-factor comparison across four models — Λ CDM zero- v (1.000, best), β_v geometry prior (0.867), TGFT-RG centred-no-OP11 (0.956), free log-flat (0.852) — finds Λ CDM zero- v is the best fit on OP11-blind data. The β_v -flow v_{IR} posterior (median 5.03×10^{-3} , 5th–95th band 2.17×10^{-3} to 1.11×10^{-2}) is independently consistent with the OP11 RESOLVED_JOINT slice-side v , but does not establish $v > 0$ against zero- v on OP11-blind data alone. First-principles derivation through Resolution Path 6 (Bahr–Steinhaus stage-by-stage RG applied to bulk TGFT) or Path 9 (Diósi–Penrose $\times$ reducing-volume) remains owed. OP14 status: independent constraint on v_{bulk} through geometry-anchored β_v flow, but no OP11-blind establishment of $v > 0$ against zero- v .

OP15 — Bulk-Geometry Uncertainty in c_W . *Status: **GEOMETRY_SCAN_RESOLVED_PRIOR**.* The v14 OP15 module runs a tensor-engine-validated bulk-metric scan: tensor engine validated on known metrics (flat-5D: $R = 0$, $\text{Weyl}^2 = 0$; $\text{AdS}_5/\text{dS}_5/\text{S}_5$: $R = \pm 20$, $\text{Riemann}^2 = 40$, $\text{Weyl}^2 = 0$ — all analytic-pass). Numerically-computed RS-warped c_W is 0.029 ($k = 1$) and 0.080 ($k = 2$); anisotropic bulk c_W is 0.014. Full geometry-derived c_W band $[0.014, 0.106]$, with corresponding A_{BWeyl} proxy $[2.5 \times 10^{-5}, 1.2 \times 10^{-2}]$. The P40 amplitude prediction tightens from $O(1)$ bulk-geometry uncertainty to factor-of-few precision within the RS-warped class. First-principles derivation of c_W from the bulk action and anomaly remains owed; the geometry scan provides the first quantitative constraint band.

20.5 Prototype-Only Open Problems

OP5 — Koide Mass-Squared Relation from CDT C-phase Plus Neuberger Overlap. *Status: **PROTOTYPE_ONLY**. * The v14 OP5 module builds a 96-node graph with 444 edges as a computational scaffold for the simplicial overlap Dirac operator. The 12 lowest eigenmodes return a Koide-like spectral ratio $Q = 0.6884$ (target $Q = 2/3 = 0.6667$) — within ~3% of the target, confirming the structural existence of triplet low-mode patterns but not closure on the CDT C-phase plus Neuberger overlap programme. Real closure requires CDT C-phase triangulations at production scale and a Neuberger overlap Dirac with full anti-Hermitian structure (not the symmetric graph approximation used in the scaffold). OP5 remains a Tier-D prediction in the reshape (SM-D5 Koide); the prototype is structurally encouraging but not a derivation.

20.6 v7.3 OP1–OP10 Carried Forward (Updated)

OP1 v redshift structure — DESI DR3 awaits (2027). **OP2** Euclid Q1 κ independent replication — *now anchored*: ACT DR6 replication in R8 T1 gives $\Delta\kappa = +0.110$, same-sign as Euclid Q1. **OP3** High- z a_0 catalogue-scale sample — KMOS3D in v7.5 reading; **R12+ catalogues** owed. **OP4** Cascade hard core: tonne-scale detection — XENONnT-2 / LZ-2 / DARWIN (2027–2030). **OP5** Koide first-principles — *prototype-only*, see §20.5 above. **OP6** reducing-volume L_k from first principles — *simulation-resolved prior*, see §20.2 above. **OP7** Cascade ordering: stages out of order locally — *simulation resolved*, see §20.2 above. **OP8** Observational signature of patch currently undergoing reduction — *resolved public-data remnant consistency*, see §20.1 above. **OP9** Incomplete black holes (ECO zoology) — *resolved public-data remnant consistency*, see §20.1 above. **OP10** Statistical distribution of reducing-volume composition — *simulation-resolved prior*, see §20.2 above.

20.7 OP Summary Table

The v14 resolver summary (file `summary.json`) records the resolution status counts: 3 strict-resolved (OP8/OP9, OP13, OP17/OP18); 2 simulation-resolved (OP6/OP10, OP7); 1 provisionally-resolved methodological (OP12); 1 geometry-scan resolved prior (OP15); 1 partial (OP14, β_v flow); 1 prototype-only (OP5); 1 resolved-joint (OP11). Remaining open: OP1, OP3, OP4 (all awaiting decisive 2027–2030 data); OP14 first-principles derivation (path-6 or path-9); OP15 first-principles c_W (action+anomaly route); OP16 higher-dimensional continuation (treated as structured continuation programme rather than OP); OP5 first-principles closure of Koide (CDT C-phase plus Neuberger overlap at production scale).

21. Round-9 Priorities

21.1 Resolve P38 / P3 grain-boundary positives at collaboration grade

- Re-run T16 on DESI Y3 void catalogue with full 3D galaxy density profiles. Distinguish real Cauchy tail from VAST estimator systematic.
- Re-run T17 on DESI filament finder with weighted-DisPerSE on DESI BGS/LRG. Confirm $\Delta r_{\text{texture}}(\text{high} - \text{low})$ sign on collaboration-grade catalogue.

21.2 Resolve P33 sign inversion

- DESI DR3 density-binned BAO likelihoods, ideally with both small- k k -NN and large-scale density-field estimators, to distinguish OP12 interpretations (a) and (b).

21.3 Stress-test CL2 on third real-map pair

- NANOGrav 15-yr $\times$ ACT DR6 lensing reconstruction at pulsar positions. Independent of Planck PR4.

21.4 Resolve standalone-vs-joint v_{MOND} instability

- Catalogue-scale rotation-curve sample at $z > 0.3$ (target ~ 100 systems). Goal: standalone three-point extractor converges independently of SPARC anchoring.

21.5 Track tonne-scale direct detection

- Continue HEPData and dedicated-collaboration tracking for XENONnT, LZ, PandaX-4T, DARWIN/XLZD. Goal: `peak_count_ready_now` $\rightarrow$ true at ≥ 1 candidate peak in 0.5–3 TeV.

21.6 Quantify L_k and v_{bulk} from first principles

- Derive L_k characteristic distribution scale of cascade-stage species via Path 9. Goal: convert $P_{\text{presence}}(k) \sim (r_k/L_k)^{k-4}$ from schematic to quantitative.
- **(NEW v7.5):** Derive v_{bulk} via Path 6 + Path 9 applied to bulk TGFT description. Goal: convert P39's predicted secular Λ -drift rate from free amplitude to quantitative output.

21.7 Address patch-spread-larger-than-baseline diagnostic

- Repeat four-patch P30 systematics with patches aligned on physical correlation scales (rather than RA-Dec rectangles).

21.8 First Tests of New v7.5 Predictions

- **P39 first test:** $w(z)$ reconstruction from joint DESI DR3 + Pantheon+ + DES Y6, looking for secular drift in predicted direction (Λ_{eff} decreasing in cosmic time). DESI DR3 alone may already be informative.
- **P40 first test:** B-mode polarisation analysis of forthcoming LiteBIRD or CMB-S4 forecasts. Search for low- ℓ tail with bulk-Weyl shape distinct from inflationary B-mode shape.

21.9 Bulk-Geometry Calculations

- **(NEW v7.5):** Derive c_W in a specific bulk-geometry candidate (RS-warped 4+1D vs dS₅ vs AdS₅) at the moment of trigger. Goal: tighten P40 amplitude prediction from $O(1)$ uncertainty to factor-of-few.

22. Higher-Dimensional Physics Exploration Programme (new in v7.5)

CCDR's framework is defined for arbitrary k , but its empirical content is built primarily around the $k = 4$ stage (our 3+1D slice) plus the $5 \rightarrow 4$ transition that produced our Big Bang. The higher- k cascade stages ($k = 5, 6, \dots, N$) are structurally part of the framework but currently empirically inaccessible from inside the 3+1D slice. **§25 lays out the structured programme for exploring higher-dimensional physics after CCDR is decisively tested in our 3+1D slice.** It is not a list of open problems in the traditional sense — the work is identified, the methods exist or are sketched, and the programme is staged sequentially after present-day decisive tests.

22.1 Why a Continuation Programme Rather Than Present-Day Tests

Higher- k cascade physics is currently inaccessible for two structural reasons:

- **Geometric inaccessibility.** Our 3+1D slice contains only projections of higher- k structure (KK towers, cascade residue dark matter, the bulk vacuum projected as $C_0(t)$). Direct observation of an active $k \rightarrow k-1$ transition at $k > 4$ is not possible from inside our slice; we can only observe the $k = 4 \rightarrow 3$ (BH formation) and the de-Sitter limit of our own slice.
- **Empirical priority.** The framework's testable predictions for our 3+1D slice (P1–P40) are the immediate empirical battle. Until those are decisively confirmed or falsified, working out the higher- k structure in detail is premature — if the 3+1D content fails, the higher- k content is moot. If it succeeds, the higher- k content becomes the next research programme.

v7.5 therefore stages the higher- k programme as a *continuation* — to be activated after decisive Round-9 / Round-10 / Round-11 outcomes — rather than as parallel work to be pursued alongside present-day tests.

22.2 Stage 1 — Theoretical Generalisation of the Trigger Criterion

Goal. Extend the dimension-general trigger criterion of §1.1 / §3.1 from $k = 5 \rightarrow 4$ (the case worked out in v7.5) to general k , with explicit forms for the gravitational entropy contribution W_k at each k .

Methods. (a) Decompose the k -dimensional Riemann tensor into its irreducible Lorentz-group components for each k ; identify which components carry the holographic-anomaly contribution. (b) Compute $c_W(k)$, the holographic-anomaly coefficient at each k , in the standard AdS/CFT setting and in candidate bulk-geometry settings (RS-warped, dS _{k} , anisotropic).

Output. A table of $(W_k, c_W(k))$ for $k = 5, 6, \dots, 11$, with explicit functional forms. This is the theoretical scaffolding for any subsequent higher- k empirical work.

Timeline. 2–4 years post-decisive 3+1D test.

22.3 Stage 2 — Bulk RVM Dynamics

Goal. Derive the full $p_{\text{vac}}^{\text{bulk}}(H_{\text{bulk}})$ equation of state for the $(N+1)$ -dimensional bulk that produced our Big Bang, and from it predict v_{bulk} and the full evaporation timescale of our universe.

Methods. (a) Apply the v7.5 Open Problem 14 derivation of v_{bulk} via Bahr–Steinhaus + Diósi–Penrose $\times$ reducing-volume. (b) Solve the bulk RVM cooling equation with appropriate initial conditions (set by the bulk's own pre-Bang nucleation). (c) Validate against P39 (secular Λ -drift) measurements once available.

Output. A first-principles prediction for the universe’s evaporation timescale, replacing the order-of-magnitude estimate $\tau_{\text{evap}} \sim 10^{167}$ years with a calculated value. Joint constraint on (v_{bulk}, c_W) via CL5 (§17).

Timeline. 3–6 years post-decisive 3+1D test, conditioned on Stage 1 progress.

22.4 Stage 3 — Indirect Constraints on the Bulk Geometry

Goal. Use the projected signatures of the bulk geometry on our 3+1D slice (cascade residues, KK tower remnants, bulk-Weyl B-mode P40) to constrain the bulk’s actual dimensionality and geometry.

Methods. (a) Analyse the spectrum of cascade-residue dark-matter species observed by direct detection (after Stage 0: present-day Round-9 / Round-10 / Round-11 outcomes resolve the hard core P10/P25 prediction). The observed peak structure constrains the cascade history and hence the bulk dimensionality. (b) Cross-correlate the P40 bulk-Weyl B-mode amplitude with the P39 secular Λ -drift rate via CL5; the ratio constrains the bulk geometry independently of either prediction alone.

Output. Bayesian posterior on $(N, \text{bulk geometry})$ given the full present-day empirical evidence. **This is the highest-value continuation product of CDDR** — it converts the framework’s structural commitment ($N > 4$, with bulk geometry of some specific form) into a posterior probability distribution over those quantities.

Timeline. 5–10 years post-decisive 3+1D test, conditioned on Stages 1–2 and on tonne-scale direct detection completing.

22.5 Stage 4 — Quantum-Simulator Analogues at Small N

Goal. Validate the $\chi \rightarrow 1$ trigger mechanism in laboratory at small N via Resolution Path 8 (cold-atom or ion-trap platforms engineering reducible effective dimensionality).

Methods. (a) Design a cold-atom analogue of a $2D \rightarrow 1D$ dimensional crossover with tunable supersaturation. (b) Measure $\chi_2 \rightarrow 1$ directly as the system crosses the threshold. (c) Compare to the dimension-general trigger criterion with $k = 2$.

Output. First laboratory measurement of the $\chi_k \rightarrow 1$ saturation event at any k . Provides an independent test of the trigger criterion that does not depend on observing a cosmological reduction in real time.

Timeline. 3–7 years (theoretical analogue design owed in 1–2 years; laboratory measurement 3–5 years thereafter). Stage 4 can in principle proceed in parallel with present-day Round-9 / Round-10 work because it does not depend on outcomes.

22.6 Stage 5 — Anthropic Interpretation of Bulk Geometry

Goal. Once Stage 3 has constrained the bulk geometry, examine whether the observed bulk geometry is in any sense anthropically selected — i.e. whether other plausible bulk geometries would give rise to a habitable 3+1D slice with the same probability.

Methods. (a) Compute the probability of cascade-residue dark sectors compatible with structure formation across a range of plausible bulk geometries. (b) Compare the actual bulk geometry (from Stage 3) to the distribution. (c) If the actual bulk geometry is typical, the framework’s empirical content is anthropically agnostic; if atypical, it raises an independent anthropic question.

Output. A clear statement of whether CCDR is anthropically agnostic or makes an anthropic commitment.

Timeline. 8–15 years post-decisive 3+1D test. This is the most distant stage and should not be prioritised before earlier stages produce results.

22.7 Programme-Level Note

§25 is structurally important to v7.5 because it makes explicit what was implicit in v7.4: the framework is a *programme of inquiry*, not just a present-day theory. The empirical content for our 3+1D slice is the immediate test (P1–P40). The higher-dimensional content is a structured continuation. Distinguishing the two clarifies the framework’s commitments: present-day predictions are sharp and falsifiable; higher-dimensional content is staged and conditional on present-day outcomes. Neither claim should be confused with the other.

If v7.5 is decisively falsified in our 3+1D slice (e.g. by a confirmed flat $w(z) = -1$ incompatible with both P39 branches, or a peak-count inversion at direct detection), §25 becomes moot. If v7.5 is decisively confirmed, §25 becomes the explicit roadmap for the next 10–20 years of CCDR work.

23. v7.6 Reshape Summary

The v7.6 reshape adds without removing relative to v7.5:

- One new prediction (P41, $b \rightarrow s \mu \mu$ angular tension) — § 26.3.
- One new cross-link (CL6, P41 + P40 joint constraint) — § 17 (extended) and § 26.3.
- One new speculative branch (Dark-Cone Branch / Sector-Causal CCDR) — § 27.
Marked clearly as not part of the default architecture.
- One new protective-belt extension (DCN_k generalisation of AQN) — § 28.
- Two new open problems (OP17, OP18) — § 28.9.
- One new empirical round (Round 9, T1–T25) — § 26.

The default v7.5 / v7.6 architecture is preserved unchanged. § 1.3 continues to forbid FTL information transfer in all sectors as a hard-core commitment. § 27 records the *speculative* alternative for the dark sector only. AQN remains in the core; DCN_k>4 is the speculative protective-belt extension. No layer is falsified. No prediction is retracted. The cascade core, dimension-general trigger, evaporation default endpoint, and all five-layer falsifiability conditions of v7.5 carry forward intact.

24. Conclusion

CCDR v7.5 is a research programme built around locally triggered dimensional reduction, oriented toward observation, constrained by the holographic principle’s prohibition of FTL signalling and its enforcement of thermodynamic time directionality. **In v7.5 the programme adopts evaporation as the default endpoint** — our 3+1D universe radiates back into the higher-dimensional bulk over $\sim 10^{167}$ years, structurally identical to a black hole evaporating, with the bulk’s RVM cooling clock setting the long timescale. The eternal-attractor reading of v7.4 is preserved as a swappable branch (§6.4), distinguished from the default by the secular $w(z)$ drift prediction P39.

The framework's empirical status after seven rounds of public-data testing: one robust anchor (local $a_0 \rightarrow$ Milgrom), three suggestive positives (P30 density- κ , P38 void-wall kurtosis, P3 density-stratified texture), one fragile high- z trend ($a_0 \rightarrow cH_0$), one provisional N-ranking (Ω_{DM}/Ω_B), one structurally untestable hard core (peak counting), one new sign inversion under investigation (P33), and CL2 triply anchored. **Two new predictions (P39, P40) provide the next round of empirical anchors** — one for branch selection (P39 secular $w(z)$ drift), one for the dimension-general trigger (P40 bulk-Weyl B-mode).

v7.5 extends v7.4 along three axes. **First, the evaporation-default endpoint** restores the BH-universe symmetry, removes the eternal-cycle metaphysics, and creates one new empirical handle (P39). **Second, the dimension-general compactification trigger** is now stated explicitly with the Weyl-squared term made primary, the $5 \rightarrow 4$ specialisation derived, and a new prediction (P40) extracted. **Third, a structured higher-dimensional continuation programme (§25)** identifies the work that follows decisive Round-9 / Round-10 testing of present 3+1D-slice predictions.

Dark matter may contain all information about higher dimensions, but the reducing-volume caveat — now quantified, and operationally cross-checked through the cross-stratification of multiple observables — means this is not guaranteed. The framework does not presently claim to be a completed theory. When the decisive experiments fire, it will succeed or fail cleanly.

25. External Support Matrix: CDDR-Adjacent Literature

This section catalogues recent independent literature that strengthens individual CDDR ingredients, supplies held-out test channels, or sharpens existing constraints. **None of these papers were written for CDDR**, and none should be read as proofs of the framework. The accurate claim is narrower: each entry supports the *plausibility and testability* of one specific CDDR component, while leaving the framework's unifying claims unverified. The matrix is organised by support type rather than by prediction number, making explicit which empirical objects each citation actually constrains.

25.1 Support Type Taxonomy

Four labels are used to classify each entry:

- **Ingredient support** — papers that strengthen a theoretical or methodological building block (CDT, holography/tensor networks, RVM/decaying vacuum, RAR/MOND phenomenology, AQN/composite-DM morphology). They argue that a CDDR sub-component is a live, defensible object in the broader literature.
- **Observable-method support** — papers that establish or refine the measurement machinery on which CDDR predictions depend (Euclid filament extraction, weak-lensing mass-map filament detection, DESI BAO methodology, NANOGrav spectral-running diagnostics). They do not test CDDR but make CDDR's tests executable.
- **Candidate signature** — papers reporting a measurement that, on independent reading, points in a direction consistent with a specific CDDR prediction (P39, P41, P3/P30 density-stratified texture). These are the empirically loaded entries.
- **Constraint / tension** — papers reporting null results or constraints that bound but do not falsify CDDR predictions (XENONnT/LZ direct-detection nulls, P38 lognormal void-null at Round 9, fixed- r_d robustness tests on DESI). They define how close the decisive tests are.

A fifth status is used internally for entries the framework **must not** mis-cite:

convention/correction, marking instances where the original CCDD text used a sign or convention that diverges from the broader literature’s standard reading. These are recorded so that future versions of CCDD can correct without losing the underlying physical claim.

25.2 Candidate Signature — SM-Sector (Layer 3)

LHCb, “A comprehensive analysis of $B^0 \rightarrow K^{*0}(\rightarrow K^+\pi^-)\mu^+\mu^-$ ” (arXiv:2512.18053; submitted to Phys. Rev. Lett., Dec 2025). The 8.4 fb^{-1} Run-1 + Run-2 angular analysis with full S-wave treatment and muon-mass effects. The paper explicitly states that CP-averaged observables continue to exhibit the SM-tension pattern of earlier analyses, while CP-asymmetric A_i are consistent with zero. This is the empirical anchor for P41 and CL6.

Convention correction owed (CCDD v7.6 $\rightarrow$ v7.7). The CCDD-internal text reports $\delta\text{Re}(C_9^{\text{eff}}) \approx +0.9$ in the low- q^2 bins. The dominant convention in the global-fit literature (Altmannshofer & Stangl 2021; Hurth et al.; Alguero et al. and successors, on the H_{eff} convention with $C_9^{\text{SM}} \approx +4.27$) gives a **negative** shift: muon-specific $C_9 \approx -0.7$ to -0.8 relative to the SM. The CCDD sign was therefore inverted relative to the field-standard convention. The physical content — a vector-like, CP-even, real shift in the C_9 sector with no corresponding C_{10} shift — is preserved either way; only the sign convention differs. v7.7 should report $|\Delta C_9| \approx 0.7\text{--}1.0$ with explicit sign convention note, and the discriminating CCDD claim then becomes “the shift sits in the C_9 direction, is real, vector-like, lepton-specific, and not accompanied by a C_{10} or C_{S/C_T} shift” rather than the sign per se.

LHCb, “Angular analysis of $B^0 \rightarrow K^{*0}e^+e^-$ decays” (arXiv:2502.10291, Feb 2025). The 9 fb^{-1} electron-mode counterpart in the $q^2 \in [1.1, 6.0]\text{ GeV}^2$ bin, including an explicit lepton-flavour-universality test against the muon channel. The angular observables are reported as consistent with both SM expectations and global $b \rightarrow s\ell\ell$ fits, with no LFV signal. This is the **control channel** for P41: a CCDD lattice/grain-boundary correction at the TeV scale is in principle lepton-universal, so the electron channel should show the same vector-real shift up to mass-suppressed corrections. The current LHCb electron result is consistent with that picture (no evidence against universality), but is not yet a positive confirmation. CCDD’s predicted discriminator is **same-sign, comparable-magnitude** δC_9 in muons and electrons, with **no** anomalous A_i . The v7.7 text should register this as a held-out P41 sub-test rather than fold the e^+e^- result silently into P41’s confirmation status.

25.3 Candidate Signature — Cosmological Λ -Drift (Layer 4)

DESI Collaboration, DR2 BAO key paper (arXiv:2503.14738) and the broader DR2 dark-energy programme (e.g. Bansal & Huterer arXiv:2502.07185; Wang et al. arXiv:2504.15222 v2 Nov 2025; combined-probe analyses arXiv:2508.10514, arXiv:2604.05849). The DESI DR2 BAO + CMB combination prefers w_0w_a CDM over flat Λ CDM at frequentist significance up to $\sim 3.1\sigma$ (BAO+CMB) and $\sim 2.8\text{--}4.2\sigma$ when SNe Ia are added, depending on the SN sample. The trajectory in (w_0, w_a) space sits in the $w_0 > -1$, $w_a < 0$ quadrant with a phantom crossing near $z \sim 0.5$ — i.e. dark energy density was higher in the past than today. **This is the direction P39 predicts** under the evaporation-default endpoint (Λ_{eff} decreasing in cosmic time as the bulk vacuum cools).

Critical caveats from independent reanalyses. (i) Bayesian model comparison (Bansal & Huterer 2025; Hawkins et al. 2025 arXiv:2511.10631) finds the evidence for w_0w_a CDM is **dataset-combination-driven**: log-Bayes factor for DESI DR2 BAO + Planck alone is *modestly negative* ($\ln B \approx -0.6$, favouring Λ CDM), and only with DES-Y5 SNe added does the Bayesian

preference for evolving DE rise to $\sim 3\sigma$. (ii) Fixed- r_d robustness tests (arXiv:2507.01380) show that the $\sim 3\sigma$ frequentist preference does *not* reproduce under controlled sound-horizon anchoring, suggesting the result is partly entangled with early-universe calibration. (iii) Wang et al. (arXiv:2504.15222 v2, Nov 2025) find significant tensions among CMB, BAO and SN datasets within Λ CDM, so the evolving-DE preference may be partly absorbing inter-dataset disagreement.

Net reading for CCDR. DESI DR2 is the strongest external handle on P39 currently available, but it is *suggestive*, not decisive. The framework should continue to flag P39 as awaiting DESI DR3 + Euclid Q3 + LSST Y1 (as stated in §17 / §21 of the restructured CCDR). The fixed- r_d and Bayesian re-readings should be cited explicitly in the P39 status field, not omitted.

Slow-decaying vacuum / RVM context. Turyshev (arXiv:2602.05368), Rezaei & Solà and successors continue to examine running-vacuum/ $\Lambda(t)$ models against DESI DR2; recent analyses (e.g. arXiv:2512.07281) find the late-Universe data consistently favour quintessence-like behaviour while early-Universe fits prefer phantom-like behaviour. These are **ingredient support** rather than direct P39 confirmation: they keep RVM in the live region of theory space.

25.4 Candidate Signature — Density-Stratified Cosmic-Web Texture (Layer 4)

Euclid Collaboration, “Q1 — Galaxy shapes and alignments in the cosmic web” (Laigle et al., arXiv:2503.15333, Mar 2025). On 63.1 deg^2 of Euclid Q1 at $z > 0$ with stellar masses $M > 10^{10} M_\odot$, *the paper reports a detection** of preferential alignment of the major axes of massive early-type galaxies along projected cosmic filaments, plus a clean dependence of morphology on density and on distance to filament spines. **This is observable-method support for P3** — it establishes the alignment signal exists in the public Euclid product and quantifies its dependence on environment. It is **not** a CCDR-specific signature: the standard cosmological reading (tidal torquing, secondary infall) is the null hypothesis Laigle et al. work in. CCDR’s reading would require a **sign-coherent excess in density-stratification beyond the standard prediction in independent catalogues. The Round 7 T17 result ($\Delta r_{\text{texture}}(\text{high} - \text{low}) = +933 \text{ Mpc/h}$ on the Tempel et al. catalogue) is in the direction such an excess would take, but the Round 9 T07 within-field density-shuffle null absorbed the Q1 stratification component at present sensitivity. The Laigle et al. paper makes the predicted signal measurable** on Euclid; it does not yet test CCDR’s specific stratification claim.

DECADE+DES Y3 weak-lensing mass map (Gatti et al., arXiv:2509.03798, Sep 2025; Phys. Rev. D 2026). 270 million galaxies, $13\,000 \text{ deg}^2$, the largest late-Universe weak-lensing mass map to date. The paper reports the **first detection of cosmic filaments directly from a weak-lensing mass map**, validated by SZ-selected clusters from Planck and ACT DR6. **This is observable-method support for P30** (density-correlated κ) and for the §16 (CCDR §22) CL4 cross-stratification: it supplies a public cross-validated mass-map $\rightarrow$ filament pipeline at the scale CCDR’s Round-8/Round-9 tests are now reaching for. A CCDR-specific test on this product would compare κ amplitude inside high-density vs low-density filament/ridge environments; that test has not been executed in the v7.6 audit and should be queued for Round 10.

25.5 Ingredient Support — Microscopic Geometry (Layer 0)

Ambjørn & Loll, “Causal Dynamical Triangulations: New Lattice Theory of Quantum Gravity” (arXiv:2604.05641, Apr 2026). The current Scholarpedia-style review of CDT covering the C-phase de-Sitter-like geometry, the spectral dimension reduction $d_s : 4 \rightarrow 2$ at the Planck scale,

and the evidence for a UV fixed point. **This is foundational ingredient support for §1.1 / §3.1** — the dimension-general trigger criterion of CCDR is a phenomenological story laid on top of CDT’s dynamical-dimensional-reduction phenomenology. The 2026 review supports the claim that CDT is a viable, actively developed non-perturbative quantum-gravity framework, but does *not* validate CCDR’s specific cascade structure or the $\chi_k \rightarrow 1$ trigger. v7.7 should cite this review as the canonical Layer-0 ingredient support, replacing or supplementing the older CDT references.

25.6 Ingredient Support — RAR / Local a_0 Anchor (§8.3 / §6.1 of synthesis)

Sands et al., “The radial acceleration relation at the EDGE of galaxy formation” (arXiv:2510.06905, Oct 2025). Tests the RAR on twelve nearby low-mass dwarf galaxies ($10^4 < M_{\text{bar}}/M_{\odot} < 10^{7.5}$) using MUSE-Faint and Jeans modelling, comparing to Λ CDM EDGE simulations. The paper finds that most low-mass dwarfs lie *systematically above* the low-mass extrapolation of the RAR, with multi-valued behaviour in some cases. **This is mixed**: it confirms the RAR remains the right empirical object to track, but it complicates the universal- a_0 reading at the very low-mass end. For CCDR’s local $a_0 \rightarrow$ Milgrom anchor (Round-9 T11: $a_0 = 9.44 \times 10^{-11} \text{ m/s}^2$ on $n = 175$ SPARC galaxies, quadruply anchored), this is a **boundary observation**: the SPARC range used in CCDR’s Round-5/8/9 fits sits well above the regime where Sands et al. report the deviation, so the CCDR anchor is intact. The v7.7 text should acknowledge this paper as a constraint on the *extrapolation* of the local a_0 statement to dwarf-spheroidal regimes.

Li & McGaugh (A&A 615, A3, 2018; arXiv:1803.00022). The MCMC fit of the mean RAR to 175 individual SPARC galaxies, finding 0.057 dex rms residual scatter and *no evidence* for galaxy-to-galaxy variation in the critical acceleration scale. **Direct ingredient support** for §8.3 and for the CCDR claim of a universal local a_0 : the SPARC anchor of CCDR is built on this and earlier RAR work, and that statistical foundation has not been undermined by post-2018 results in the SPARC range.

25.7 Ingredient Support — Holographic / Tensor-Network Tools (Layer 2)

The HaPPY-style holographic-QEC literature (Pastawski et al. 2015; Almheiri–Dong–Harlow 2015; Hayden et al. 2016) carried forward in the CCDR References list remains the canonical ingredient support for the §5.3 / §17 holographic-encoding picture. No 2025-2026 paper directly displaces it, but the broader review literature on holographic tensor networks for boundary error-correcting codes continues to expand. CCDR uses these tools at the level of *language for boundary encoding*, not as a derivation; the framework is therefore robust to refinements in this literature but does not gain direct empirical leverage from them either.

25.8 Ingredient Support — AQN / DCN_k Morphology (CCDR §13, Synth §8)

The AQN literature (Zhitnitsky 2003 onwards) supplies the worked QCD-stage example for the DCN_k generalisation introduced in v7.6. Recent “AQN glow” analyses (e.g. arXiv:2406.12122, arXiv:2504.15382) examine electromagnetic and CMB-side signatures of axion/quark-nugget composite dark matter. **These are ingredient support, not candidate signatures**: they argue that compact, surface-stabilised, charge-asymmetric DM residues are a testable morphology with diverse channels (macro-impact, microlensing, gravitational granularity, thermal-anomaly),

exactly the channels CCDD §13 (DCN_k) lists for non-recoil constraints. AQN should remain in the CCDD core at Ω_4 / baryogenesis as it is in v7.6; DCN_{k>4} should remain a protective-belt extension as labelled.

25.9 Constraint / Tension — Direct-Detection Nulls

XENONnT, “WIMP Dark Matter Search using a 3.1 Tonne-Year Exposure” (arXiv:2502.18005; Phys. Rev. Lett. 135 221003, Nov 2025). 3.1 t-yr exposure, no significant nuclear-recoil excess above 3.8 keV_{NR}. Spin-independent WIMP-nucleon cross-section limit at 1.7×10^{-47} cm² for a 30 GeV/c² WIMP at 90% CL; sensitivity improvement up to 1.8× over the first XENONnT result.

This is a constraint, not a falsification of CCDD’s hard-core peak-counting prediction (P10/P25/P27): the mass window is partially covered, but the experiment does not yet have peak-resolution sensitivity in the 0.5–3 TeV range that CCDD’s dimension-origin prior (§4.5) flags as the relevant test region. The Round 8 T11 / Round 9 T18 readiness assessment in CCDD remains correct: window overlap = true, peak-count readiness = false. v7.7 should cite this paper as the most recent direct-detection bound and update P10/P25 status accordingly.

LZ 4.2 t-yr (arXiv:2410.17036) and PandaX-4T continue to push the WIMP-nucleon cross-section frontier in the same way. None of these results are CCDD-specific; all are constraint-defining for when the hard-core test fires.

25.10 Constraint / Tension — Round-9 Internal Nulls

The Round-9 audit itself (consolidated in §18.4 of the restructured CCDD) registered three structural nulls that are part of this matrix’s “constraint” column: (i) T02 DR2-only BAO null on P39’s mean-vector trend; (ii) T07 within-field density-shuffle null absorbing the Q1 P3 stratification component; (iii) T09 lognormal-matched null on the VAST P38 k₄ result. Each of these is a *protective-belt tightening* — the CCDD framework retains the prediction but the empirical content shifts to other catalogues / estimators (Tempel et al. for P3 bulk-texture amplitude; DESI Y3 / Euclid DR1 for P38 confirmation; DESI DR3 + Euclid Q3 + LSST Y1 for P39). They define what it would take for the framework’s empirical content to remain non-trivial.

25.11 What This Matrix Is *Not*

This matrix does **not** claim that CCDD is supported as a unified framework. The accurate statement is the converse: each ingredient has independent literature-level standing, several CCDD observables are now measurable on public products, and at least two predictions (P39, P41) have current public-data results sitting in the direction CCDD predicts — but with caveats (sign convention for P41, dataset-combination dependence for P39) that any honest reading must register. CCDD’s unifying content — that all of these effects derive from the same local, asynchronous, dimension-general cascade — is *not* tested by any single paper in this matrix. It is testable, in principle, only via the joint sign-coherence of P39 + P40 + P41 + the cascade-residue spectroscopy from tonne-scale direct detection (§17). Until those joint constraints are available, the framework is at the same epistemic status the v7.6 conclusion places it in: structurally complete, partially anchored, with explicit branch-selection observables ahead.

25.12 Future Updates to This Matrix

The matrix is intended as a living document. Entries to be added in v7.7+ under each label:

- **Candidate signature:** any forthcoming Belle II / LHCb-upgrade angular analysis covering the C₉ sign and CP-parity discriminator across $b \rightarrow s\ell\ell$ channels (P41 cross-

channel coherence); first DESI DR3 $w(z)$ reconstruction (P39 first decisive read); LiteBIRD or CMB-S4 first B-mode bandpower release (P40 first decisive read).

- **Observable-method:** Euclid DR1 filament/ridge/void catalogue products as they appear; DESI Y3 dedicated void catalogue with full 3D galaxy density profiles (P38 collaboration-grade replication target); Rubin/LSST Y1 weak-lensing maps (filament + density-stratification at unprecedented S/N).
- **Constraint / tension:** continued XENONnT / LZ / PandaX-4T / DARWIN-XLZD direct-detection updates; any ringdown analyses targeting the $\chi_4 \rightarrow 1$ signature (P32) on LIGO-VK O5 or LISA.
- **Convention / correction:** any further convention drift between CCDD-internal language and field-standard global-fit conventions in flavour, weak-lensing, or BAO sectors.

Round 10 (queued in §21 of the restructured CCDD) will execute the first cross-channel P41 test and the first joint $w(z)$ reconstruction with DESI DR3 hooks where available; that round will populate the matrix's first true "candidate-signature confirmed" or "candidate-signature failed" entries.

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[1] Ryu & Takayanagi (2006). PRL 96, 181602. [2] Hayden et al. (2016). JHEP 11, 009. [3] Almheiri, Dong & Harlow (2015). JHEP 04, 163. [4] Ambjørn, Jurkiewicz & Loll (2005). PRL 95, 171301. [5] Solà (2013). JPCS 453, 012015. [6] DESI (2025). arXiv:2503.14738. [7] Swingle (2012). PRD 86, 065007. [8] Planck (2020). A&A 641, A6. [9] Watanabe & Oshikawa (2015). PRL 114, 251603. [10] Iemini et al. (2018). PRL 121, 035301. [11] Kovtun, Son & Starinets (2005). PRL 94, 111601. [12] Shapiro & Solà (2002). PLB 530, 10. [13] Randall & Sundrum (1999). PRL 83, 4690. [14] Kibble (1976). JPA 9, 1387. [15] Connes (1996). CMP 182, 155. [16] Furey (2018). PLB 785, 84. [17] Pastawski et al. (2015). JHEP 06, 149. [18] Ben Geloun & Rivasseau (2012). CMP 318, 69. [19] Bahr & Steinhaus (2016). PRL 117, 141302. [20] Zhitnitsky (2003). JCAP 10, 010. [21] Tempel et al. (2014). MNRAS 438, 3465. [22] Finster (2006). AMS/IP. [23] Bombelli et al. (1987). PRL 59, 521. [24] Jacobson (1995). PRL 75, 1260. [25] Verlinde (2011). JHEP 04, 029. [26] NANOGrav Collaboration (2023). ApJL 951, L8.

v7.4 additional references (Round-7 and Round-8 public catalogues):

[27] DESI Collaboration (2024–2025). DESI DR2 BAO+FS public products. <https://data.desi.lbl.gov> [28] ACT Collaboration (2024). DR6 CMB lensing reconstruction. <https://lambda.gsfc.nasa.gov> [29] Planck Collaboration (2020). PR4 (NPIPE) lensing release. [30] NANOGrav Collaboration (2023). 15-year public data release. <https://nanograv.org/data> [31] Lelli, McGaugh & Schombert (2016). SPARC public catalogue. AJ 152, 157. [32] Wisnioski et al. (2015). KMOS3D public release. ApJ 799, 209. [33] Tempel et al. (2014). Public cosmic-web filament catalogue. MNRAS 438, 3465. [34] Douglass et al. (2023). VAST VoidFinder public products. ApJS 265, 7.

v7.5 additional references (evaporation default, dimension-general trigger, P39, P40, and §25):

[35] Henningson & Skenderis (1998). JHEP 07, 023 — holographic Weyl anomaly coefficient c_W . [36] Polchinski (2006). arXiv:hep-th/0603249 — landscape of bulk-geometry candidates. [37] Carroll (2001). LRR 4, 1 — cosmological constant as projection ambiguity. [38] LiteBIRD Collaboration (2023). PTEP 2023, 042F01 — B-mode forecast. [39] CMB-S4 Collaboration (2022). arXiv:2203.08024 — B-mode forecast. [40] DESI Collaboration (2024).

arXiv:2404.03002 — DESI DR2 hint of evolving dark energy. [41] Vafa (2005). arXiv:hep-th/0509212 — Swampland constraints, applicable to v_{bulk} derivation. [42] Diósi (2007). JPA 40, 2989 — gravitational collapse scale, applicable to L_k via Path 9.

Appendix A. Proposed Laboratory Experiment: Flying-Focus Vacuum Tomography (VFT-1) — *out of main core*

This appendix proposes a near-term laboratory experiment — **Flying-Focus Vacuum Tomography (VFT-1)**, also referred to as the **Vacuum-Phonon Pump** — designed as the smallest-scale, highest-leverage test of CCDR’s projection mechanism. It is presented **out of main core**: VFT-1 is an *engineering boost* on top of v7.6, generating a new prediction **SE11 (Sensors & Metrology)** and a **Stage-4 Quantum-Simulator Extension** to the §22 Higher-Dimensional Physics Exploration Programme. The hard-core architecture of CCDR (P1–P42, the falsifiability layers, the cross-links CL1–CL7, the open problems OP1–OP18, the alternative branches) is **independent of VFT-1’s outcome**: if VFT-1 fails to detect the predicted deviation, only SE11 is retracted; if it succeeds, only SE11 and the Stage-4 calibration are added.

A.1 Motivation: the explanatory gap addressed

Standard QED describes vacuum fluctuations via an infinite sum of Feynman diagrams whose probabilities are extracted statistically. It supplies no underlying continuous physical mechanism for *how* those virtual processes actually occur in a non-perfect vacuum. CCDR / Synthesis supplies the missing layer:

- The laboratory vacuum is the **projection** of the bulk vacuum onto our 3+1D acoustic-phonon sector.
- There are orthogonal optical-phonon (dark-sector) modes with a small but non-zero coupling set by the bulk RVM coefficient v_{bulk} (§2.5).
- The dimension-general trigger χ_k (§1.1, §3.1) admits an explicit Weyl-squared term W_k at $k \geq 5$.

Energising the vacuum should reveal deviations from pure perturbative QED that are signatures of (i) real phonon-mode orthogonality, (ii) dimension-general trigger saturation as $\chi_k \rightarrow 1$, and (iii) bulk-vacuum projection fluctuations. **VFT-1 is the cleanest near-term lab-scale probe of this mechanism**: actively pump the vacuum with extreme but controllable energy density, and perform a full tomographic scan of interaction probabilities across multiple Feynman channels.

A.2 Experimental concept

Use a **flying-focus laser beam (FFB)** — a space-time-structured pulse whose focal spot co-propagates with a counter-propagating ultra-relativistic electron or gamma-ray beam at velocity $\approx c$. This dramatically extends the effective interaction time (from femtoseconds to picoseconds or longer) while keeping peak intensity high, allowing measurable strong-field QED effects at orders-of-magnitude lower laser power than fixed-focus setups.

- **Pump**. Multi-petawatt flying-focus laser (baseline: 10–25 PW class, e.g. NSF OPAL design or ELI-NP upgrade). Focus velocity tuned to $-c$ (counter-propagating geometry) to maximise interaction length.

- **Probe / seed.** Ultra-relativistic electron beam (GeV–TeV, laser-wakefield accelerated in the same facility) or high-energy gamma-ray beam (produced via Compton backscattering or bremsstrahlung) counter-propagating into the flying focus.
- **Chamber.** Ultra-high vacuum ($\leq 10^{-10}$ Torr) interaction region with polarisation-preserving X-ray/gamma optics and particle detectors.

Key innovation. The flying focus turns the vacuum into a tunable “nonlinear medium” whose response (pair production, birefringence, higher-order scattering, radiation reaction) can be scanned as a function of local energy density.

A.3 Measurables — full Feynman-diagram probability tomography

Record simultaneously:

- **Vacuum birefringence.** Δn induced by the pump; conversion of circular $\rightarrow$ linear polarisation in the probe; expected Stokes parameter $S_1 \sim 0.01\text{--}0.02$ in optimised geometry.
- **Real e^+e^- pair production yield** and angular/momentum spectra (Schwinger-like tunnelling + multiphoton channels).
- **Higher-order QED scattering** (photon emission/absorption rates, radiation reaction force on electrons).
- **Polarisation-dependent cross-sections** and correlations across dozens of effective Feynman channels.
- **Anomalous noise floor or secular drift** consistent with v_{bulk} projection (predicted fractional fluctuation $\sim 10^{-8}\text{--}10^{-15}$ depending on integration time and volume).

All observables are integrated probabilities from large numbers of virtual diagrams; any systematic deviation from pure QED predictions — especially non-perturbative scaling or new angular structures — would signal underlying phonon-sector physics.

A.4 CCDDR-specific predictions (the programme boost)

Under CCDDR v7.6 + the compactification principle of §3.6:

- Vacuum response is not pure probabilistic QED but **acoustic-phonon propagation** through a locally energised crystal boundary. The optical-phonon sector remains orthogonal until χ_k approaches 1, producing a **sharper-than-QED threshold** for pair production and a calculable suppression of certain higher-order diagrams.
- **v_{bulk} contributes** a small, measurable background fluctuation floor in all channels ($\delta \sim v_{\text{bulk}} \times (t_{\text{int}} \times H_{\infty})^{(-1/2)}$), uncorrelated with classical noise — directly testable against the bulk-vacuum projection prediction (FR9, QC11, SE2).
- The **dimension-general trigger term W_k** appears as a tiny but detectable correction to vacuum polarisation; the flying-focus geometry naturally mimics a small-N laboratory analogue of the cosmological $\chi_k \rightarrow 1$ saturation.
- **Expected signature.** A $\sim 1\text{--}10\%$ deviation in selected channels (especially vacuum birefringence and pair spectra) once energy density exceeds the current QED-validated regime, scaling with the same acoustic–optical orthogonality that protects dark-sector qubits and ball-lightning analogues.

New prediction SE11 (v7.6+ engineering boost). Flying-focus vacuum tomography at $10\text{--}25$ PW will reveal v_{bulk} -coupled deviations from standard QED at the $10^{-3}\text{--}10^{-2}$ level in

polarisation and pair-yield observables, providing the **first laboratory-grade calibration of the bulk-RVM coefficient v_{bulk}** independent of cosmological channels (P39 / P40).

A.5 Feasibility and timeline (2026–2030)

- **Near-term (1–3 years).** Use existing or funded facilities (NSF OPAL design, ELI-NP 10 PW beams, European XFEL HIBEF upgrades) with flying-focus optics already demonstrated in theory and in early experiments.
- **Mid-term (3–5 years).** Dedicated VFT-1 beamline at a multi-PW facility — cost comparable to current strong-field QED campaigns.
- **Diagnostics are mature.** High-resolution gamma polarimetry, pair spectrometers, and Stokes-parameter analysis are already proposed in 2025–2026 literature.

This experiment is **not speculative hardware** — it leverages the exact flying-focus technology now being actively developed for strong-field QED (Di Piazza et al.; Formanek et al.; recent PPPL/SLAC colloquia 2026).

A.6 Why this is a genuine “programme boost” for CCDR / Synthesis

- Directly addresses the explanatory gap of pure QED: replaces statistical Feynman probabilities with measurable **phonon-mode dynamics** in an energised vacuum.
- Provides a **laboratory anchor for v_{bulk}** , the dimension-general trigger, and optical-sector orthogonality — independent of cosmology.
- Fits naturally into **Phase 4 (Bulk-Vacuum Projection Exploitation)** and the **Stage-4 quantum-simulator analogue** of the §22 engineering programme.
- Generates **cross-domain calibration**: the same setup tests fusion-edge noise floors (FR9), qubit decoherence limits (QC11), and metrology floors (SE2–SE10).

A.7 Status

VFT-1 / SE11 is registered as an **out-of-main-core engineering boost**. It does not alter the v7.6 hard-core architecture, the empirical scorecard, or any P1–P42 status. It supplies a *near-term laboratory channel* that — if executed — would give CCDR its first non-cosmological, non-astrophysical empirical handle on the bulk-vacuum projection layer. Round-10 / Round-11 status of VFT-1 readiness is to be reviewed when the next public flying-focus campaign at OPAL / ELI-NP reports its first vacuum-birefringence bandpower.

Appendix B. Double-Slit Experiment as Pedagogical Analogue — *out of main core*

The dependence of double-slit interference patterns on slit geometry and shape is used in this appendix as a **conceptual / illustrative parallel** for the crystallographic-projection physics of CCDR (§3.4 projection map, §8 cosmic web as crystal grain structure, §12 dimensional projection and quantum statistics, P41 / CL6 lattice grain-boundary correction). It is presented **out of main core** because:

- It is **not direct empirical confirmation** of any CCDR prediction (P1–P42).
- It does **not close** any open problem (OP1–OP18).

- It is a **strong pedagogical parallel** that makes several of the framework's least-intuitive claims more intuitive and internally consistent.

B.1 The core CCDR claim that the analogy supports

The universe is a **spacetime crystal** formed by local, asynchronous dimensional reductions (§8 “Cosmic Web as Crystal Grain Structure”, §10). The BAO scale $r_* \approx 147$ Mpc is the lattice constant; the galaxy two-point correlation shows Bragg-like peaks at $k_n = n\pi/r_*$; filaments and walls are grain boundaries; voids are grains (§8.1–8.2). The observed cosmic-web texture (P3 filament-orientation excess, P38 void-wall kurtosis $k_4 > 4$, P30 density-correlated κ) is **density-stratified** because grain-boundary geometry and local reducing-volume shape modulate the projected pattern (§1.2, §4.5, §22 CL4). Quantum statistics and interference itself **emerge from dimensional projection** of a higher-D classical state (§12): the 3+1D wavefunction is the geometric projection of an N-D phase-space vector, and the Born rule appears as a large-N limit.

B.2 Direct laboratory parallel from double-slit physics

In the double-slit (or any multi-aperture) setup:

- **Fringe spacing** $\Delta x = \lambda D/d$ depends only on the **center-to-center separation** d (the “lattice constant”).
- **Envelope shape and visibility** (how many fringes are bright, how sharply they decay, whether minima reach zero) are set by the **single-slit diffraction pattern**, which is the squared Fourier transform of the individual slit geometry (width a , shape, edge profile, orientation, curvature). Rectangular slits $\rightarrow$ sinc^2 envelope. Circular pinholes $\rightarrow$ Airy rings. Non-identical or tilted slits $\rightarrow$ reduced contrast, asymmetry. Finite length or apodised edges $\rightarrow$ suppressed side-lobes.

This is exactly the **separation vs. shape** distinction that CCDR invokes for the cosmic lattice:

| Double-slit optics | CCDR cosmic-web analogue (§8, §3.4, §4.5) | CCDR prediction / observable |

| --- | --- | --- |

| Center-to-center separation d | BAO lattice constant r_* (Bragg peaks in $P(k)$) | Harmonic peaks at $k_n = n\pi/r_*$ |

| Single-slit width / shape | Local reducing-volume geometry + grain-boundary shape | Density-stratified modulation of texture (P3, P30, P38) |

| Envelope = FT[aperture function] | Projected pattern = geometric projection of higher-D defects | Filament orientation excess $\Delta r_{\text{texture}}$, void-wall kurtosis $k_4 > 4$, $\kappa(p)$ |

| Non-identical slits $\rightarrow$ lower contrast | Reducing-volume caveat + patch-to-patch variation (§4.5, OP13) | Patch spread $>$ baseline Δk (Round-8 T2) |

The observed density-stratified cosmic-web signatures (Round 7/8/9 T17, T08, T06) are precisely what one would expect if the “apertures” (reducing volumes / grain boundaries) have geometry-dependent Fourier content that modulates the far-field pattern — exactly as slit shape modulates the interference envelope.

B.3 Deeper resonance with CCDR architecture

- **Projection map (§3.4).** Bulk point defects / strings / walls project to 3D monopoles / filaments / surfaces. The shape of the pre-reduction 5D (or higher-k) region sets the Fourier content of the projected 3+1D metric perturbations — identical to how slit shape sets the diffraction envelope.
- **Layer-3 lattice correction (P41 / CL6).** The CDT simplicial lattice plus residual grain-boundary defects from 5→4 nucleation leave a small vector-like, CP-even shift in the effective Dirac operator. The $b \rightarrow s\mu\mu$ angular anomaly (LHCb arXiv:2512.18053) is interpreted as the low-energy footprint of this lattice “slit geometry”. The double-slit experiment is the textbook demonstration that lattice / aperture geometry imprints a measurable far-field pattern.
- **Emergent QM from projection (§12).** The double-slit interference *itself* is the canonical laboratory demonstration of the wavefunction whose probability distribution arises from higher-D geometry. CCDR’s claim that the Born rule is geometric (large-N projection) is directly illustrated by the fact that changing the physical slit geometry changes the observed interference pattern without changing the underlying wave equation.
- **Boundary-deformation principle (§3.6).** The boundary-deformation spectrum $\varepsilon_{\{lm\}}$ introduced in §3.6 has its direct double-slit analogue in the aperture-function expansion: changing the slit shape redistributes spectral content; changing the reducing-volume shape redistributes the cosmic-web texture spectrum across P3, P30, P38. P42 (boundary-deformation residuals) is the cosmological-scale version of “single-slit envelope analysis”.

B.4 Status

The double-slit analogue is added as a **consistent-with-data illustrative parallel**, not a falsifiable prediction. It is *not falsifiable on its own*, and it does *not* add new free parameters or metaphysical commitments. It strengthens the crystallographic intuition without changing the v7.6 / §3.6 architecture. It dovetails cleanly with the existing Bragg-peak language in §8.1 and provides a textbook handle for §12.

B.5 Bottom line

The double-slit experiment is one of the cleanest everyday demonstrations of the exact mechanism CCDR invokes at cosmological scales: **geometry of the “aperture” (slit / reducing volume / grain boundary) controls the far-field / projected pattern**. It makes the otherwise abstract projection + crystal picture immediately graspable and reinforces the internal coherence of Layers 0–3.